

DECLARATION

I declare that this project work/thesis is my own work. It is being submitted for the degree of Doctor of Philosophy in Mathematics. in the University of Mines and Technology (UMaT), Tarkwa. It has not been submitted for any degree or examination in any other University.

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ABSTRACT

In this thesis, two new probabilistic distributions are developed: modified extended Chen and equilibrium renewal Burr XII. The two new distributions are both uni-variate and continuous probabilistic distributions. The Burr-Hatke differential equation and equilibrium renewal process were used in developing the two novel distributions. Statistical properties of the proposed distributions were derived and inferences were made on them. Monte Carlo simulations were performed to investigate the properties of the maximum likelihood, ordinary and weighted least squares estimators, for the parameters of the developed distributions. In order to compare the developed distributions, the Akaike Information Criterion, Consistent Akaike Information Criterion, Bayesian Information Criterion, Hannan Quinn Information Criterion were used, along with three other main statistic sources: Cramér-Von Mises, Anderson-Darling and Komolgorov-Smirnov tests. The results from the simulation study revealed that the average bias and root mean square error of the maximum likelihood estimators from the simulation study decrease in terms of overall trend as the sample size increases indicating consistency and asymptotic unbiasedness of the estimators. It was evidenced that the proposed distributions have shown good fit when applied to real data sets in the fields of meteorology and health science, when in comparison to other competing models.

DEDICATION

To my father Peter Anafo and mother Janet Awuni.

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LIST OF ABBREVIATIONS

A	Anderson-Darling Test
AB	Average Bias
AIC	Akaike Information Criterion
BIC	Bayesian Information Criterion
CAIC	Consistent Akaike Information Criterion
CDF	Cumulative Distribution Function
CV	Coefficient of Variation
EB	Extended Burr XII
ECDF	Emperical Cumulative Distribution Function
ER-BurrXII	Equilibrium Renewal Burr XII
GIW	Generalised Inverse Weibull
GUM	Gumbel
HQIC	Hannan-Quinn information criterion
IW	Inverse Weibull
KS	Kolmogorov-Smirnov Test
LL	Log-Logistic
MEC	Modified Extended Chen
MLE	Maximum Likelihood Estimation
NAK	Nakagami
ODE	Ordinary Differential Equation
OLS	Ordinary Least Square
RHR	Reversed Hazard Rate

RMSE	Root Mean Square Error
W	Cramér-Von Mises Test
WLS	Weighted Least Square

CHAPTER 1

INTRODUCTION

1.1 Background

Statistical probability distributions have been extensively used to model random phenomena in various fields of study. Their usage cannot be ignored since all parametric statistical analyses heavily depend on distributional assumptions. For instance, the classical regression model makes use of the assumption that the error term is normally distributed. Thus, when such normality assumption is violated inferences resulting from the estimated regression model would be misleading. To overcome some of these challenges the development of new probability distributions for modelling dataset is very imperative so far as parametric statistical analyses are concerned. Alternatively, one may suggest the use of non-parametric methods but they also come with their drawbacks such as loss of power and information when the underlying distribution of the dataset is well known (Allison, 2018; Blossfeld, 2001; Nasiru, 2018a).

Statistical distributions have been employed to model datasets in several areas of study. Actuaries utilise exponential distributions to model natural disasters, enabling them to profitably price insurance policies. Gamma and Beta distributions are commonly utilised in Bayesian statistics, as a great number of posterior distributions are expressed in these forms. Statistical distributions aids researchers to acquire estimates of the likelihood that an event may occur or estimate the variability of occurrences. An in depth knowledge of the technicalities of the concept of statistical probability distributions are necessary for researchers to appropriately apply these distributions to model random events which

enhances humans ability to make decisions under uncertainty.

The quest to develop flexible probability distributions have become an issue of interest to myriad of researchers owing to the usefulness of these distributions in modeling datasets and making inference in areas such as engineering, finance, actuarial, economics and biological modeling among others. The theory underlying the construction of these distributions is mathematically rigorous and ensuring that they are not dubious models. The use of different techniques in constructing distributions has increased during the last decades and so has data sets become complex. One most important approach is by the use of differential equations. Ordinary Differential Equation (ODE) is an important tool applied in many fields of study. In the field of statistics, ODE is used to establish the mode and estimations of parameters of various probability distributions amongst others. Differential equations generate what are classified as systems of frequency in most studies. Some of the most prominent ones include: the Pearson system developed in the early 20th Century by Pearson (1894) and Burr system of distributions (Burr, 1968).

Another approach that is not common but has been used to develop statistical distributions is from the field renewal theory. Ordinary, modified and equilibrium renewal processes are all models generated from renewal theory. Ordinary renewal process is also a process that relates to replacement of devices or components based on failure times (Cox, 1963). Equilibrium renewal process can be employed in constructing density functions. For instance, Voda (2008) in a short paper established the Weibull distribution and other related distributions as generators through the application of the equilibrium renewal process.

Hence, the aim of this study is to develop statistical distributions that provide good parametric fit to datasets obtained from different fields using Burr-Hatke differential equation

and equilibrium renewal process. However, it is worth noting that no single probability distribution can provide good fit to all kinds of datasets. This makes it imperative to develop new ones in addition to the existing ones.

1.2 Research Problem

Myriad of parametric statistical models and tests heavily depends on distributional assumptions. For instance, the classical regression model requires that the response variable should be normally distributed so that the normality assumption of the error term can be achieved. Other regression models such as the beta regression, gamma regression, logistic regression, negative binomial regression, and Poisson regression require that response variable to follow the beta distribution, gamma distribution, binomial distribution, negative binomial distribution, and Poisson distribution respectively. Statistical tests such as the z-test and t-test requires the data to follow normal and t-distributions respectively. Identifying the appropriate distribution for a given parametric statistical analysis, therefore, enhances the quality of the inference and the power of the test in relation to the data. The data generating process in certain fields such as medicine, meteorology, actuarial, finance and others produce data that exhibit some complex features such as kurtosis, skewness, bimodality, monotonic and non-monotonic (bathtub, upside-down bathtub or modified bathtub) failure rates (Bhatti *et al.*, 2018; Nasiru, 2018b). Typical examples of these non-monotonic failure rates include the human mortality life cycle and the machine life cycle. The existing standard distributions such as exponential, normal and gamma among others do not really provide good parametric fit to data sets that exhibit some of these characteristics. Thus, these existing distributions are often used as approximations to model the data sets. This could result in data loss and incorrect statistical inferences.

Because of these challenges, researchers are constantly working to build new statistical distributions for modeling data sets with some of these properties using applicable approaches.

However, there is no single statistical distribution that provides a good fit for all kinds of data sets. It is therefore not trivial to develop new statistical distributions for modeling data sets that show some of these characteristics. With this motivation, this study seeks to develop new distributions using the Burr-Hatke differential equation and the equilibrium renewal process for modeling data sets.

1.3 Research Questions

The study seeks to answer the following research questions.

- i. What are the desirable properties of the modified Chen distribution?
- ii. What are the desirable properties of the Burr XII equilibrium renewal distribution?
- iii. Which of the estimation techniques is suitable for estimating the parameters of the developed distributions?
- iv. How best can the developed distributions fit given data sets?

1.4 Objectives

The general objective of the study is to develop, study the statistical properties and demonstrate the applications of new statistical distributions based on the Burr-Hatke differential equation and equilibrium renewal process. The study seeks to achieve the following specific objectives:

- i. Develop the modified Chen distribution.
- ii. Develop the Burr XII equilibrium renewal distribution.
- iii. Derive the statistical properties of the developed distributions.
- iv. Develop estimators for the parameters of the developed distributions.
- v. Investigates the behavior of the estimators through simulations.
- vi. Illustrate the applications of the developed distributions using data sets.

1.5 Organization of Thesis

There are six chapters in this thesis. Chapter 1 gives the background of the study, research problem, research questions and objectives of the study. Chapter 2 can be regarded as a review of literature on different techniques in proposing statistical distributions over the past decades. Chapter 3 presents some basic concepts and methods used in the research. Chapter 4 presents the development of the modified extended Chen and equilibrium Burr XII distribution. Chapter 4 also includes the statistical properties and estimation techniques of the developed distributions. Chapter 5 covers the simulation study and applications of the developed distribution to data sets. Chapter 6 presents the conclusions and recommendations.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

In this chapter, a critical review of some methods for developing extensions and modifications of distributions are provided.

2.1.1 Method Based on Differential Equations

The method of differential equations is the oldest approach applied in proposing statistical distributions . The method was first developed by Karl Person in the late 19th Century. Pearson (1894) prepared a system of theoretical curves with wide different shapes for adequate representation of observed data as an alternative to the normal distribution using the differential equation approach. In an attempt to derive an alternative distribution to the normal distribution, Pearson's derivation led to the well-known Differential Equation,

$$\frac{1}{p(y)} \frac{dp(y)}{dy} = \frac{y + a}{b_0 + b_1 y + b_2 y^2}, \quad (2.1)$$

where a , b_0 , b_1 and b_2 are real parameters, and solutions $p(y)$ of are different types of curves. Initially, Pearson (1894) developed six types of curves and they include Types I, II, III, IV, V and VI based on equation (2.1). In a final revision done by Pearson (1916), the diagram enabled the classification of six (6) different types of frequency curves (Stuart and Ord, 1994). The Pearson system consists of various important statistical distributions such as Beta distribution (Pearson Type I), Gamma distribution (Pearson Type III) and the normal and student -distributions (Pearson Type VII). In the literature of

Stavroyiannis (2014), there is a generalization of the Pearson differential equation that can be used to construct the vast majority of continuous probability density functions. This is known as the generalized Pearson system of continuous probability distributions. Although many distributions have been developed through the Pearson differential equation approach, they are flexible for fitting specific real world scenarios. Majority of families derived from the Pearson differential equation method such as gamma, beta, normal, hyper-geometric and binomial do not have simple closed form expressions either for the failure rate function to extract an important identity connecting the two.

After Karl Pearson developed the system of distributions through a differential equation approach, Burr (1968) proposed a system of twelve types of distribution functions centred on generating a differential equation to fit cumulative frequency data which was further studied by Gomes *et al.* (2015), Rodriguez (1977), Cordeiro *et al.* (2018). This system of distributions is appealing for model fitting because it combines a simple mathematical equation for the cumulative frequency function with skewness-kurtosis plane coverage. Some important distributions derived from the Burr system include the Uniform Distribution (Burr Type I), Generalized Logistic Distribution (Burr Type II), Generalized Rayleigh Distribution (Burr Type X) and Burr Type XII Distribution which is related to various other distributions, namely, Exponential-Gamma, Weibull-Gamma, Weibull-Exponential and Log-Logistic. In addition some distributions in recent times derived through the use of the Burr-Hatke differential equation include the Extended Chen (Bhatti *et al.*, 2019), Burr-Hatke Exponential (Yadav *et al.*, 2021), and modified exponential distribution (Kim, 2020). According to Jamal *et al.* (2020) despite the numerous advantages of the Burr XII distribution, one limitation of the Burr XII distribution is that it lacks flexibility on the hazard rate function, it exhibits a monotonic feature when

the scale parameter is > 2 , ≤ 2 or between $(1, 2)$ which contributes to a limitation for modelling some lifetime data which require non-monotonic shapes for the hazard rate function like the bathtub shape, the unimodal or modified unimodal shape and many others need various monotonic shapes.

2.1.2 Method Based on Transformation

In an attempt to propose flexible systems based on the drawbacks of the Pearson, and Burr systems, Johnson (1949) through the method of translation developed a system for generating statistical distributions using Normalization Transformation with the general form,

$$Z = \gamma + \sigma f\left(\frac{x - \rho}{\lambda}\right),$$

where $f(\cdot)$ is the transformation function and a monotonic function of and $\gamma, \sigma > 0$ are shape, is a scale parameter, is a location parameter and is a standardized normal random variable. The system has the practical and theoretical advantages of being able to transform these curves to the normal distribution. Johnson (1949) proposed three transformation functions and defined the Log-Normal system (SL), the bounded system of distributions (SB) and unbounded system of distributions (SU). The Johnson system is able to closely approximate many of the standard continuous distributions through one of the three functional forms and hence highly flexible. The Johnson system consists of many distributions such as Normal, Log-normal, Gamma, Beta, and Exponential distributions. Although sub-models like Gamma, Log-normal and Beta distributions obtained from the Johnson system are flexible, they do lead to a wide diversity of distribution shapes. They still do not provide the degree of generality that is frequently desirable.

2.1.3 Method Based on Percentile functions

Hastings Jr *et al.* (1947) and Tukey (1960) developed the Lambda Distribution through the Percentile functions approach. Initially, the Lambda Family of Distributions were proposed through the Percentile function of the form $Q(p)$,

$$Q(p) = \frac{(p^\lambda - (1-p)^\lambda)}{\lambda}, \lambda \neq 0,$$

where p denotes the probabilities $p \in [0, 1]$. Consequently, a four-parameter quantile function lambda distribution was developed by Ramberg and Schmeiser (1972) which is a generalization of the lambda distribution. Although, the lambda distribution can fit the common distributions such as normal, lognormal and Weibull with a good accuracy, Ramberg *et al.* (1979) noted that the proposed generalized lambda distribution is not defined for certain combinations of the parameters. In an attempt to make the Generalized lambda distributions more flexible and adequate for modelling, especially in the reliability of life-time of devices, an extended generalised lambda was proposed by Karian and Dudewicz (2003).

2.1.4 Method of Generating Skew Distribution

Another approach in regard to the development of classical probability distributions was propounded through a mechanism of combining two symmetric distributions. The approach was proposed by Azzalini (1985). In his study, he initiated a method which introduced the skew-normal distribution. The skew normal distributions is a continuous probability distribution which produces thinner tails compared to the normal curves, the distribution can be classified as an extension of the normal distribution. Hence

(1986) proposed a probabilistic representation that explains the skew normal class in terms of a normal random variable and a truncated normal random variable. Gupta and Chen (2004) provided another possible modification of the univariate skew normal model into the vector skew normal models. The extended skew normal distribution family was adapted to the matrix variate case by Ning and Gupta (2012). The epsilon-skew normal family of distributions was presented by Mudholkar and Hutson (2000). In their work, an additional parameter was introduced to control the magnitude of skewness. Elal-Olivero (2010) through the intuition of the skew normal method proposed the log-skew-normal distribution which makes use of the log function in making the distribution more flexible. However, until Bolfarine *et al.* (2011) suggested a bimodal version of the log-skew-normal distribution, the log-skew-normal distribution did not fit bimodal data sets.

2.1.5 Method of Adding Parameters

The method of introducing an additional method is one of the most popular approach in recent times proposing distributions. Mudholkar and Srivastava (1993) pioneered this method in which an introduction of the exponentiated Weibull distribution was proposed by including an additional parameter to the Weibull distribution. The CDF of exponentiated family has the following form

$$H(x; \rho, \xi) = G(x; \xi)^\rho,$$

where $\xi > 0$, $x \in \mathbb{R}$ and ρ is an extra shape parameter. The developed exponentiated distributions are more flexible than other models due to the addition of an extra parameter. The exponentiated distributions include the exponential, gamma, Weibull, Gumbel and Fréchet distributions (Nadarajah and Kotz, 2006b). Using equation (2.1.5), a large

number of modified existing distributions have been developed, some of which can be found in Gupta and Kundu (2001a), Nadarajah (2005), Shirke and Kakade (2006), and Nadarajah and Kotz (2006a). Consequently, Marshall and Olkin (1997) introduced another concept of including a single parameter to an existing family of distributions. Many academics have used this strategy to extend well-known distributions in recent years (Ahmad *et al.*, 2019). Through their approach, they developed the Marshall-Olkin family of distributions. Some special cases of the Marshall Olkin family include the Marshall-Olkin extensions of the Weibull distribution (Ghitany *et al.*, 2007), Pareto distribution (Ghitany *et al.*, 2005) and gamma distribution (Ristić and Nadarajah, 2014).

2.1.6 Transformed-Transformer Method

Development of the transformed-transformer technique was first introduced with the concept of the beta generator proposed by Eugene *et al.* (2002). Later, a generalized version of the beta generated method was developed by Alzaatreh *et al.* (2013) and referred to as transformed-transformer family of distributions. Using the generalized version of the beta generated concept, several new classes of distributions such as the Exponentiated family (Alzaghal *et al.*, 2013), logistic-G family (Torabi and Montazeri, 2014), Gamma-G Type-1 family (Zografos and Balakrishnan, 2009) and most recently the new Weibull family by (Ahmad *et al.*, 2018). Consequently, (Nasiru, 2018a) introduced a new generalisation of the transformed-transformer family. In his work many family distributions such as the EG gamma-X, EG Weibull-X, EG beta-exponential-X, exponentiated gamma-X, exponentiated Weibull-X, exponentiated beta-exponential-X, beta family, Kumaraswamy family and exponentiated family were established.

2.1.7 Method Based on Equilibrium Renewal Process

The equilibrium renewal process model arises from the theory of renewal process. Cox (1963) and Feller (1968) provided an extensive theoretical aspect of the equilibrium renewal process. In their work, they perceived the idea that from a population of failure times distributed according to has the selection of any unit in the population is proportional to its size, the random variable which is the failure time of the component whose life fails in the sample, has density function equation (3.2). The literature on reliability analysis is heavily based on continuous models and the use of the equilibrium renewal theory as an approach in developing statistical distributions has not been properly investigated or exploited. In a recent study, Voda (2008) pioneered the use of the concept of equilibrium renewal process model as a generator of density functions. In his work, the PDF of the Weibull distribution was used as a generator. The new generator consists of the generalized exponential and generalized half-normal distributions as sub-models. Other families such as the Gamma and power distributions were also modified using similar approach.

2.2 Summary of Literature Review

In this chapter, we discussed some methods used in proposing statistical distributions over the decades. Although significant amount of distributions has been developed using varied methods and due to the complexity of emerging data sets from various fields, it is important for the developments of new statistical distributions using these methods in deriving more flexible statistical distributions to meet the growing complexity of data.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter presents some important definitions, basic concepts and methods used in the thesis. The topics discussed include the Burr differential equation, equilibrium renewal process, extended Chen distribution, Burr XII distribution, estimation techniques, goodness-of-fit test, information criteria, data sets and software package.

3.2 Burr-Hatke Differential Equation

Ordinary differential equations (ODEs) are widely used in various fields in science to understand and model real-life problems. In statistical and probabilistic models, ODEs can be useful. It can be used to propose statistical distributions. Pearson (1894) and Burr (1968) used the differential equation method in constructing systems of distributions. Burr (1968) proposed the Burr system of distributions using the concept of differential equations, this differential equation is popularly referred to as the Burr-Hatke differential equation and is given by

$$\frac{d[\xi(y)]}{dy} = \xi(y)[1 - \xi(y)]w(y, \xi(y)), \quad (3.1)$$

such that $\forall y, \xi(y) \geq 0$ and $w(y, \xi(y))$ is a non-negative function for $y \in [0, 1]$. Different functional forms of $w(y, \xi(y))$ yields different solutions.

3.3 Equilibrium Renewal Process

Researchers in the field of reliability analysis are usually interested in probability density functions related to failure phenomena. The notion of renewal theory provided researchers with concepts connected to the failure and replacement of components. The use of the equilibrium renewal process model was employed as a technique in constructing PDFs in the reliability context by Voda (2008). Suppose $\{y_1, y_2, y_3, \dots, y_n\}$ are ordinary renewal process, then a modified density function in reliability context using the equilibrium renewal process is given as;

$$E_p(y) = \frac{1 - G(y)}{\mu}, \quad (3.2)$$

where $G(y)$ is the underlining Cumulative Distribution Function (CDF) and μ is the average lifetime of the underlining CDF (Cox, 1963).

3.4 Extended Chen Distribution

The extended Chen (EC) distribution introduced by Bhatti *et al.* (2019), is derived from the generalized Burr-Hatke differential equation. This is a Burr-Hatke differential equation-based extension of the Chen distribution. The CDF of the EC distribution is given by,

$$G(y) = 1 - \left[1 + k \left(e^{y^\beta} - 1\right)\right]^{-c}, \quad y > 0, \quad (3.3)$$

where $c > 0$, $\beta > 0$ are shape parameters and $k > 0$ is a scale parameter. The PDF corresponding to equation (3.3) is given by

$$g(y) = c\beta k y^{\beta-1} e^{y^\beta} \left[1 + k \left(e^{y^\beta} - 1\right)\right]^{-c-1}, \quad y > 0.$$

The study intends to propose a new distribution by modifying the extended Chen distribution dubbed the modified extended Chen distribution via the use of Burr-Hatke differential equation as a generator by Burr (1968). The modified distribution seeks to exhibit higher flexibility as well as wider applicability.

3.5 Burr XII Distribution

The Burr XII distribution was first proposed by Burr (1968). The CDF of the Burr XII distribution is given by

$$G(y) = 1 - \left(1 + \left(\frac{y}{s}\right)^c\right)^{-k}, \quad y > 0, \quad (3.4)$$

where $c > 0$, $s > 0$ and $k > 0$ are parameters respectively. The corresponding PDF of the Burr XII distribution is given by

$$g(y) = y^{c-1} c k s^{-c} \left(1 + \left(\frac{y}{s}\right)^c\right)^{-k-1}, \quad y > 0. \quad (3.5)$$

The Burr XII distribution have been used in a wide variety of applications, such as actuarial, biological science and economics. The study seeks to modify the Burr XII distribution through the use of the equilibrium renewal process from the concept of renewal theory. The modification seeks to improve the flexibility of the Burr XII distribution in modeling data sets in varied fields of study.

3.6 Estimation Technique

Estimation of the parameters of statistical distributions is a primary area in statistics. Selecting an appropriate estimator is an important task. In practice, constructive techniques of parameter estimations are required, different techniques used to estimate the unknown parameters have been suggested and implemented by researchers. Some of these methods used frequently include the maximum likelihood estimation, ordinary and weighted least square estimation. In this section, we shall discuss the maximum likelihood, moments, weighted and ordinary least squares estimation technique.

3.6.1 Maximum Likelihood Estimation

The method of maximum likelihood estimation is one of the most extensively used and general estimation approaches. The likelihood function is at the foundation of this method for estimating parameters of a probability distribution. Suppose $y_1, \dots, y_n$ are independent and identically distributed random sample of size n with PDF $g(y; \theta)$, where θ is a vector of parameters that characterises $g(y; \theta)$. The joint density is treated as a function of the parameters, and the likelihood function is defined on it as follows;

$$\begin{aligned} L(\theta|y_1, \dots, y_n) &= g(y_1, \dots, y_n; \theta) \\ &= \prod_{i=1}^n g(y_i; \theta) \end{aligned}$$

Then the joint PDF and log-likelihood function may be expressed as $g(y; \theta)$ and $\ell(\theta; y)$ respectively. It is usually difficult to maximise $L(\theta|y)$. Since $\ln(\cdot)$ is a monotone function

with random sampling, the log-likelihood has the particular simple form;

$$\begin{aligned}\ell(\theta|y) &= \ln L(\theta|y) \\ &= \sum_{i=1}^n \ln g(y_i; \theta).\end{aligned}$$

The idea behind maximum likelihood estimation is to find a parameter estimator that maximizes the likelihood function. Thus, if there exist a function $\hat{\theta} = \hat{\theta}(y_1, y_2, \dots, y_n)$ of the sample values which maximises log-likelihood function values for variation in the estimator, then $\hat{\theta}$ is taken as the θ . Setting and solving the initial partial derivatives of the log-likelihood function yields the estimates with respect to $\theta_1, \dots, \theta_k$ to zero. Thus,

$$\frac{\partial \ell(\theta; y)}{\partial \theta_i} = 0,$$

where $i = 1, 2, \dots, k$.

3.6.2 Ordinary and Weighted Least Square

Legendre was the first to propose the least squares method in 1805. Swain *et al.* (1988) were the first to apply the least squares method to estimate the parameters of the Beta distributions. Recently, Gupta and Kundu (2001b), Kundu and Raqab (2005), Alkasabeh (2009), and Al-Mofleh *et al.* (2020) have employed the use of both ordinary and weighted least square methods in estimating parameters of various proposed statistical distributions.

Suppose $Y_1, Y_2, \dots, Y_q$ is a random sample of size q from a distribution function $G(\cdot)$ and $Y_{(1)} < Y_{(2)} < \dots < Y_{(q)}$ denotes the order statistics of the observed sample. The

expectation and variance is given by,

$$E \left(G(y_{(i)}) \right) = \frac{i}{q+1}$$

and

$$V \left(G(y_{(i)}) \right) = \frac{i(q-i+1)}{(q+1)^2(q+2)}$$

where, $i = 1, 2, 3, \dots, n$. respectively. Applying the expectation and variance. The least squares estimators are obtained by minimizing,

$$OLS = \sum_{i=1}^n \left\{ \left[G(y_{(i)}) - \frac{i}{q+1} \right] \right\}^2$$

with respect to the unknown parameter(s). Hence, in relation to distributional analysis one can find the least squares estimates and root mean square error of unknown parameter(s) by using extensive simulation techniques.

The weighted least squares estimators can be derived by minimizing,

$$WLS = w_i \sum_{i=1}^n \left\{ \left[G(y_{(i)}) - \frac{i}{q+1} \right] \right\}^2$$

with respect to the unknown parameters, where

$$w_i = \frac{(q+1)^2(q+2)}{i(q-i+1)}, \quad i = 1, 2, 3, \dots, q.$$

As a result, utilizing comprehensive simulation techniques, one may discover the weighted least squares estimates and root mean square error of unknown parameters for any distribution.

3.7 Methods of Evaluating Parameter Estimators

Assume $Y_1, Y_2, \dots, Y_n$ is a random sample of size n from the sampling model $g(x|\theta)$, where θ is an unknown parameter. An estimator of θ obtained by method such as maximum likelihood estimation, ordinary and weighted least square estimation, is a function of the sample, that is a statistic $\hat{\theta} = T(Y_1, Y_2, \dots, Y_n)$. To study the performance of the estimator, Root Mean Square Error (RMSE) and Average Bias (AB) are employed.

3.7.1 Root Mean Square Error

Let $\hat{\theta}$ be the estimator of the unknown parameter θ from the random sample $Y_1, Y_2, \dots, Y_n$. Then the deviations from $\hat{\theta}$ to the true parameter, measures the quality or performance of the estimator. That is, the RMSE of an estimator $\hat{\theta}$ defined by

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{\theta}_i - \theta)^2},$$

where N is the sample size.

3.7.2 Average Bias

The AB of an estimator $\hat{\theta}$ of the parameter θ is the difference between the expected value of $\hat{\theta}$ and θ . That is,

$$AB = \frac{1}{N} \sum_{i=0}^N (\hat{\theta}_i - \theta),$$

where N is the sample size.

3.8 Return Period and Level

Return periods are estimated by statistical analysis applied to extreme observations. The area of statistics on which analytical methods are based is called extreme value analysis. Extreme value analysis allows us to estimate the probability of events that are more extreme than those in the data set. The return level can be defined as the level which is associated to be exceeded one every $1/S(y_T)$ period, where $S(y_T)$ is the exceedance probability which is obtained from the survival function, known as a return period. The return level, say y_T , exceeded on average once in T years is defined as

$$T = \frac{1}{S(y_T)}$$

and the return period is given by

$$y_T = F^{-1} \left(1 - \frac{1}{T} \right).$$

3.9 Goodness-Of-Fit Tests

A statistical model's goodness-of-fit (GOF) test determines how well it fits a given data set. The disparity between observed values and expected values under a statistical model is summarized by GOF values. The study seeks to employ the Kolmogorov-Smirnov Test (KS), Anderson-Darling Test (A) and Cramér-Von Mises Test (W) tests.

3.9.1 Kolmogorov-Smirnov Test

The Kolmogorov-Smirnov (KS) test is used to determine whether a sample belongs to a population with a particular distribution. The KS test is based on the Empirical

Cumulative Distribution Function (ECDF). Suppose that we have observations $y_1, \dots, y_n$ which we consider from a distribution P . Suppose the CDF $G(y)$ of a random variable Y , is

$$G(y) = P(Y \leq y).$$

Given, observations, $y_1, \dots, y_n$ the ECDF G_{obs} gives the proportion of the data that lies below y ,

$$G_{obs}(y) = \frac{y_i}{n},$$

where, y_i is the number of samples and n is the number of observations. To compare the EDF of the data, G_{obs} , with the CDF associated with the null hypothesis G_{exp} (expected CDF). The KS statistic is given by,

$$KS = \{ |G_{exp}(y_i) - G_{obs}(y_i)|, |G_{exp}(y_i) - G_{obs}(y_i)| \}$$

where $i = 1, 2, \dots, n$.

3.9.2 Cramér-Von Mises Test

The Cramér-Von Mises (W) test is used to test if a random sample $y_1, y_2, \dots, y_n$ came from a specific distribution. The W test basically relies on the specific distribution to calculate the critical values. Let $G(y; \theta)$ be the CDF and W be the Cramér-Von Mises. The W test is derived as follows;

- i. Compute the distribution function $v_i = G(y_i, \theta)$, where the data are ordered in an ascending manner are in ascending order and then compute $y_i = \phi^{-1}(\cdot)$ is the quantile function of the standard normal $N(0, 1)$.

ii. Compute the standard values of y_i , which can be obtained as

$$p_i = (y_i - \bar{y})/s_y,$$

where $(\bar{y})$ is the mean of y_i , $i = 1, 2, \dots, n$ and s_y is the standrad. deviation of y_i .

iii. Then the W test statistic can be written as

$$W^2 = \sum_{i=1}^n \left\{ p_i - \frac{(2i-1)}{2n} \right\}^2 + \frac{1}{12n}.$$

iii. To compare models we employ the use of the modified version of the statistics given by

$$W = W^2 \left(1 + \frac{0.5}{n} \right).$$

When comparing between two models by these tests, we select the model with smaller value.

3.9.3 Anderson-Darling Test

The Anderson-Darling (A) test was proposed by Anderson and Darling (1952). The A test, like the W test, is used to determine whether a random sample originated from a given distribution. The A test was created to emphasize distribution fit in the tails. (Stephens, 1974). Let $G(y; \theta)$ be the CDF and A be the Anderson-Darling. The A test is computed as follows;

- i. Compute the distribution function $v_i = G(y_i, \theta)$, where the data are in ascending order and then compute $y_i = \phi^{-1}(\cdot)$ is the quantile function of the standard normal $N(0, 1)$.

ii. Compute the standard values of y_i , which can be obtained as

$$p_i = (y_i - \bar{y})/s_y,$$

where $(\bar{y})$ is the mean of y_i , $i = 1, 2, \dots, n$ and s_y is the standard. deviation of y_i .

iii. Then the A test statistic can be written as

$$A^2 = -n - \frac{1}{n} \sum_{i=1}^n \{(2i - 1) \log(u_i) + (2n + 1 - 2i) \log(1 - u_i)\}$$

where $\log(\cdot)$ is the natural logarithm.

iii. To compare different models we employ the use of the modified version of the statistics, A^2 given by

$$A = A^2 \left(1 + \frac{0.75}{n} + \frac{2.25}{n^2} \right).$$

When comparing between two models by these tests, we select the model with smaller value.

3.10 Information Criteria

When comparing multiple models, we consider criteria like the Akaike Information Criterion (AIC), Consistent Akaike Information Criterion (CAIC), Hannan-Quinn information criterion (HQIC) and Bayesian Information Criterion (BIC). These criteria methods are used to choose a model from a group of models for a given set of data. The better model corresponds to lesser AIC, CAIC, BIC and HQIC values.

3.10.1 Akaike Information Criterion

Akaike (1974) provided an information theory interpretation of the likelihood function and extended it to define a criterion that is used to test the goodness of assumed models. His criterion popularly known as the AIC in majority literature is a criterion for selecting linear models. Many researchers such as Sakamoto *et al.* (1986) applied the AIC to almost all statistical problems. Let Δ be the maximum value of the model's likelihood function and the number of estimated parameters in the model is given by k . The AIC estimate for the model is thus given by,

$$AIC = -2 \log \Delta(\hat{\theta}) + 2k.$$

Suppose for a given data set, the AIC value of several competing models may be calculated, The model with the lowest AIC value is preferred.

3.10.2 Bayesian Information Criterion

Schwarz (1978) proposed the Bayesian Information Criterion (BIC). The BIC is a model selection criterion for a class of models with varying numbers of parameters; the model with the lowest BIC is preferred. It has a lot of similarities with AIC. The penalty for additional parameters is higher in the BIC than it is in the AIC. The BIC is determined by,

$$BIC = -2 \log \Delta(\hat{\theta}) + k \log(n), \tag{3.6}$$

where $\Delta(\cdot)$ is the number of estimated parameters, k is the value of the Log-likelihood function and n is the sample size.

3.10.3 Corrected Akaike Information Criterion

The Corrected Akaike Information Criterion (CAIC) introduced by Hurvich and Tsai (1989), was designed as an estimator of the expected Kullback-Leibler discrepancy between the model generating the data and a fitted candidate model. Hurvich and Tsai (1989) derived CAIC which includes a criterion for small sample sizes. CAIC is similar to AIC, but with a higher penalty for additional parameters. The formula for the CAIC is given as,

$$CAIC = AIC + \frac{2k(k+1)}{n-k-1},$$

where k is the number of estimated parameters in the model.

3.10.4 Hannan-Quinn information criterion

The HQIC is an alternative of the BIC, AIC and CAIC. The formula for the HQIC is given by,

$$HQIC = -L_{max} + 2k \log(\log(n)),$$

where L_{max} is the log-likelihood, k is the number of parameters, and n is the number of sample (Hurvich and Tsai, 1989). The HQIC is meant to have the smallest rising penalty term while maintaining great consistency in the information criterion.

3.11 Data Sets

To demonstrate the applicability of the distributions developed, a total of eleven secondary data sets from the field of meteorology and health sectors are used.

3.11.1 Annual Maximum Rainfall Data

The annual maximum of the average monthly rainfall data from five locations namely, Koforidua, Accra, Cape coast, Bolgatanga and Ho in Ghana are considered for the period 1980-2019. Monthly average rainfall in centimeter(cm) data used in the study were retrieved from (<https://www.globalclimatemonitor.org/>) using geographic coordinates in simple decimal standard format of the locations. The names and geographic locations (simple decimal standard) applied in the research are displayed in Table 3.1. From Table 3.1, it can be observed that all geographic locations can be found within the geographic coordinates of Ghana.

Table 3.1: Geographic locations

Locations	Latitude	Longitude
Accra	5.55602	-0.1969
Koforidua	6.09408	-0.2591
Ho	6.60084	0.4713
Cape Coast	5.10535	-1.2466
Bolgatanga	10.7875	0.8580

3.11.2 Remission and survival times data

The remission of patients suffering from bladder cancer data in months consist of a random sample of 128 patients reported by Lee and Wang (2003). The data was previously studied by Lemonte (2013), Zea *et al.* (2012), and Lemonte and Cordeiro (2013). The data is listed in Table A1 in Appendix A.

The survival times data sets, consists of the survival times in weeks of 33 patients suffering from acute myelogenous leukaemia. Feigl and Zelen (1965) introduced these data, which were then analyzed by Mead *et al.* (2017) and Mustafa *et al.* (2016). This data

set is known for exhibiting a bathtub hazard shaped. The data is listed in Table A2 in Appendix A.

3.11.3 Length of stay data

The length of stay data consist of monthly length of stay data of patients in the maternity wards of public hospitals from four district in the Northern region of Ghana from the period of 2014-2020. Monthly length of stay of patients in the maternity ward in hours data used in the study were retrieved from (GHS,2021).

3.12 Software Package

The R software is a statistical software and language developed by Ross and Robert in 1996. At present, it is freely available in the website <http://www.R-project.org/>. Several statistical packages such as SPSS, SAS, S- PLUS, MINITAB, MATLAB and MATHEMATICA are also available in R. The R software programming is extensively used through out the study. In the study, the following packages are used for different computations;

- i. bbmle: For optimization of log likelihood function and to estimate the parameter estimation.
- ii. goftest: For calculating the Anderson-Darling (A), Cramér-von Mises (W) and Kolmogorov-Smirnov Test (KS) statistics.
- iii. hypergeo: For calculating the hypergeometric function.
- iv. numDeriv and Matrix: For computing the algorithm for Monte Carlo Simulation Study.

3.13 Summary of Methodology

This chapter provided definitions of the Burr-Hatke differential equation, equilibrium renewal process, extended Chen distribution and Burr XII distribuion. Methods of parameter estimations, such as the method of maximum likelihood, ordinary and weighted least squares estimation are discussed together with the methods of evaluating the parameters are also presented. Goodness-of-fit tests, thus, Cramér-Von Mises (W) test, Anderson-Darling (A) test and Kolmogorov-Smirnov (KS) test are presented. Model selection criteria like the Akiake Information Criterion (AIC), Corrected Akiake Information Criterion (CAIC), Hannan-Quinn Information Criterion (HQIC) and the Bayesian Information Criterion (BIC) are also defined. Description of secondary data sets used in the study are explained in relation to their sources. Finally, software packages used for computations are discussed.

CHAPTER 4

THEORETICAL RESULTS

4.1 Introduction

This chapter presents the derivation of the modified extended Chen and equilibrium renewal Burr XII distributions. Useful expansion of the PDF of the proposed modified extended Chen distribution is derived as a series representation of the modified extended distribution. For both the modified extended Chen and equilibrium renewal Burr XII Distributions, statistical properties such as the quantile function, moments, incomplete moments, moment generating function, function, entropy, mean deviations, order statistics, stress strength, inequality measures, and stochastic ordering have been derived and discussed. Methods of estimations for the proposed modified extended Chen and equilibrium renewal Burr XII are also presented.

4.2 Formulation of the Modified Extended Chen

Let the random variable Y follow the Modified Extended Chen (MEC) distribution with positive parameters $a > 0$, $b > 0$ and $\rho > 0$. Let $y \in \mathbb{R}^+$, consider the general solution of the Burr-Hatke differential equation ,

$$F(y) = \left[\exp \left\{ - \int w[y, \xi(y)] dy \right\} + 1 \right]^{-1}. \quad (4.1)$$

Suppose the function $w[y, \xi(y)]$ defined as the quotient of the inverse of the extended Chen PDF to the inverse of its CDF is given by

$$\frac{ab\rho y^{-b-1}e^{y^{-b}} \left[1 + \rho \left(e^{y^{-b}} - 1\right)\right]^{a-1}}{1 - \left[1 + \rho \left(e^{y^{-b}} - 1\right)\right]^a}, \quad (4.2)$$

the CDF of the random variable Y is obtained by substituting equation (4.2) into equation (4.1) and simplifying the integral. Hence, the CDF of the MEC distribution is given by

$$F(y) = \left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-a}, y \geq 0, \quad (4.3)$$

where $a > 0$ and $b > 0$ are the shape parameters and $\rho > 0$ is a scale parameter. The shape parameters a and b controls the shape of the distribution of the random variable Y and scale parameter ρ controls the variability of the distribution.

Figure 4.1 displays the CDF of the MEC distribution for different parameter values. It can be observed that for some parameter values, as y gets closer to zero the CDF approaches zero and as y gets closer to one the CDF approaches one.

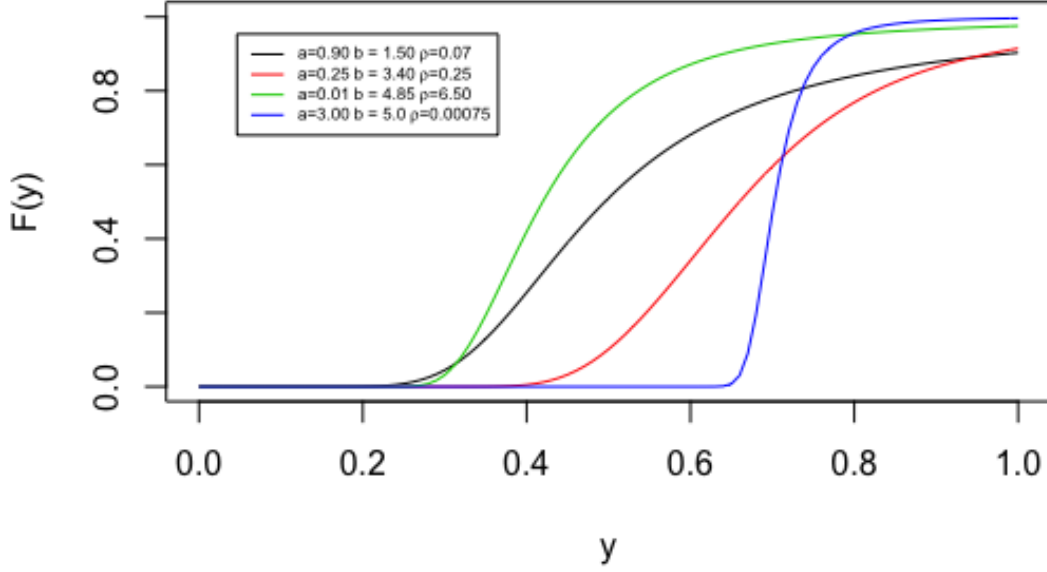


Figure 4.1: CDF plots of the MEC distribution.

The PDF of the MEC distribution is obtained by differentiating equation (4.3) with respect to y . Hence, the PDF is given by

$$f(y) = ab\rho y^{-b-1} e^{y^{-b}} \left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-a-1}, y > 0.$$

In Figure 4.2, the density plots of the MEC distribution are shown. The density plots for various parameter values exhibit different kinds of shapes, most of the plots are uni-modal in shape with different degrees of kurtosis. For the various parameter values that were used, it can be observed that most of the plots exhibits right skewed shape. In addition, for some parameter values the density exhibits a reverse J shape.

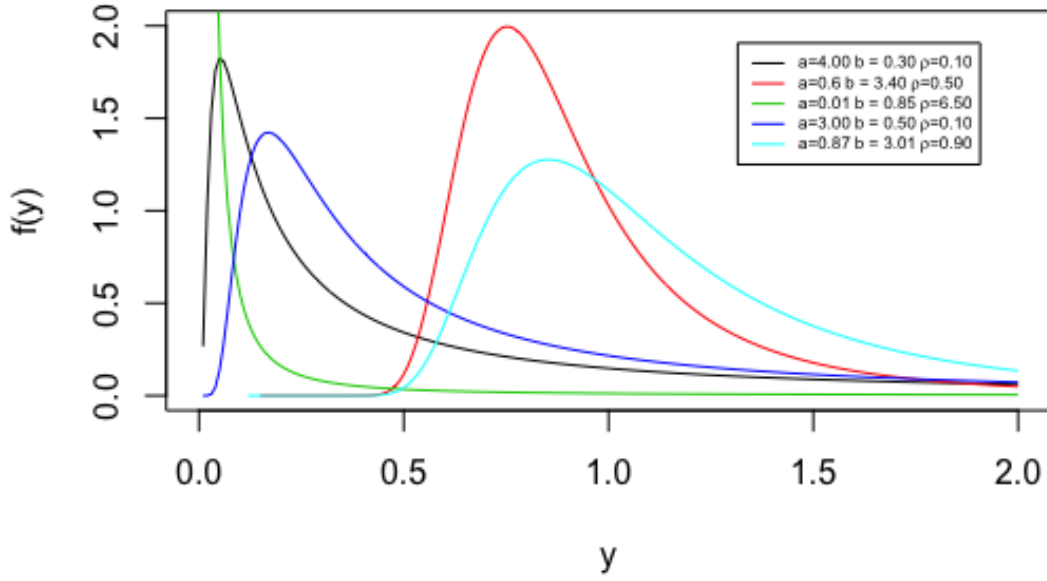


Figure 4.2: PDF plots of the MEC distribution.

The survival function is the complement of the CDF. The survival function of the MEC distribution is given by,

$$S(y) = 1 - \left(\rho \left(e^{y^{-b}} - 1 \right) + 1 \right)^{-a}.$$

For some parameter values, Figure 4.3 shows the various shapes of the survival function for the MEC distribution. Figure 4.3 shows that for some parameter values the survival function decays very fast as it approaches 1 which indicates a low reliability. In addition, for some parameter values the survival function decays slowly as it approaches one indicating a high reliability.

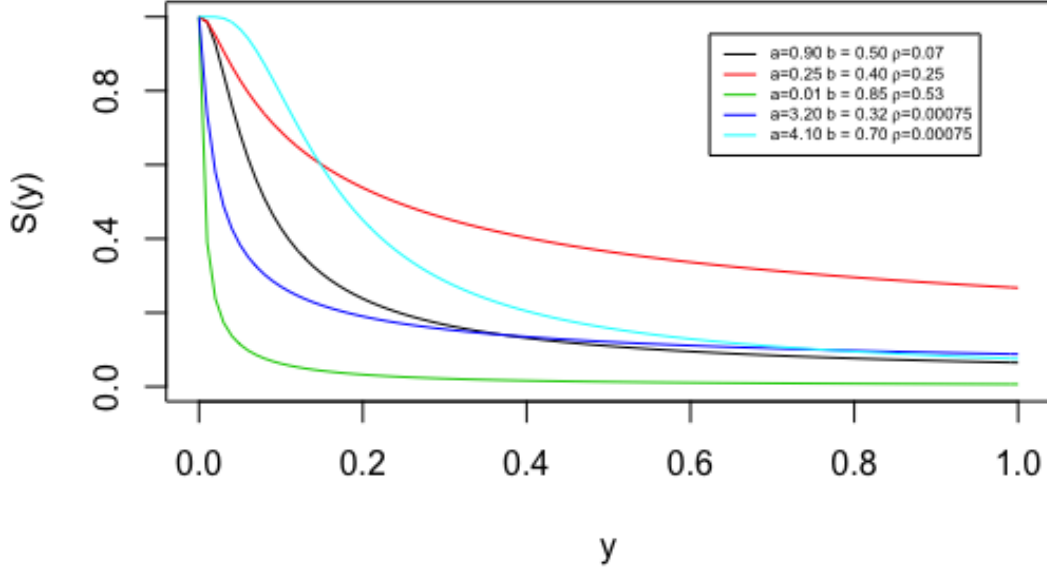


Figure 4.3: Survival function plots of the MEC distribution.

The hazard function, is a primary tool in the field of data analysis because the nature of the hazard could determine the type of distribution that could be used in molding data sets. The hazard function of the MEC distribution is given by,

$$h(y) = \frac{f(y)}{S(y)} = \frac{ab\rho y^{-b-1}e^{y^{-b}} \left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-a-1}}{1 - \left(\rho \left(e^{y^{-b}} - 1 \right) + 1 \right)^{-a}},$$

where $y > 0$.

Figure 4.4 exhibits the plots of the hazard function of the MEC distribution for different parameter values. We can observe that the hazard function has an inverted J shape or an exponential decay for specific parameter values. Furthermore, for a few selected parameter values, the hazard plot shows an upside down bathtub shape, which is an indication that the MEC distribution is appropriate for modeling lifetime data.

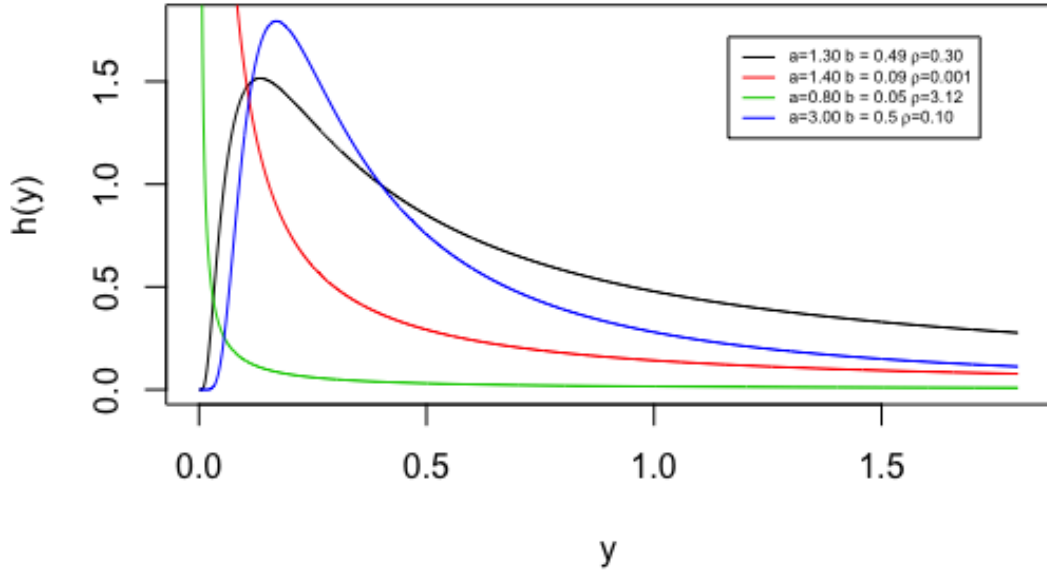


Figure 4.4: Plot of hazard rate function of the MEC distribution for some values of b , ρ and a .

In the fields of engineering and medicine, the Reversed Hazard Rate (RHR) function is quite useful. The function is typically used to assess waiting times and hidden failures.

The RHR of the MEC distribution is defined as,

$$RHR(y) = \frac{f(y)}{F(y)} = \frac{ab\rho e^{y^{-b}} y^{-b-1}}{\rho(e^{y^{-b}} - 1) + 1},$$

where $y > 0$. Figure 4.5 exhibits the RHR function plots of the MEC distribution for varied parameter values. It is observed that the function decays faster near 2.0 on the y-axis for particular parameter values. Also, it can be clearly observed that the RHR exhibits an upside down bathtub feature.

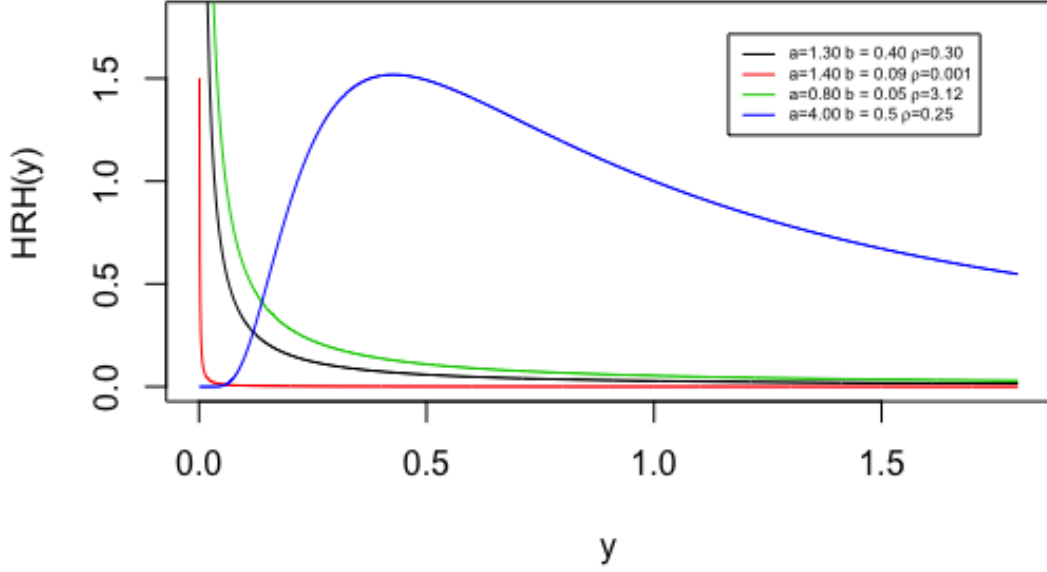


Figure 4.5: RHR function plots of the MEC distribution.

4.2.1 Useful Expansion

We provide an alternative form for the MEC distribution's PDF. Through the concept of power series, the density function of the MEC distribution is expressed in a mixture form in order to obtain certain statistical features of the distribution.

Lemma 4.1. The density function of the MEC distribution can be expressed in a series representation as

$$f(y) = ab\rho \sum_{i=0}^{\infty} \sum_{j=0}^0 \varpi_{ij} y^{-b-1} e^{-(j-i-1)y^{-b}}, \quad (4.4)$$

where $\varpi_{ij} = \frac{(-1)^{i+j} \rho^i (a+1)^{(i)}}{i} \binom{i}{j}$.

Proof. Given the density function

$$f(y) = ab\rho y^{-b-1} e^{y^{-b}} \left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-a-1} \quad y > 0.$$

Using the Taylor series expansion,

$$Z^{-\alpha} = \sum_{i=0}^{\infty} \frac{(-1)^i \alpha^{(i)}}{i!} (Z-1)^i,$$

where $\alpha^{(i)} = \alpha(\alpha+1)(\alpha+2)\dots(\alpha+i-1)$ is the rising factorial. Thus,

$$[1 + \rho(e^{y^{-b}} - 1)]^{-(a+1)} = \sum_{i=0}^{\infty} \frac{(-1)^i (a+1)^{(i)}}{i!} \rho^i (e^{y^{-b}} - 1)^i.$$

Further,

$$\sum_{i=0}^{\infty} \frac{(-1)^i (a+1)^{(i)}}{i!} \rho^i (e^{y^{-b}} - 1)^i = \sum_{i=0}^{\infty} \frac{(-1)^i \rho^i (a+1)^{(i)}}{i!} e^{iy^{-b}} (1 - e^{-y^{-b}})^i.$$

Employing the fact that $0 < (1 - e^{-y^{-b}})^i < 1$ and applying the binomial expansion, we have

$$\sum_{i=0}^{\infty} \frac{(-1)^i (a+1)^{(i)}}{i!} \rho^i (e^{y^{-b}} - 1)^i = \sum_{i=0}^{\infty} \sum_{j=0}^i \frac{(-1)^{i+j} \rho^i (a+1)^{(i)}}{i!} \binom{i}{j} e^{-(j-i)y^{-b}}.$$

The density function can therefore be written as

$$f(y) = ab\rho \sum_{i=0}^{\infty} \sum_{j=0}^i \frac{(-1)^{i+j} \rho^i (a+1)^{(i)}}{i!} \binom{i}{j} y^{-b-1} e^{-(j-i-1)y^{-b}}.$$

This completes the proof.

4.2.2 Statistical Properties of the Modified Extended Chen distribution

In this section, the structural properties of the MEC distribution such as the quantile function, moments, moment generating function, incomplete moment, entropy, order statistic, stochastic ordering and inequality measures are derived and discussed.

Quantile Function

The quantile function is the inverse of the distribution function. Through the use of the quantile function, various characteristics such as the skewness, kurtosis and median of the random variable can be computed (Chen, 2000).

Given a random variable Y with continuous and strictly monotonic PDF $f(y)$, a quantile function of a statistical distribution is expressed mathematically as,

$$Q_f(p) = \{y : Pr(Y \leq y) = p\}, \quad (4.5)$$

where $p \in (0, 1)$.

Proposition 4.1. Given a random variable Y with CDF $F(y; a, b, \rho)$, The quantile function for the MEC distribution is given by

$$Q(p) = \left[\log \left(\frac{p^{-\frac{1}{a}} - 1}{\rho} + 1 \right) \right]^{-\frac{1}{b}}, \quad 0 < p < 1. \quad (4.6)$$

Proof. Suppose the random variable U follows the standard uniform distribution then $0 < p < 1$.

Let $p = \left[1 + \left(e^{y^{-a}} - 1\right)\rho\right]^{-b}$. The MEC distribution's quantile function is obtained by solving for y .

Table 4.1 displays the quantile values of the MEC distribution for some values $u \in (0, 1)$ and choice parameter a, b and ρ . From Table 4.1 we observe that as u increases towards one for various parameter values, the quantile also increases. The quantile values generally increases as parameter values increases. In general, as parameter values increase the quantile values generally increases in value.

Table 4.1: Some values of the quantile of the MEC distribution for some values.

$Q(u)$	$a = 0.4, b = 0.5, \rho = 0.25$	$a = 2.5, b = 3.4, \rho = 1.5$	$a = 2.5, b = 1.5, \rho = 3.25$
$Q(0.1)$	0.0196	1.1086	1.8995
$Q(0.25)$	0.0343	1.2396	2.5517
$Q(0.35)$	0.0526	1.3550	3.2054
$Q(0.45)$	0.0772	1.4717	3.9434
$Q(0.5)$	0.1128	1.5998	4.8449
$Q(0.65)$	0.1695	1.7511	6.0362
$Q(0.75)$	0.2740	1.9456	7.7774
$Q(0.85)$	0.5227	2.2300	10.7626
$Q(0.9)$	1.5987	2.7694	17.94244

The MEC distribution's quantile function can be used to simulate a random sample of size n from the distribution as follows

$$x_i = \left[\log \left(\frac{u_i^{-\frac{1}{a}} - 1}{\rho} + 1 \right) \right]^{-\frac{1}{b}}, i = 1, 2, \dots, n.$$

In instances where the moments does not exist due to the fact that some of the moments may be infinite. We employ the use of the quantile function to find certain measures such as the kurtosis, skewness and quartiles through the use of Galton's skewness and Moor's

kurtosis introduced by Galton (1883) and Moors (1988). The measure of skewness S and kurtosis K as defined by Galton (1883) and Moors (1988) is given by

$$S(p) = \frac{Q\left(\frac{6}{8}; p\right) - 2Q\left(\frac{4}{8}; p\right) + Q\left(\frac{2}{8}; p\right)}{Q\left(\frac{6}{8}; p\right) - Q\left(\frac{2}{8}; p\right)}$$

and

$$K(p) = \frac{Q\left(\frac{7}{8}; p\right) - Q\left(\frac{5}{8}; p\right) + Q\left(\frac{3}{8}; p\right) - Q\left(\frac{1}{8}; p\right)}{Q\left(\frac{6}{8}; p\right) - Q\left(\frac{2}{8}; p\right)}.$$

Table 4.2 displays numerical values of the Moors kurtosis for varied parameter values. The Moors kurtosis of the MEC distribution grows in value as the parameter values increase. Also, from Table 4.2, it can be noticed that the MEC distribution displays both platykurtic ($K < 3$) and leptokurtic ($K > 3$) characteristics.

Table 4.2: Numerical values of the Moors kurtosis of the MEC distribution various values of a , b and ρ

$\rho = 0.5$			$\rho = 1.7$			$\rho = 8.6$		
a	b	K	a	b	K	a	b	K
0.1	0.2	1.8389	0.5	1.5	2.2036	0.1	0.5	3.8613
0.2	0.2	1.8884	2.0	2.0	2.1462	0.2	0.6	2.8692
0.3	0.5	1.9489	5.8	3.5	2.1435	0.3	0.7	2.5041
0.8	1.5	2.0767	6.0	4.5	2.1422	0.4	0.8	2.3480
1.8	2.3	2.1228	6.5	5.5	2.1462	0.5	1.5	2.2712
2.5	3.6	2.1309	7.0	6.0	2.1421	0.7	2.0	2.2049
4.0	5.5	2.1371	7.1	6.2	2.1420	1.7	3.5	2.1514
5.5	7.0	2.1392	8.2	6.6	2.1420	2.5	4.0	2.1462
6.1	8.2	2.1396	9.1	7.0	2.1420	4.0	5.0	2.1435
7.2	9.2	2.1402	10.1	8.0	2.1420	9	6.1	2.1421

Table 4.3 displays numerical values of the Galtons skewness for some parameter values. The skewness of the MEC distribution reveals positive skew values for some parameter values, indicating that the right side of the distribution's tail is longer than the left side, and the majority of the values lie to the left of the mean. The MEC distribution, on the

other hand, shows negative skewed values for some parameter values, indicating that the tail on the left side of the distribution is longer than the tail on the right side.

Table 4.3: Numerical values of the Galtons skewness of the MEC distribution various values of a , b and ρ

$\rho = 0.2$			$\rho = 2.8$			$\rho = 9.6$		
a	b	S	a	b	S	a	b	S
0.2	1.6	0.2130	0.2	1.6	0.4098	0.2	1.6	0.5276
	2.5	0.1617		2.5	0.4479		2.5	0.5757
	3.5	0.1751		3.5	0.4673		3.5	0.6001
	4.0	0.1792		4.0	0.4780		4.0	0.6078
1.5	1.6	-0.0079	1.5	1.6	-1.0380	1.5	1.6	-3.8380
	2.5	0.0967		2.5	-0.5191		2.5	-2.3071
	3.5	0.1498		3.5	-0.2555		3.5	-1.5294
	4.0	0.1664		4.0	-0.1091		4.0	-1.2864
2.5	1.6	-0.1206	2.5	1.6	-2.2692	2.5	1.6	-8.0061
	2.5	0.0516		2.5	-1.3110		2.5	-4.9804
	3.5	0.1254		3.5	-0.8243		3.5	-3.4435
	4.0	0.1485		4.0	-0.6722		4.0	-2.9633
4.0	1.6	-0.2265	4.0	1.6	-4.1436	4.0	1.6	-14.4053
	2.5	-0.0260		2.5	-2.5127		2.5	-9.0789
	3.5	0.0759		3.5	-1.6844		3.5	-6.3734
	4.0	0.1077		4.0	-1.4255		4.0	-5.5280
10	1.6	-0.7677	10	1.6	-11.6976	10	1.6	-40.2794
	2.5	-0.3629		2.5	-7.3493		2.5	-25.6412
	3.5	-0.1574		3.5	-5.1406		3.5	-18.2059
	4.0	-0.0931		4.0	-4.4504		4.0	-15.8823

Moments

In this section we derive the r^{th} moment for the MEC distribution. Moments have a crucial role in statistical analysis, especially in applied applications. Important characteristics of a distribution, such as central tendency, dispersion, skewness, and kurtosis, are obtained and investigated using moments.

Proposition 4.2. The r^{th} moment of the MEC distributed random variable Y is given

as

$$\mu'_r = a\rho \sum_{i=0}^{\infty} \sum_{j=0}^i \varpi_{ij} \frac{\Gamma(1 - \frac{r}{b})}{(j - i - 1)^{1 - \frac{r}{b}}}, r < b, r = 1, 2, \dots, \quad (4.7)$$

where $\Gamma(\beta) = \int_0^{\infty} t^{\beta-1} e^{-t} dt$ is the gamma function.

Proof. By definition, the r^{th} non-central moment is given by

$$\mu'_r = \int_0^{\infty} y^r f(y) dy.$$

Applying the series representation of the PDF in Lemma (4.1), we derive

$$\begin{aligned} \mu'_r &= \int_0^{\infty} y^r ab\rho y^{-\gamma-1} \sum_{i=0}^{\infty} \sum_{j=0}^i \varpi_{ij} e^{-(j-i-1)y^{-b}} dy \\ &= ab\rho \sum_{i=0}^{\infty} \sum_{j=0}^i \varpi_{ij} \int_0^{\infty} y^{r-b-1} e^{-(j-i-1)y^{-b}} dy. \end{aligned}$$

Let $u = (j - i - 1)y^{-b}$. As $y \rightarrow 0, u \rightarrow \infty$ and as $y \rightarrow \infty, u \rightarrow 0$. Also, $y = \left(\frac{u}{j - i - 1}\right)^{-1/b}$ and $dy = \frac{-du}{b(j - i - 1)y^{-b-1}}$. Thus,

$$\begin{aligned} \mu'_r &= a\rho \sum_{i=0}^{\infty} \sum_{j=0}^i \varpi_{ij} \int_0^{\infty} \frac{u^{-r/b} e^{-u}}{(j - i - 1)^{1 - r/b}} du \\ &= a\rho \sum_{i=0}^{\infty} \sum_{j=0}^i \varpi_{ij} \frac{\Gamma(1 - \frac{r}{b})}{(j - i - 1)^{1 - \frac{r}{b}}}, r < b. \end{aligned}$$

This completes the proof.

The mean, median, variance, skewness, and kurtosis for various values of a , b , and ρ are calculated using the r^{th} raw moment of the MEC distribution. These values are presented in Tables 4.4, 4.5 and 4.6.

Table 4.4 illustrates that the median and mean are increasing functions of a for fixed ρ and b , while the skewness is decreasing. Variance and skewness are both decreasing functions of b for fixed ρ and b . According to table 4.4, the MEC distribution can be right skewed, platykurtic (kurtosis < 3), or leptokurtic (kurtosis > 3). The MEC distribution, on the other hand, is a versatile distribution that may be used to model data sets with skewed or extreme values.

Table 4.4: Mean, variance, skewness and kurtosis of the MEC distribution for $\rho = 1$ and various values of a and b .

a	b	Mean	Variance	Skewness	Kurtosis
0.5	10	0.9970	1.0135	1.0543	1.1286
	20	0.9963	0.9970	1.0026	1.0135
	30	0.9970	0.9960	0.9971	1.0002
	40	0.9976	0.9963	0.9961	0.9970
1.0	10	1.0686	1.1642	1.2980	1.4892
	20	1.0315	1.0686	1.1125	1.6423
	30	1.0204	1.0432	1.0686	1.0971
	40	1.0151	1.0314	1.0493	1.0686
2.5	10	1.1712	1.3984	1.7087	2.1485
	20	1.0798	1.1713	1.2764	1.3984
	30	1.0520	1.1089	1.1712	1.2396
	40	1.0386	1.0798	1.1239	1.1712
5	10	1.2552	1.6063	2.1037	2.8349
	20	1.1179	1.2552	1.4162	1.6063
	30	1.0766	1.1613	1.2552	1.3596
	40	1.0567	1.1179	1.1838	1.2552
20	10	1.4419	2.1196	3.1886	4.9359
	20	1.1981	1.4419	1.7436	2.1196
	30	1.1275	1.2737	1.4419	1.6357
	40	1.0940	1.1981	1.3136	1.4419

Table 4.6 indicates that, the mean, variance and skewness generally increases as the parameter values of a , b increases for a fixed ρ . The MEC distribution is asymmetric

distribution as the values of the skewness are not zero. Also, we observe that for some parameter values the MEC distribution can be moderately skewed (between 0.5 and 1) and highly skewed (greater than 1). We observe that the distribution can have lower and wider peak and thinner tails, this is because the kurtosis produces some values less than 3.

Table 4.5: Mean, variance, skewness and kurtosis of the MEC distribution for $\rho = 0.2$ and various values of a and b .

a	b	Mean	Variance	Skewness	Kurtosis
0.5	10	0.9214	0.8576	0.8082	0.7738
	20	0.9588	0.9214	0.8877	0.8576
	30	0.9721	0.9459	0.9214	0.8985
	40	0.9789	0.9588	0.9396	0.9214
1.0	10	0.9673	0.9464	0.9389	0.9483
	20	0.9823	0.9673	0.9553	0.9464
	30	0.9879	0.9710	0.9673	0.9590
	40	0.9908	0.9823	0.9745	0.9673
2.5	10	1.0350	1.0856	1.1580	1.2619
	20	1.0158	1.0350	1.0581	1.0857
	30	1.0102	1.0218	1.0350	1.0499
	40	1.0075	1.0158	1.0249	1.0350
5	10	1.0941	1.2153	1.3755	1.5942
	20	1.0442	1.0941	1.1507	1.2153
	30	1.0289	1.0601	1.0741	1.1310
	40	1.0210	1.0446	1.0680	1.0941
20	10	1.2371	1.5578	2.0042	2.6489
	20	1.1010	1.2371	1.3848	1.5578
	30	1.0716	1.1502	1.2371	1.3330
	40	1.0531	1.1091	1.1712	1.2371

Table 4.6 indicates that, the mean, variance and skewness generally increases as the parameter values of a , b increases for a fixed ρ . Here, we observe that for some parameter values, the MEC distribution can be both moderately skewed (between 0.5 and 1) and highly skewed (greater than 1). Furthermore, we observe that the distribution can be

platykurtic, with a lower and broader central peak and shorter and thinner tails, or leptokurtic, with a higher and sharper central peak and longer and fatter tails.

Table 4.6: Mean, variance, skewness and kurtosis of the MEC distribution for $\rho = 8.2$ and various values of a and b .

a	b	Mean	Variance	Skewness	Kurtosis
0.5	10	1.1541	1.3786	1.7108	2.2175
	20	1.0699	1.1541	1.2557	1.3786
	30	1.0452	1.0963	1.1541	1.2197
	40	1.0333	1.0699	1.1101	1.1541
1.0	10	1.2708	1.6613	2.2422	3.1406
	20	1.1235	1.2708	1.4475	1.6613
	30	1.0799	1.1697	1.2708	1.3849
	40	1.0591	1.1230	1.1938	1.2708
2.5	10	1.4228	2.0716	3.0981	4.7844
	20	1.1896	1.4228	1.7116	2.0716
	30	1.1221	1.2620	1.4228	1.6082
	40	1.0899	1.1896	1.3001	1.4228
5	10	1.5370	2.4127	3.8830	6.4418
	20	1.2367	1.5370	1.9202	2.4127
	30	1.1515	1.3289	1.5370	1.7817
	40	1.1114	1.2367	1.3778	1.5370
20	10	1.7760	3.2172	5.9669	11.3921
	20	1.3297	1.7760	2.3840	3.2172
	30	1.2086	1.4636	1.7760	2.1599
	40	1.1525	1.3296	1.5358	1.7760

Incomplete Moments

Incomplete moments are very useful in many fields of study. They play a vital role in the field of economics such as the income quintiles, the Lorenz curve, Pietra and Gini measures of inequality.

Proposition 4.3. The incomplete moment of the MEC distribution is

$$m_r(y) = a\rho \sum_{i=0}^{\infty} \sum_{j=0}^i \frac{\varpi_{ij} \Upsilon(1 - \frac{r}{b}, (j-i-1)y^{-b})}{(j-i-1)^{1-\frac{r}{b}}}, r < b, r = 1, 2, \dots, \quad (4.8)$$

where $\Upsilon(s, x) = \int_x^{\infty} t^{s-1} e^{-t} dt$ is the lower incomplete gamma function.

Proof. The incomplete moment of a random variable is given as

$$m_r(y) = \int_0^y x^r f(x) dx.$$

Thus, by replacing the series representation of the density function of the MEC distribution into the definition of the incomplete moment, we have

$$\begin{aligned} m_r(y) &= \int_0^y ab\rho \sum_{i=0}^{\infty} \sum_{j=0}^i \varpi_{ij} x^{r-b-1} e^{-(j-i-1)x^{-b}} dx \\ &= ab\rho \sum_{i=0}^{\infty} \sum_{j=0}^i \varpi_{ij} \int_0^y x^{r-b-1} e^{-(j-i-1)x^{-b}} dx. \end{aligned}$$

Let $u = (j-i-1)x^{-b}$. As $x \rightarrow 0, u \rightarrow \infty$ and as $x \rightarrow y, u \rightarrow (j-i-1)y^{-b}$. Also, $x = \left(\frac{u}{j-i-1}\right)^{-1/b}$ and $dx = \frac{-du}{b(j-i-1)x^{-b-1}}$. Hence,

$$\begin{aligned} m_r(y) &= a\rho \sum_{i=0}^{\infty} \sum_{j=0}^i \varpi_{ij} \int_{(j-i-1)y^{-b}}^{\infty} \frac{u^{-r/b} e^{-u}}{(j-i-1)^{1-r/b}} du \\ &= a\rho \sum_{i=0}^{\infty} \sum_{j=0}^i \frac{\varpi_{ij} \Upsilon(1 - \frac{r}{b}, (j-i-1)y^{-b})}{(j-i-1)^{1-\frac{r}{b}}}, r < b. \end{aligned}$$

This completes the proof.

Inequality Measures

The measurement of inequality mainly focuses on measuring the inequality in outcomes (Jenkins and Van Kerm, 2008). The Lorenz and Bonferroni curves are income inequality measures that can be applied in diverse fields including reliability, demography, medicine and insurance (Jenkins and Van Kerm, 2008). In this section, the derived Lorenz and Bonferroni curves for the MEC distribution is derived.

Proposition 4.4. The Lorenz curve for a random variable having the MEC distribution is,

$$L_F(y) = \frac{a\rho}{\mu} \sum_{i=0}^{\infty} \sum_{j=0}^i \frac{\varpi_{ij} \Upsilon(1 - \frac{1}{b}, (j-i-1)y^{-b})}{(j-i-1)^{1-\frac{1}{b}}}, \quad b > 1. \quad (4.9)$$

Proof. The Lorenz curve of the distribution of a random variable is defined as

$$L_F(y) = \frac{1}{\mu} \int_0^y x f(x) dx.$$

From the definition, the Lorenz curve is simply the product of the first incomplete moment and the reciprocal of the mean of the random variable. Hence,

$$L_F(y) = \frac{a\rho}{\mu} \sum_{i=0}^{\infty} \sum_{j=0}^i \frac{\varpi_{ij} \Upsilon(1 - \frac{1}{b}, (j-i-1)y^{-b})}{(j-i-1)^{1-\frac{1}{b}}}, \quad b > 1.$$

The proof is therefore complete.

Proposition 4.5. The Bonferroni curve of a random variable having the MEC distribution is

$$B_F(y) = \frac{a\rho}{\mu F(y)} \sum_{i=0}^{\infty} \sum_{j=0}^i \frac{\varpi_{ij} \Upsilon(1 - \frac{1}{b}, (j-i-1)y^{-b})}{(j-i-1)^{1-\frac{1}{b}}}, \quad b > 1. \quad (4.10)$$

Proof. The Bonferroni curve by definition is given as

$$B_F(y) = \frac{L_F(y)}{F(y)}.$$

Thus, substituting the Lorenz curve into the definition of the Bonferroni curve completes the proof.

Moment Generating Function

The moment generating function (MGF) if it exist is a special function used to compute statistical measures such as the mean and variance of a given random variable. In this sub-section the moment generating function of the MEC distribution is derived.

Proposition 4.6. Given a random variable Y having the MEC distribution, the MGF for the MEC distribution is given by

$$M_Y(t) = a\rho \sum_{r=0}^{\infty} \sum_{i=0}^{\infty} \sum_{j=0}^i \frac{t^r \varpi_{ij}}{r! (j-i-1)^{1-\frac{r}{b}}} \Gamma\left(1 - \frac{r}{b}\right), \quad r < b, \quad (4.11)$$

where $\Gamma(\beta) = \int_0^{\infty} t^{\beta-1} e^{-t} dt$ is the gamma function.

Proof. The MGF of Y is defined as,

$$M_Y(t) = E(e^{tY}) = \int_{-\infty}^{\infty} e^{ty} f(y) dy.$$

Using Taylor series expansion, the MGF can be rewritten as,

$$\begin{aligned} M_Y(t) &= \int_{-\infty}^{\infty} \sum_{r=0}^{\infty} \frac{t^r y^r}{r!} f(y) dy \\ &= \sum_{r=0}^{\infty} \frac{t^r}{r!} \mu_r' \end{aligned}$$

substituting the r^{th} moment of the MEC distribution yields

$$M_Y(t) = a\rho \sum_{r=0}^{\infty} \sum_{i=0}^{\infty} \sum_{j=0}^i \frac{t^r \varpi_{ij}}{r! (j-i-1)^{1-\frac{r}{b}}} \Gamma\left(1 - \frac{r}{b}\right).$$

The proof is therefore complete.

Characteristic Function

Characteristic functions are particularly useful in handling heavy-tailed random variables, for which the corresponding moment generating functions do not exist.

Proposition 4.7. Given a random number Y having the MEC distribution, Then characteristic function for the MEC distribution is given by

$$C_Y(t) = a\rho \sum_{r=0}^{\infty} \sum_{i=0}^{\infty} \sum_{j=0}^i \frac{(it)^r \varpi_{ij}}{r! (j-i-1)^{1-\frac{r}{b}}} \Gamma\left(1 - \frac{r}{b}\right), \quad r < b, \quad (4.12)$$

where $\Gamma(\beta) = \int_0^{\infty} t^{\beta-1} e^{-t} dt$ is the gamma function.

Proof. The characteristic function of Y is defined as,

$$C_Y(t) = E(e^{itY}) = \int_{-\infty}^{\infty} e^{ity} f(y) dy,$$

where $i = \sqrt{-1}$. Using Taylor series expansion, the characteristic function can be rewrit-

ten as,

$$\begin{aligned}
M_Y(t) &= \int_{-\infty}^{\infty} \sum_{r=0}^{\infty} \frac{i^r t^r y^r}{r!} f(y) dy \\
&= \sum_{r=0}^{\infty} \frac{t^r i^r}{r!} \int_{-\infty}^{\infty} y^r f(y) dy \\
&= \sum_{r=0}^{\infty} \frac{t^r i^r}{r!} \mu_r'
\end{aligned}$$

substituting the r^{th} moment of the MEC distribution yields

$$M_Y(t) = a\rho \sum_{r=0}^{\infty} \sum_{i=0}^{\infty} \sum_{j=0}^i \frac{(it)^r \varpi_{ij}}{r! (j-i-1)^{1-\frac{r}{b}}} \Gamma\left(1 - \frac{r}{b}\right).$$

The proof is therefore complete.

Entropy

The entropy of a random variable Y is the measure of variation of uncertainty. A higher entropy value suggests more uncertainty in the data, while a lower entropy value indicates less uncertainty.

Proposition 4.8. The Rényi entropy of the MEC distribution is given by,

$$(1 - \psi)^{-1} \log \left\{ (a\rho b)^{\psi} \sum_{i=0}^{\infty} \sum_{j=0}^i \eta_{ij} \frac{\Gamma\left(\frac{\psi-1}{b} + \psi\right)}{b(j-i-\psi)^{\frac{\psi-1}{b} + \psi}} \right\}, \quad (4.13)$$

where $\psi > 0$, $\psi \neq 0$ and where $\eta_{ij} = \frac{(-1)^{i+j} \psi^{(i)} \rho^i (a+1)^{(i)}}{i} \binom{i}{j}$.

Proof. Rényi (1960) proposed the Rényi entropy which is denoted by $I_R(\psi)$ for Y is defined as,

$$I_R(\psi) = (1 - \psi)^{-1} \log \left\{ \int_0^{\infty} f(y)^{\psi} dy \right\}, \quad (4.14)$$

where $\psi > 0$ and $\psi \neq 0$. Substituting the density of the MEC distribution into equation (4.14), we obtain

$$I_R(\psi) = (1 - \psi)^{-1} \log \left\{ \int_0^\infty \left[ab\rho y^{-b-1} e^{y^{-b}} \left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-a-1} \right]^\psi dy \right\}.$$

Let

$$v(y) = \int_0^\infty \left[ab\rho y^{-b-1} e^{y^{-b}} \left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-a-1} \right]^\psi dy.$$

applying the approach for expanding the density function in Lemma (4.1),

$$v(y) = \int_0^\infty (ab\rho)^\psi \sum_{i=0}^\infty \sum_{j=0}^i \varpi_{ij} y^{\psi(-b-1)} e^{-(j-i-\psi)y^{-b}} dy$$

where $\eta_{ij} = \frac{(-1)^{i+j} \psi^{(i)} \rho^i (a+1)^{(i)}}{i} \binom{i}{j}$.

Let $u = (j - i - \psi)y^{-b}$. As $y \rightarrow 0$, $u \rightarrow \infty$ and as $y \rightarrow \infty$, $u \rightarrow 0$. Also, $y = \left(\frac{u}{j - i - \psi} \right)^{-1/b}$ and $dy = \frac{-du}{b(j - i - \psi)y^{-b-1}}$. Thus,

$$\begin{aligned} v(y) &= (ab\rho)^\psi \sum_{i=0}^\infty \sum_{j=0}^i \eta_{ij} \int_0^\infty \frac{u^{\frac{\psi-1}{b} + \psi - 1} e^{-u}}{b(j - i - 1)^{\frac{\psi-1}{b} + \psi}} du \\ &= (a\rho b)^\psi \sum_{i=0}^\infty \sum_{j=0}^i \eta_{ij} \frac{\Gamma\left(\frac{\psi-1}{b} + \psi\right)}{b(j - i - \psi)^{\frac{\psi-1}{b} + \psi}}. \end{aligned}$$

Hence,

$$I_R(\psi) = (1 - \psi)^{-1} \log \left\{ (a\rho b)^\psi \sum_{i=0}^\infty \sum_{j=0}^i \eta_{ij} \frac{\Gamma\left(\frac{\psi-1}{b} + \psi\right)}{b(j - i - \psi)^{\frac{\psi-1}{b} + \psi}} \right\}, \quad \psi > 0 \text{ and } \psi \neq 0.$$

The proof is therefore complete.

Table 4.7 summarizes the Rényi entropy values for some value of a , b and ρ . It should

be noticed that the Rényi entropy decreases with rising parameter values. Also, as the values of a, ρ increases with a constant b , the entropy values tends to negative and increases towards zero. This indicates the flexibility of the amount of randomness of the MEC distribution.

Table 4.7: Numerical values of the Rényi entropy of the MEC distribution at different values of a , b , and ρ .

a	b	ρ	$\psi = 1.5$	$\psi = 2$	$\psi = 3$	$\psi = 5$
0.5	1.0	1.3	1.5987	1.2400	0.8904	0.6039
0.6	1.5	1.3	1.4673	1.2174	0.9618	0.7421
1.0	2.5	1.3	1.1260	0.9503	0.7618	0.5918
1.5	3.5	1.3	0.7741	0.5886	0.4219	0.2698
1.6	4.5	1.3	0.3165	0.1788	0.0247	-0.1200
4.0	3.5	5.0	1.7544	1.6058	1.4416	1.2895
4.0	4.5	6.5	1.1935	1.0569	0.9037	0.7596
4.0	5.2	7.0	0.8586	0.7273	0.5789	0.4384
4.0	6.1	8.0	0.5183	0.3919	0.2478	0.1105
4.0	7.5	9.0	0.0830	-0.0386	-0.1782	-0.3123
0.01	2.5	1.3	-1.7037	-1.8848	-2.0778	-2.2509
0.02	2.5	1.5	-1.2833	-1.4694	-1.6659	-1.8410
0.03	2.5	2.0	-1.0039	-1.2002	-1.4046	-1.5841
0.04	2.5	3.5	-0.7354	-0.9534	-1.1754	-1.3602
0.05	2.5	4.0	-0.5459	-0.7740	-1.00561	-1.2019

Stress-Strength Reliability

Stress-strength models are of importance in the field of reliability analysis and engineering applications. The stress-strength reliability is defined as the probability that the unit's strength Y_1 is greater than the stress Y_2 , the unit fails when $Y_1 \leq Y_2$.

Proposition 4.9. Suppose Y_1 and Y_2 follows the MEC distribution, then the stress-

strength reliability is given by,

$$R = ab\rho \sum_{i=0}^{\infty} \sum_{j=0}^i \Pi_{ij} \frac{1}{(j-i-1)^{1-\frac{1}{b}}},$$

where $\Pi_{ij} = \frac{(-1)^{i+j} \rho (2a+1)^{(i)}}{i} \binom{i}{j}$.

Proof. The stress-strength reliability measure of a component is defined as,

$$R = \int_0^{\infty} f(y)F(y)dy.$$

Substituting the PDF and CDF of the MEC distribution we have,

$$\begin{aligned} R &= \int_0^{\infty} ab\rho y^{-b-1} e^{y^{-b}} \left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-a-1} \left[\left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-a} \right] dy \\ &= \int_0^{\infty} ab\rho y^{-b-1} e^{y^{-b}} \left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-(2a+1)} dy \\ &= ab\rho \int_0^{\infty} y^{-b-1} e^{y^{-b}} \left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-(2a+1)} dy \end{aligned}$$

using similar concept in lemma (4.1),

$$R = ab\rho \sum_{i=0}^{\infty} \sum_{j=0}^i \Pi_{ij} \int_0^{\infty} y^{-b-1} e^{-(j-i-1)y^{-b}} dy.$$

Let $u = (j-i-1)y^{-b}$. As $y \rightarrow 0, u \rightarrow \infty$ and as $y \rightarrow \infty, u \rightarrow 0$. Also, $y =$

$\left(\frac{u}{j-i-1}\right)^{-1/b}$ and $dy = \frac{-du}{b(j-i-1)y^{-b-1}}$. Hence,

$$\begin{aligned} R &= ab\rho \sum_{i=0}^{\infty} \sum_{j=0}^i \Pi_{ij} \int_0^{\infty} \frac{e^{-u}}{(j-i-1)^{1-\frac{1}{b}}} du \\ &= ab\rho \sum_{i=0}^{\infty} \sum_{j=0}^i \Pi_{ij} \frac{1}{(j-i-1)^{1-\frac{1}{b}}}. \end{aligned}$$

This completes the proof.

To study the behaviour of the stress-strength reliability for the MEC distributed random variable, we present some values of the stress-strength reliability of the MEC distribution in Table 4.8 for some parameter values of a , b and ρ . It can be observed that as a and b increases with a fixed $\rho = 0.5$ and $\rho = 3.5$, the stress-strength reliability increases. However, it is observed that, with an increase in a and b values with $\rho = 5$, the stress strength reliability decreases keeping to a fixed value of 0.5000.

Table 4.8: Numerical values of the stress-strength reliability of the MEC distribution at different values of a, b , and ρ .

$\rho = 0.5$			$\rho = 3.5$			$\rho = 5$		
a	b	R	a	b	R	a	b	R
0.07	0.002	0.0600	0.07	0.002	0.0600	0.07	0.020	0.4700
	0.003	0.0600		0.003	0.1000		0.5	0.4999
	0.004	0.1000		0.004	0.1000		1	0.5000
	0.017	0.5000		0.017	0.4600		2	0.5000
	0.020	0.5000		0.020	0.4810		5	0.5000
0.5	0.002	0.1000	0.5	0.002	0.0600	0.07	0.020	0.4810
	0.003	0.3000		0.003	0.0002		0.5	0.5000
	0.004	0.3000		0.004	0.1000		1	0.5000
	0.017	0.4700		0.017	0.4700		2	0.5000
	0.020	0.4960		0.020	0.4860		5	0.5000
1	0.002	0.3000	1	0.002	0.0300	1	0.020	0.4600
	0.003	0.3000		0.003	0.0600		0.5	0.5000
	0.004	0.3000		0.004	0.0600		1	0.5000
	0.017	0.4920		0.017	0.4500		2	0.5000
	0.020	0.4960		0.020	0.4700		5	0.5000
1.7	0.002	0.1000	1.7	0.002	0.0040	1.7	0.020	0.4400
	0.003	0.3000		0.003	0.0080		0.5	0.5000
	0.004	0.3000		0.004	0.0200		1	0.5000
	0.017	0.4860		0.017	0.4100		2	0.5000
	0.020	0.4930		0.020	0.4600		5	0.5000
2.5	0.002	0.0600	2.5	0.002	0.0005	2.5	0.020	0.4100
	0.003	0.1000		0.003	0.0020		0.5	0.5000
	0.004	0.2000		0.004	0.0040		1	0.5000
	0.017	0.4800		0.017	0.3800		2	0.5000
	0.020	0.4900		0.020	0.4300		5	0.5000

Order Statistics

Order statistics are most commonly found in statistical theory. The characteristics and uses of these ordered random variables, as well as functions incorporating them, are the subject of order statistics.

Let $Y_{(1)}, Y_{(2)}, \dots, Y_{(n)}$ denote the order statistics of a random sample $Y_1, Y_2, \dots, Y_n$ from a continuous population with CDF $F_Y(y)$ and PDF $f_Y(y)$, then the PDF of the p^{th} order statistics Y_p is given by,

$$f_{p:n} = U_{r:n} [F(y)]^{p-1} [1 - F(y)]^{n-p} f(y),$$

for $r = 1, 2, \dots, n$, where $U_{r:n} = \frac{n!}{(p-1)!(n-p)!} = [B(p, n-p+1)]^{-1}$ is the beta function.

The PDF of the p^{th} order MEC random variable $Y_{(p)}$ is defined as,

$$\begin{aligned} f_{p:n}(y) = & U_{r:n} \left[\left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-a} \right]^{p-1} \left[1 - \left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-a} \right]^{n-p} \\ & \times ab\rho y^{-b-1} e^{y^{-b}} \left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-a-1}. \end{aligned}$$

Therefore, the PDF of the n^{th} MEC statistic $Y_{n:p}$ is given by,

$$f_{n:p}(y) = n \left[\left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-a} \right]^{n-1} ab\rho y^{-b-1} e^{y^{-b}} \left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-a-1}$$

and the PDF of the first MEC statistic $Y_{1:p}$ is given by,

$$f_{1:p}(y) = n \left[1 - \left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-a} \right]^{n-1} ab\rho y^{-b-1} e^{y^{-b}} \left[\rho \left(e^{y^{-b}} - 1 \right) + 1 \right]^{-a-1}.$$

Stochastic Ordering

A stochastic order measures the notion of one random variable being "larger" than another in probability theory and statistics. These are frequently partial orders, with one random variable X not being greater than, less than, or equal to another random variable Y . Stochastic ordering have important applications in many fields for instance, defining notions of positive and negative aging, bounding system reliabilities and availability, and

comparing maintenance policies (Jenkins and Van Kerm, 2008).

Let Y_1 and Y_2 be two continuous random variable with density function $f_{Y_1}(y)$ and $f_{Y_2}(y)$ respectively, such that $f_{Y_1}(y)/f_{Y_2}(y)$ is non-decreasing over the union of the supports of Y_1 and Y_2 . Then Y_2 is said to be smaller than Y_1 in the likelihood ratio order, denoted as $Y_2 \leq_{lr} Y_1$. The likelihood ratio order is more powerful than the hazard rate and typical stochastic orders, which are described as follows: Suppose Y_1 and Y_2 are two random variables with CDFs $F_{Y_1}(y)$ and $F_{Y_2}(y)$ and hazard rates $h_{Y_1}(y)$ and $h_{Y_2}(y)$, respectively. Then

- i. Y_2 is said to be stochastically lesser than Y_1 , defined by $Y_2 \leq_{st} Y_1$, if $F_{Y_2}(y) \geq F_{Y_1}(y) \forall y$.
- ii. Y_2 is said to be lesser than Y_1 in hazard rate order, defined by $Y_2 \leq_{hr} Y_1$, if $h_{Y_2}(y) \geq h_{Y_1}(y) \forall y$.
- iii. An immediate consequences and well known fact is that $Y_2 \leq_{lr} Y_1 \Rightarrow Y_2 \leq_{hr} Y_1 \Rightarrow Y_2 \leq_{st} Y_1$.

Proposition 4.10. Suppose Y_1 follows the MEC distribution with parameters a_1, b, ρ and Y_2 also follows the MEC distribution with parameters a_2, b, ρ . Then Y_2 is less than Y_1 in likelihood ratio order ($Y_2 \leq_{lr} Y_1$).

Proof. The density functions of Y_1 and Y_2 are respectively given by

$$f_{Y_1}(y) = a_1 b \rho y^{-b-1} e^{y^{-b}} \left[1 + \rho(e^{y^{-b}} - 1) \right]^{-(a_1+1)}, y > 0,$$

and

$$f_{Y_2}(y) = a_2 b \rho y^{-b-1} e^{y^{-b}} \left[1 + \rho(e^{y^{-b}} - 1) \right]^{-(a_2+1)}, y > 0.$$

The ratio of the two densities is

$$\frac{f_{Y_1}(y)}{f_{Y_2}(y)} = \left(\frac{a_1}{a_2}\right) \left[1 + \rho(e^{y^{-b}} - 1)\right]^{a_2 - a_1}, y > 0.$$

The derivative of the ratio of the densities is

$$\frac{d}{dy} \left(\frac{f_{Y_1}(y)}{f_{Y_2}(y)} \right) = (a_2 - a_1) \left(\frac{a_1}{a_2}\right) b^{-b-1} e^{y^{-b}} \left[1 + \rho(e^{y^{-b}} - 1)\right]^{a_2 - a_1 - 1}.$$

Hence, if $a_1 < a_2$, then

$$\frac{d}{dy} \left(\frac{f_{Y_1}(y)}{f_{Y_2}(y)} \right) > 0.$$

This implies that $Y_2 \leq_{lr} Y_1$ and this completes the proof.

4.2.3 Estimation of Parameters of the MEC Distribution

The estimators of the parameters for the MEC distribution are derived by employing three estimation approaches namely the Maximum Likelihood Estimation (MLE) method, method of Ordinary Least Square (OLS) and Weighted Least Square (WLS).

Maximum Likelihood Estimation

The method of MLE in the context of the MEC is described in this subsection. Considering $y_1, y_2, y_3, \dots, y_n$ as random sample from the MEC distribution. Then the likelihood function of the MEC is given by,

$$L(y; a, b, \rho) = a^n b^n \rho^n \prod_{i=1}^n y_i^{-b-1} e^{y_i^{-b}} \left[\rho \left(e^{y_i^{-b}} - 1 \right) + 1 \right]^{-a-1}.$$

The log likelihood equation is given by,

$$\begin{aligned} \ln L(y; a, b, \rho) = & n \ln a + n \ln b + n \ln \rho + (-a - 1) \ln \sum_{i=1}^n \left[\rho \left(e^{y_i^{-b}} - 1 \right) + 1 \right] \\ & + (-b - 1) \ln \sum_{i=1}^n y_i + \sum_{i=1}^n y_i^{-b}. \end{aligned} \quad (4.15)$$

Differentiating equation (4.15) with respect to a , b and ρ in turn and equating to zero, we obtain the equations,

$$\frac{\partial \ln L(y; a, b, \rho)}{\partial a} = \frac{n}{a} - \ln \sum_{i=1}^n \left[\rho \left(e^{y_i^{-b}} - 1 \right) + 1 \right] = 0, \quad (4.16)$$

$$\begin{aligned} \frac{\partial \ln L(y; a, b, \rho)}{\partial b} = & \frac{n}{b} + (-a - 1) \sum_{i=1}^n \frac{-e^{y_i^{-b}} y_i^{-b} \ln y_i \rho \left(e^{y_i^{-b}} - 1 \right)}{\left(e^{y_i^{-b}} - 1 \right) + 1} - \sum_{i=1}^n \ln y_i \\ & - \sum_{i=1}^n y_i^{-b} \ln(-y_i) = 0, \end{aligned} \quad (4.17)$$

$$\frac{\partial \ln L(y; a, b, \rho)}{\partial \rho} = \frac{n}{\rho} + (-a - 1) \sum_{i=1}^n \frac{e^{y_i^{-b}}}{\rho \left(e^{y_i^{-b}} - 1 \right) + 1} = 0, \quad (4.18)$$

The estimators of the MEC distributions can be obtained by solving equation (4.16), (4.17) and (4.18) in relation to a , b and ρ simultaneously using a numerical procedure.

Ordinary Least and Weighted Least Squares Methods

Suppose $F(Y_{(j)})$ denotes the MEC distribution function of the ordered random variables $Y_{(1)} < Y_{(2)} < \dots < Y_{(n)}$ where $\{Y_1, Y_2, \dots, Y_n\}$ is a random sample of size n from a distribution function $F(y; a, b, \rho)$. Therefore, the OLS estimators of a , b and ρ , say $\hat{a}_{OLS}$, $\hat{b}_{OLS}$ and $\hat{\rho}_{OLS}$ of the MEC distribution can be derived by minimizing

$$OLS(a, b, \rho) = \sum_{i=1}^n \left[F(y_{i:n} | a, b, \rho) - \frac{i}{n+1} \right]^2, \quad (4.19)$$

with respect to a, b , and ρ where $F(y_{i:n}|a, b, \rho)$ is the CDF of the MEC distribution.

Partially differentiating equation (4.19) and equating to zero with respect to a, b , and ρ yields,

$$\frac{\partial OLS(a, b, \rho)}{\partial a} = 2 \sum_{i=1}^n \left[\left(\rho \left(e^{y_{i:n}^{-b}} - 1 \right) + 1 \right)^{-a} - \frac{i}{n+1} \right] \Omega_1(y_{i:n}, |a, b, \rho) = 0, \quad (4.20)$$

$$\frac{\partial OLS(a, b, \rho)}{\partial b} = 2 \sum_{i=1}^n \left[\left(\rho \left(e^{y_{i:n}^{-b}} - 1 \right) + 1 \right)^{-a} - \frac{i}{n+1} \right] \Omega_2(y_{i:n}, |a, b, \rho) = 0, \quad (4.21)$$

$$\frac{\partial OLS(a, b, \rho)}{\partial \rho} = 2 \sum_{i=1}^n \left[\left(\rho \left(e^{y_{i:n}^{-b}} - 1 \right) + 1 \right)^{-a} - \frac{i}{n+1} \right] \Omega_3(y_{i:n}, |a, b, \rho) = 0, \quad (4.22)$$

where

$$\begin{aligned} \Omega_1(y_{i:n}, |a, b, \rho) &= \left[-\ln \left(\rho \left(e^{y_{i:n}^{-b}} - 1 \right) + 1 \right) \left(\rho \left(e^{y_{i:n}^{-b}} - 1 \right) + 1 \right)^{-a} \right], \\ \Omega_2(y_{i:n}, |a, b, \rho) &= \left[\rho a e^{y_{i:n}^{-b}} y^{-b} \ln(y_{i:n}) \left(\rho \left(e^{y_{i:n}^{-b}} - 1 \right) + 1 \right)^{-a-1} \right], \\ \Omega_3(y_{i:n}, |a, b, \rho) &= \left[-\rho - a \left(e^{y_{i:n}^{-b}} - 1 \right) \left(\rho \left(e^{y_{i:n}^{-b}} - 1 \right) + 1 \right)^{-1} \right]. \end{aligned}$$

The WLS estimators of the MEC distribution, can be obtained by minimizing,

$$WLS(a, b, \rho) = \sum_{i=1}^n \frac{(n+1)^2 (n+2)}{i(n-i+1)} \left[F(t_{i:n}|a, b, \rho) - \frac{i}{n+1} \right]^2 \quad (4.23)$$

Differentiating equation (4.23) with respect to a, b and ρ in turn and equating to zero, we

obtain the equations

$$\frac{\partial WLS(a, b, \rho)}{\partial a} = 2 \sum_{i=1}^n \frac{(n+1)^2 (n+2)}{i(n-i+1)} \left[\left(\rho \left(e^{y_{i:n}^{-b}} - 1 \right) + 1 \right)^{-a} - \frac{i}{n+1} \right] \quad (4.24)$$

$$\times \Omega_1(y_{i:n}, |a, b, \rho) = 0$$

$$\frac{\partial WLS(a, b, \rho)}{\partial b} = 2 \sum_{i=1}^n \frac{(n+1)^2 (n+2)}{i(n-i+1)} \left[\left(\rho \left(e^{y_{i:n}^{-b}} - 1 \right) + 1 \right)^{-a} - \frac{i}{n+1} \right] \quad (4.25)$$

$$\times \Omega_2(y_{i:n}, |a, b, \rho) = 0$$

$$\frac{\partial WLS(a, b, \rho)}{\partial \rho} = 2 \sum_{i=1}^n \frac{(n+1)^2 (n+2)}{i(n-i+1)} \left[\left(\rho \left(e^{y_{i:n}^{-b}} - 1 \right) + 1 \right)^{-a} - \frac{i}{n+1} \right] \quad (4.26)$$

$$\Omega_3(y_{i:n}, |a, b, \rho) = 0.$$

Solving equations (4.27), (4.25) and (4.26) simultaneously we derive the WLS estimates of the MEC distribution.

4.3 Formulation of the Equilibrium Renewal Burr

XII

Let the random variable Y follow the Equilibrium Renewal Burr XII (ER-BurrXII) distribution with positive parameters $a > 0$, $c > 0$ and $k > 0$. Let $y \in \mathbb{R}^+$, suppose the CDF and mean of the Burr-XII distribution are given by,

$$G(y) = 1 - \left(1 + \left(\frac{y}{a} \right)^c \right)^{-k}, \quad (4.27)$$

and

$$\mu = kaB \left[\frac{1}{c} + 1, k - \frac{1}{c} \right] \quad (4.28)$$

respectively, where $a, c, k > 0$ (Okasha and Matter, 2015) and $B(\alpha, \beta)$ is the beta function defined by

$$B(\alpha, \beta) = \int_0^1 t^{\alpha-1} (1-t)^{\beta-1} dt$$

(Abramowitz and Stegun, 1972). The PDF of the random variable Y is obtained by replacing equation (4.27) and (4.28) into the equilibrium renewal process defined in equation (3.2). Hence, the PDF of the ER-BurrXII distribution is given by

$$g(y) = \frac{\left(1 + \left(\frac{y}{a}\right)^c\right)^{-k}}{kaB\left[\frac{1}{c} + 1, k - \frac{1}{c}\right]}, \quad (4.29)$$

where $c > 0$ and $k > 0$ are the shape parameters and $a > 0$ is a scale parameter. The density plots of the ER-Burr XII distribution for fixed parameter c and k values and varying values of parameter a are displayed in Figure (4.6). Figure (4.6) shows the effect of the scale parameter a on the shape of the ER-BurrXII density, while the shape parameters, c and k are constant. It is observed that as a increases, the ER-BurrXII density curve decreases exponentially.

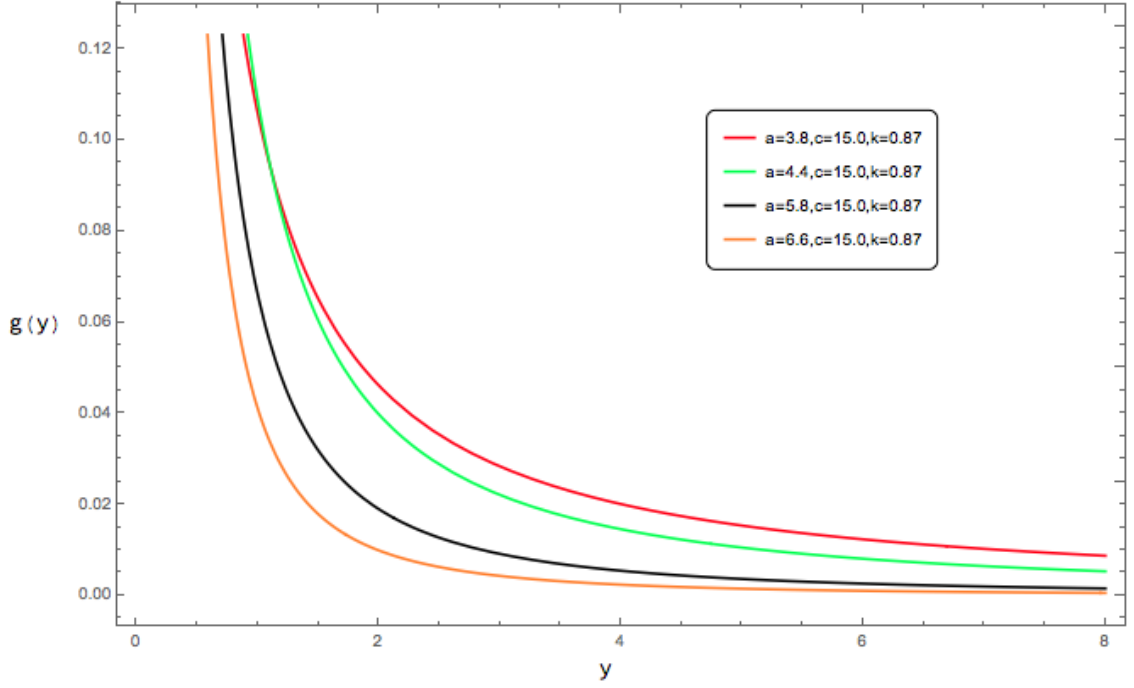


Figure 4.6: PDF plots of the ER-BurrXII distribution as a increases with a constant c and k .

The density plots of the ER-Burr XII distribution for fixed parameter a and c values and increasing values of parameter k are displayed in Figure (4.7). Figure (4.7) demonstrates the shape parameter affects k on the shape of the ER-BurrXII density, while the scale parameter, a and shape parameter k are constant. It is observed that density of ER-BurrXII can be right skewed or constant depending on the shape parameter c and k .

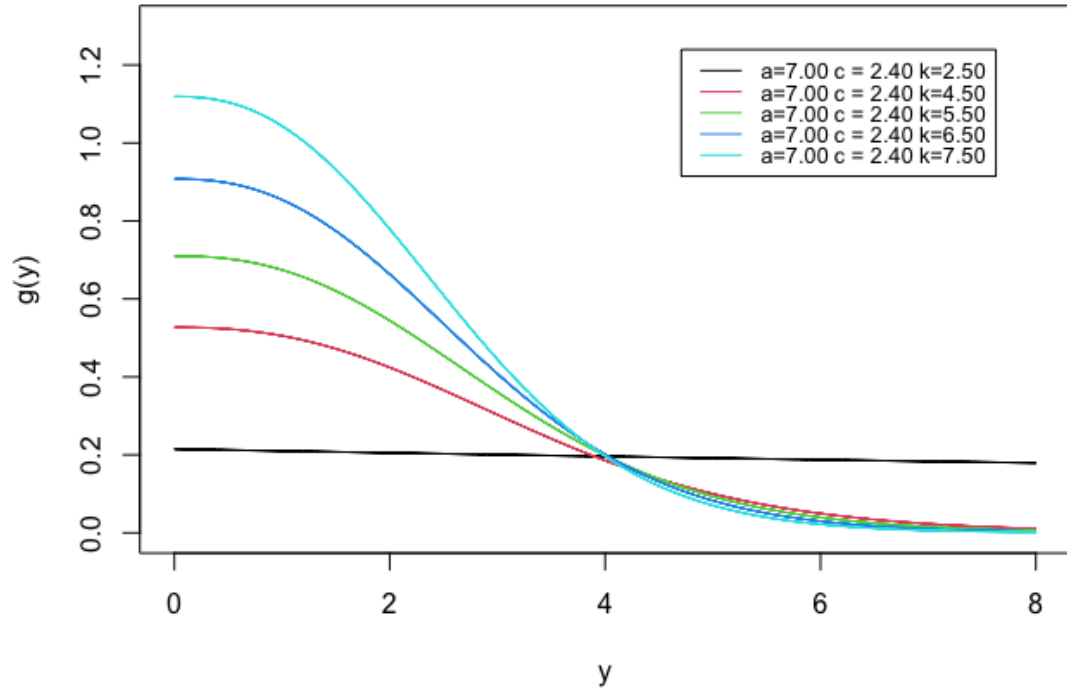


Figure 4.7: PDF plots of the ER-BurrXII distribution as k increases with a constant a and c .

The density plots of the ER-Burr XII distribution for fixed parameter c and k values and varying values of parameter a are displayed in Figure (4.8). It is observed that as the PDF of the ER-BurrXII exhibit an inverted S shape.

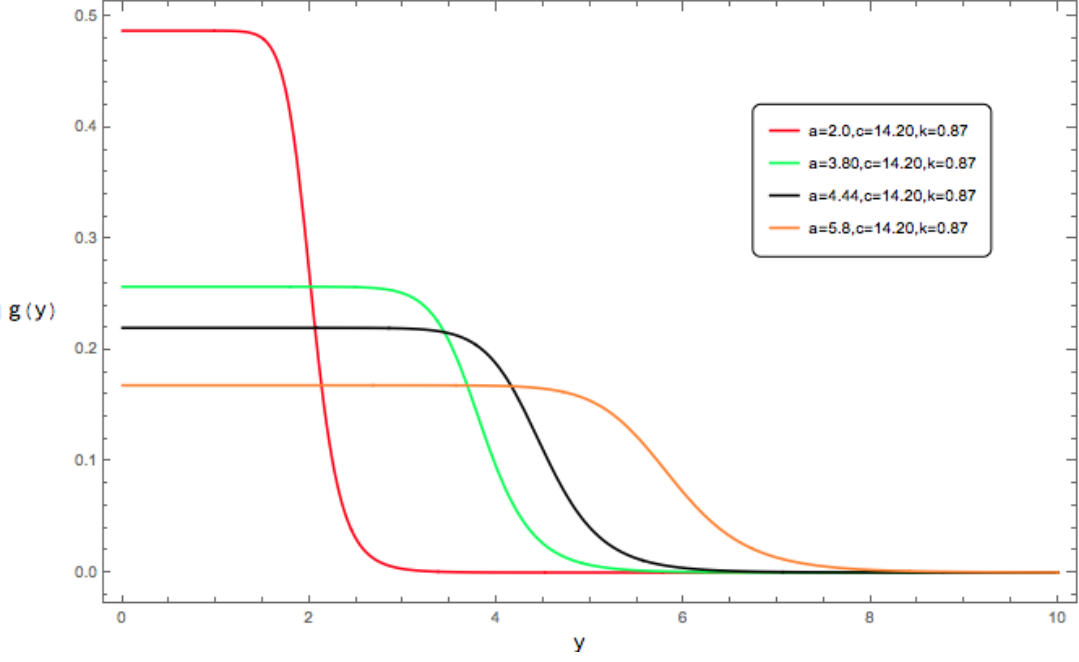


Figure 4.8: PDF plots of the ER-BurrXII distribution as c increases with a constant k and c .

The CDF of the ER-BurrXII is obtained by integrating equation (4.29) with respect to y . Hence the CDF is given by

$$G(y) = \frac{yF_{(1,2)}\left(\frac{1}{c}, k, 1 + \frac{1}{c}; -\left(\frac{y}{a}\right)^c\right)}{kaB\left[\frac{1}{c} + 1, k - \frac{1}{c}\right]}, \quad (4.30)$$

where $a, c, k > 0$ and $F_{(1,2)}(a, b, c; z)$ is the Gauss hypergeometric function is defined by,

$$F_{(1,2)}(a, b, c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 \frac{t^{b-1}(1-t)^{c-b-1}}{(1-tz)^a} dt.$$

The plots of ER-BurrXII CDF in Figure 4.3.1 displays a shape bounded between 0 and 1 which is typical of any CDF. It can be observed that for some parameter values, as y gets closer to zero the CDF approaches zero and as y gets closer to one the CDF approaches one.

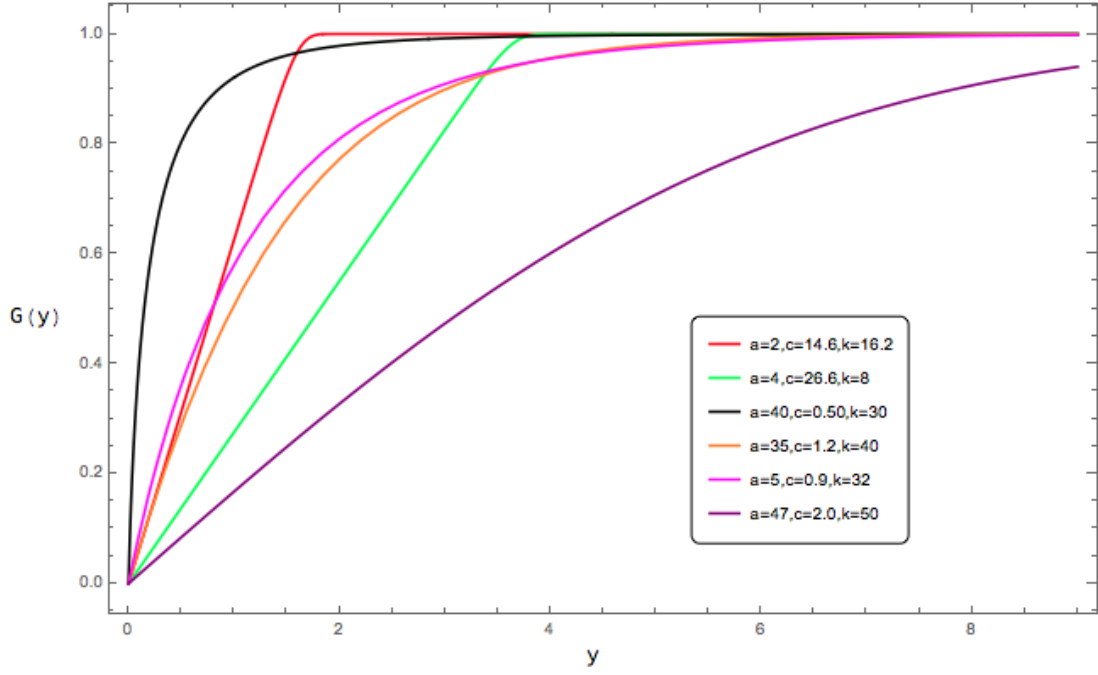


Figure 4.9: CDF plots of the ER-BurrXII distribution.

The survival function of the ER-BurrXII is defined as,

$$S(y) = 1 - \frac{yF_{(1,2)}\left(\frac{1}{c}, k, 1 + \frac{1}{c}; -\left(\frac{y}{a}\right)^c\right)}{kaB\left[\frac{1}{c} + 1, k - \frac{1}{c}\right]}.$$

Figure 4.10 shows plots of the survival function of the ER-BurrXII distribution. Figure 4.10 shows that for some parameter values the survival function declines fast as y approaches 2. In addition, the survival function declines slowly as y approaches 2 indicating a high reliability. They are monotone, non-increasing functions equal to one at zero and zero as the time approaches infinity. The plots indicates that the ER-BurrXII is quite flexible for modeling survival data.

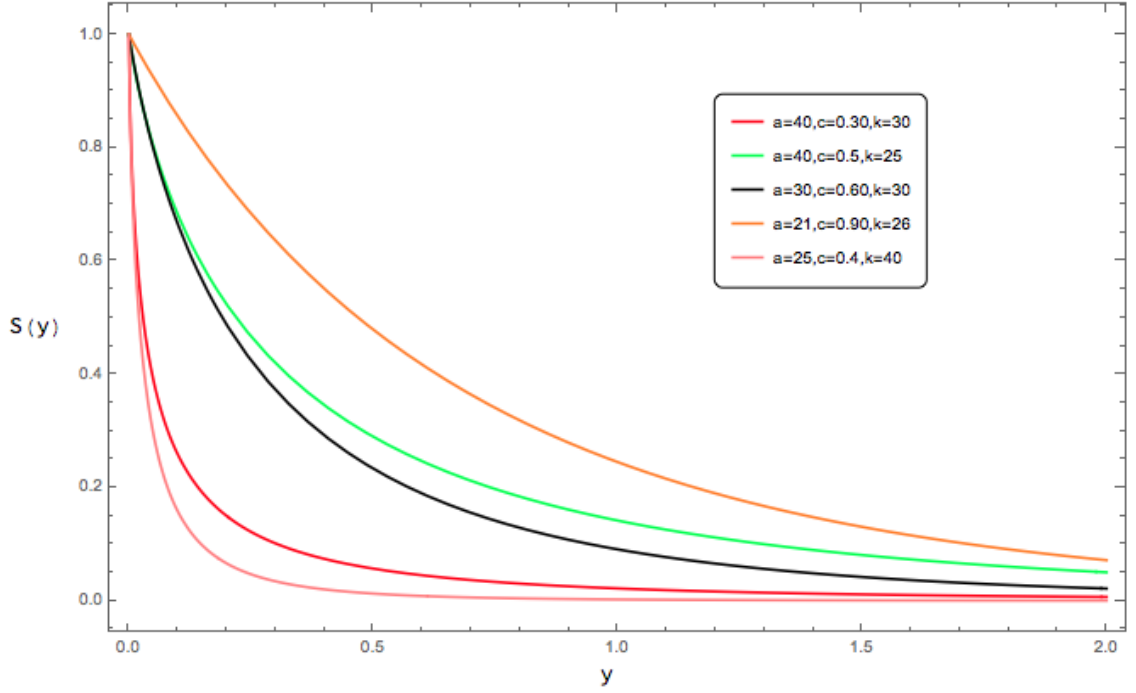


Figure 4.10: Survival function plots of the ER-BurrXII distribution.

The hazard function of the ER-BurrXII is given by,

$$h(y) = \frac{\left(1 + \left(\frac{y}{a}\right)^c\right)^{-k}}{kaB\left[\frac{1}{c} + 1, k - \frac{1}{c}\right] - yF_{(1,2)}\left(\frac{1}{c}, k, 1 + \frac{1}{c}; -\left(\frac{y}{a}\right)^c\right)}.$$

For increasing values of parameter c and fixed values of a and k , figure 4.11 shows different potential forms of the ER-BurrXII distribution's hazard function. It can be observed that for all parameter values the hazard function exhibits a reverse J shape. The hazard function of the ER-BurrXII distribution increases as the shape parameter c increases.

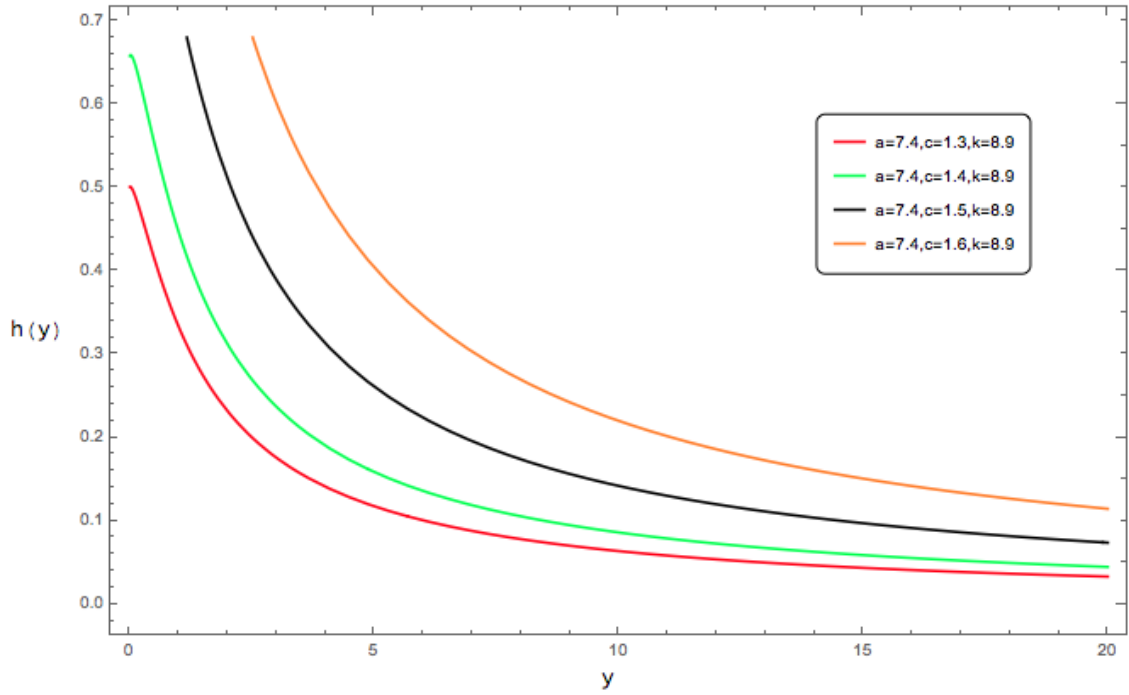


Figure 4.11: Hazard function plots of the ER-BurrXII distribution as c increases with a constant a and k .

For increasing values of parameter a and fixed values of c and k , figure 4.12 displays some of the various forms of the ER-BurrXII distribution's hazard function. It can be observed that for all parameter values the hazard function increase exponentially.

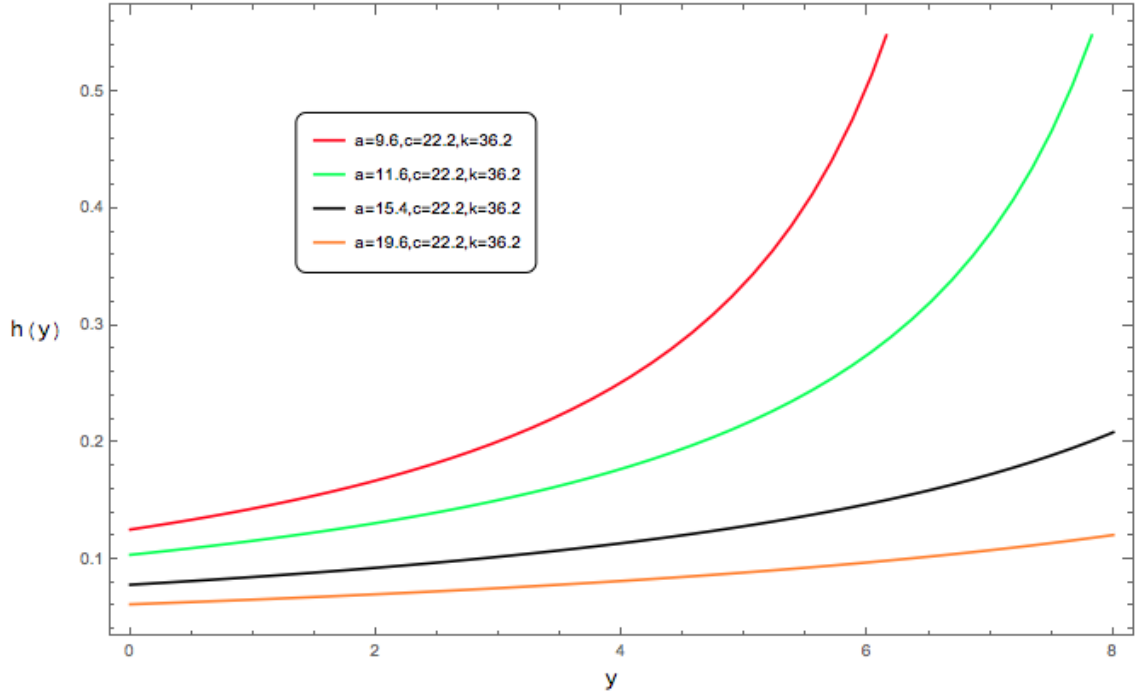


Figure 4.12: Hazard function plots of the ER-BurrXII distribution as a increases with a constant c and k .

Figure 4.13 shows the forms of the ER-BurrXII distribution's hazard function for rising values of parameter c and fixed values of a and k . We can also observe that for all parameter values the hazard plot exhibits an upside down bathtub shape, it also displays both monotonically decreasing and unimodal features.

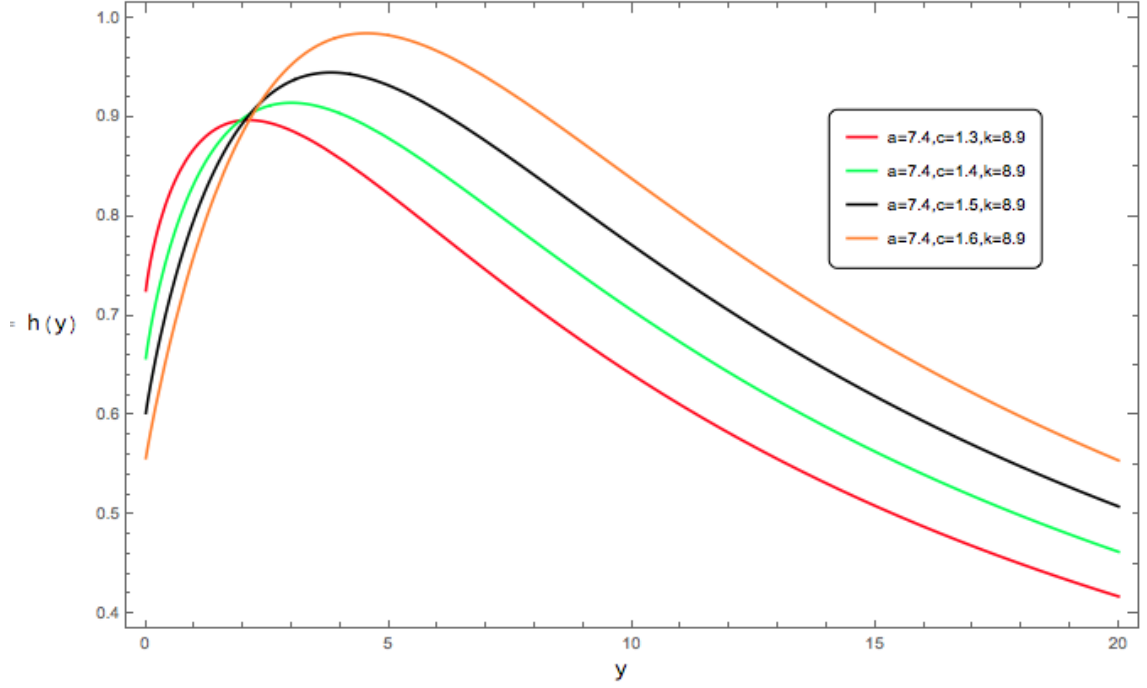


Figure 4.13: Hazard function plots of the ER-BurrXII distribution as c increases with a constant a and k .

The reversed hazard (RHR) function of the ER-BurrXII is given by,

$$RHR(y) = \frac{\left(1 + \left(\frac{y}{a}\right)^c\right)^{-k}}{yF_{(1,2)}\left(\frac{1}{c}, k, 1 + \frac{1}{c}; -\left(\frac{y}{a}\right)^c\right)}.$$

Figure 4.14 exhibits the RHR function plots of the ER-BurrXII distribution for varied parameter values. It is observed that the RHR function reduces quicker for some particular parameter values.

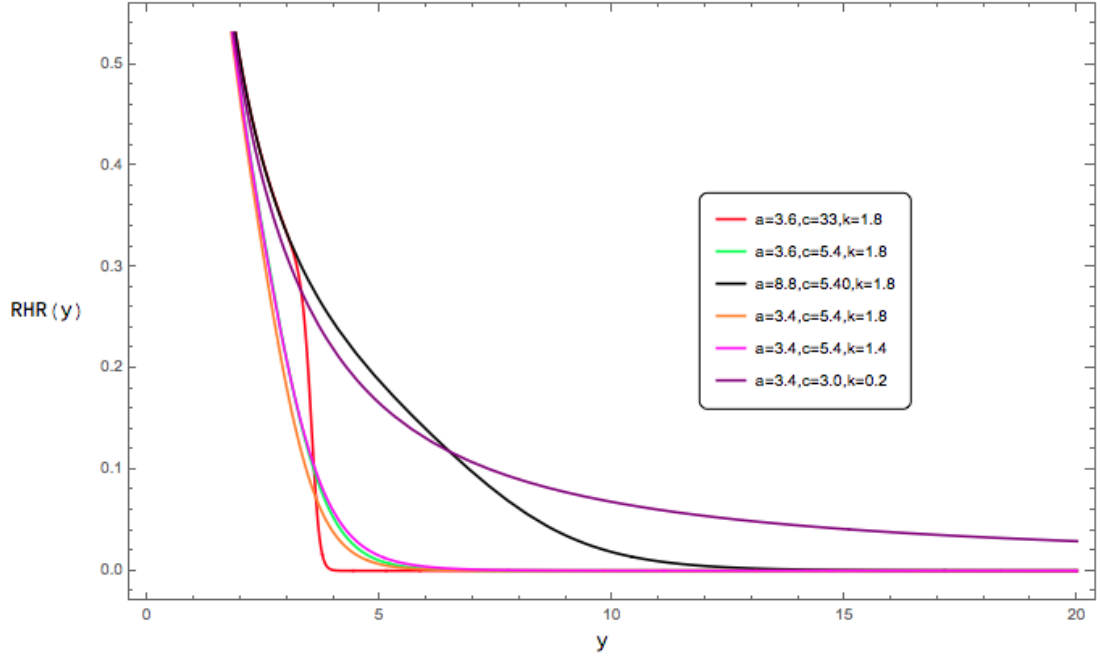


Figure 4.14: RHR function plots of the ER-BurrXII distribution.

4.3.1 Mathematical Properties of the Equilibrium Renewal BurrXII distribution

In this section, we study some mathematical properties of the ER-BurrXII distribution. Some mathematical properties such as ordinary central moments, incomplete moments, moment generating function, characteristic function, mean deviations, Beforroni and Lorenze curves, order statistics and entropy are investigated.

Quantile Function

Given a random variable Y with continuous and strictly monotonic CDF $G(y)$, a quantile function $Q(f)$ assigns to each probability p attained by g the value y for which $Pr(Y \leq y) = p$. Given a random variable Y with CDF $G(y; a, c, k)$. The quantile function of the

ER-BurrXII distribution is obtained by letting

$$p = \frac{yF_{(1,2)}\left(\frac{1}{c}, k, 1 + \frac{1}{c}; -\left(\frac{y}{a}\right)^c\right)}{kaB\left[\frac{1}{c} + 1, k - \frac{1}{c}\right]},$$

and solving for y . However, it is clear that the quantile function of the CDF of the ER-BurrXII distribution does not exhibit a closed form. Hence, numerical approximations will be applied to solve for the quantile function.

Table 4.9 displays the quantile values of the ER-BurrXII distribution for some values $p \in (0, 1)$ and choice parameter a , c and k . From Table 4.9 we observe that as p increases towards one for various parameter values, the quantile also increases. In general, we observe from Table 4.9 that as parameter values increase the quantile values generally increases in value.

Table 4.9: Some values of the quantile of the ER-BurrXII-Standard Uniform distribution for some values.

$Q(u)$	$a = 10, c = 2.4, k = 50$	$a = 5, c = 6, k = 40.9$	$a = 7.8, c = 5.1, k = 50.9$
0.1	0.1865	0.2505	0.3326
0.25	0.4654	0.6262	0.8313
0.35	0.6433	0.8766	1.1634
0.45	0.8168	1.1264	1.4938
0.5	0.9005	1.2507	1.6576
0.65	1.1372	1.6173	2.1369
0.75	1.2817	1.8502	2.4381
0.85	1.4151	2.0657	2.7153
0.9	1.4776	2.1646	2.9624

Moments

The moments of a statistical distribution are critical in any statistical analysis since they are used to study characteristics of a distribution. The r^{th} moment of the ER-BurrXII distribution is derived.

Proposition 4.11. The r^{th} moment of the ER-BurrXII distributed random variable Y

is defined as

$$\mu_r' = \frac{a^{r+1} \Gamma(\frac{r+1}{c}) \Gamma(k - \frac{r-1}{c})}{ckaB \left[\frac{1}{c} + 1, k - \frac{1}{c} \right] \Gamma(k)}, \quad r = 1, 2, \dots,$$

where $c < r + 1, r - 1$ and $\Gamma(\beta) = \int_0^\infty t^{\beta-1} e^{-t} dt$ is the gamma function.

Proof. By definition,

$$\begin{aligned} \mu_r' &= \int_0^\infty y^r g(y) dy \\ &= \int_0^\infty y^r \frac{\left(1 + \left(\frac{y}{a}\right)^c\right)^{-k}}{kaB \left[\frac{1}{c} + 1, k - \frac{1}{c} \right]} dy \\ &= \frac{1}{kaB \left[\frac{1}{c} + 1, k - \frac{1}{c} \right]} \int_0^\infty y^r \left(1 + \left(\frac{y}{a}\right)^c\right)^{-k} dy. \end{aligned}$$

Let $u = \left(1 + \left(\frac{y}{a}\right)^c\right)^{-1}$. As $y \rightarrow 0, u \rightarrow 1$ and as $y \rightarrow \infty, u \rightarrow 0$. Also, $y = a \left(\frac{1-u}{u}\right)^{1/c}$ and $dy = -\frac{a^c}{cy^{c-1}u^2} du$. Thus,

$$\begin{aligned} \mu_r' &= \frac{1}{kaB \left[\frac{1}{c} + 1, k - \frac{1}{c} \right]} \frac{a^{r+1}}{c} \int_0^1 (1-u)^{\frac{r+1}{c}-1} u^{k-\frac{r-1}{c}-1} du \\ &= \frac{1}{kaB \left[\frac{1}{c} + 1, k - \frac{1}{c} \right]} \frac{a^{r+1}}{c} B\left(\frac{r+1}{c}, k - \frac{r-1}{c}\right), \end{aligned}$$

where $B(u, v) = \int_0^1 t^{u-1} (1-t)^{v-1} dt$ is the beta function. Employing the beta function defined by the gamma function, we define

$$B(u, v) = \int_0^1 t^{u-1} (1-t)^{v-1} dt = \frac{\Gamma(u)\Gamma(v)}{\Gamma(u+v)},$$

where $u > 0$ and $v > 0$ (Andrews *et al.*, 1999). Hence, the r^{th} non-central moment of the

ER-BurrXII is given by

$$\mu_r' = \frac{a^{r+1} \Gamma(\frac{r+1}{c}) \Gamma(k - \frac{r-1}{c})}{ckaB [\frac{1}{c} + 1, k - \frac{1}{c}] \Gamma(k)}.$$

This completes the proof.

From proposition 4.3.1.2, the mean and variance of the ER-BurrXII distribution is obtained as follows,

$$E(X) = \frac{a^2 \Gamma(\frac{2}{c}) \Gamma(k)}{ckaB [\frac{1}{c} + 1, k - \frac{1}{c}] \Gamma(k)}$$

and

$$Var(X) = \left(\frac{a^2 \Gamma(\frac{3}{c}) \Gamma(k - \frac{1}{c})}{ckaB [\frac{1}{c} + 1, k - \frac{1}{c}] \Gamma(k)} \right) - \left(\frac{a^2 \Gamma(\frac{2}{c}) \Gamma(k)}{ckaB [\frac{1}{c} + 1, k - \frac{1}{c}] \Gamma(k)} \right)^2$$

respectively.

The r^{th} moment of the ER-BurrXII distribution are used to compute the mean, median, variance, skewness and kurtosis for varied parameter values of the Er-BurrXII distribution. These values are presented in Table 4.10, 4.10 and 4.11.

Table 4.10 displays that, for fixed a and c , both the variance and mean are increasing functions of k . The skewness increases as parameter values of c and k increases. In fact, from Table 4.10 the skewness values are all greater than 0 this which implies that the ER-BurrXII distribution can be quite asymmetric in nature, indicating a positively skewed characteristics. In addition, the ER-BurrXII shows positive kurtosis values, which indicates that the distribution can be heavy tailed in nature.

Table 4.10: Mean, variance, skewness and kurtosis of the ER-BurrXII distribution for $a = 5$ and various parameter values of c and k

Parameters			Moments			
a	c	k	Mean	Variance	Skewness	Kurtosis
5	10	0.5	0.3254	0.3254	0.5000	100.0000
		1.2	0.6087	0.4239	0.3424	0.3050
		1.5	0.7321	0.4881	0.3750	0.3150
		2.5	1.1331	0.6973	0.4911	0.3750
		3.0	1.3281	0.7974	0.5472	0.4066
5	12	0.5	0.3047	0.2686	0.3047	0.5000
		1.2	0.6036	0.4130	0.3246	0.2886
		1.5	0.7314	0.4836	0.3657	0.3000
		2.5	1.1473	0.7111	0.5019	0.3824
		3.0	1.3504	0.8203	0.5670	0.4226
5	20	0.5	0.2753	0.2067	0.1792	0.1715
		1.2	0.5978	0.4000	0.3032	0.2470
		1.5	0.7343	0.4822	0.3584	0.2859
		2.5	0.9592	0.6166	0.4482	0.3493
		3.0	1.4019	0.8775	0.6206	0.4702

Table 4.11 indicates that, for fixed a and c , both the variance and mean are increasing functions of k . The skewness and kurtosis exhibits an intermittent behaviour as the parameter values increases for a fixed a . Table 4.10 shows that the ER-BurrXII distribution is positively skewed (skewness > 0). In addition, the proposed distribution can be platykurtic (kurtosis < 3) in nature.

Table 4.11: Mean, variance, skewness and kurtosis of the ER-BurrXII distribution for $a = 0.5$ and various parameter values of c and k

Parameters			Moments			
a	c	k	Mean	Variance	Skewness	Kurtosis
0.5	10	0.5	0.0814	0.0407	0.0313	3.0000
		1.2	0.1522	0.0530	0.0214	0.0095
		1.5	0.1830	0.0610	0.0234	0.0098
		2.5	0.2823	0.0872	0.0307	0.0117
		3.0	0.3320	0.0997	0.0342	0.0127
0.5	12	0.5	0.0762	0.0336	0.0190	0.0156
		1.2	0.1509	0.0516	0.0203	0.0087
		1.5	0.2868	0.0889	0.0314	0.0119
		2.5	0.0119	0.0889	0.3137	0.0119
		3.0	0.3376	0.1025	0.0354	0.0132
0.5	20	0.5	0.0688	0.0258	0.0112	0.0054
		1.2	0.1495	0.5000	0.0190	0.0077
		1.5	0.1836	0.0603	0.0224	0.0089
		2.5	0.2954	0.0935	0.0335	0.0128
		3.0	0.3505	0.1097	0.0388	0.0147

Incomplete Moments

Proposition 4.12. The r^{th} incomplete moment of the ER-BurrXII distribution is defined

as,

$$m_r = \frac{B^*\left([1 + (\frac{y}{a})^c]^{-1}; \frac{r+1}{c}, k - \frac{r-1}{c}\right)}{ckB\left(\frac{1}{c} + 1, k - \frac{1}{c}\right)}, \quad r = 1, 2, \dots, \quad (4.31)$$

where $c < r + 1, r - 1$ and $B^*(y; u, v)$ is the upper incomplete beta function and $B(u, v)$ is the beta function.

Proof. By definition,

$$m_r(y) = \int_0^y x^r g(x) dx. \quad (4.32)$$

Substituting the PDF of the ER-BurrXII, into the r th incomplete moments of the ER-

BurrXII gives us,

$$m_r(y) = \frac{1}{kaB \left[\frac{1}{c} + 1, k - \frac{1}{c} \right]} \int_0^y x^r \left(1 + \left(\frac{x}{a} \right)^c \right)^{-k} dx. \quad (4.33)$$

Let $u = \left(1 + \left(\frac{x}{a} \right)^c \right)^{-1}$. As $x \rightarrow y, u \rightarrow \left(1 + \left(\frac{y}{a} \right)^c \right)^{-1}$ and as $y \rightarrow \infty, u \rightarrow 1$. Also, $y = a \left(\frac{1-u}{u} \right)^{1/b}$ and $dy = -\frac{a^c}{cy^{c-1}u^2} du$. Thus,

$$\begin{aligned} m_r(y) &= \frac{1}{kaB \left[\frac{1}{c} + 1, k - \frac{1}{c} \right]} \frac{a^{r+1}}{c} \int_{\left(1 + \left(\frac{y}{a} \right)^c \right)^{-1}}^1 (1-u)^{\frac{r+1}{c}-1} u^{k-\frac{r-1}{c}-1} du \\ &= \frac{1}{kaB \left[\frac{1}{c} + 1, k - \frac{1}{c} \right]} \frac{a^{r+1}}{c} B^* \left(\left[1 + \left(\frac{y}{a} \right)^c \right]^{-1}; \frac{r+1}{c}, k - \frac{r-1}{c} \right), \end{aligned}$$

where $B^*(a, b; y)$ is defined by

$$B^*(u, v; y) = \int_y^1 t^{u-1} (1-t)^{v-1} dt = B(u, v) - B_y(u, v)$$

where $B(u, v)$ is the beta function and $B_y(u, v)$ is the incomplete beta function. (Abramowitz and Stegun, 1972). We may call $B^*(u, v; y)$ defined by (Bílková (2014) as cited by Alshkaki (2020)) as the upper incomplete function.

Moment Generating Function

Proposition 4.13. Given a random number Y having the ER-BurrXII distribution, then the MGF for the ER-BurrXII distribution is given by

$$M_Y(t) = \sum_{r=0}^{\infty} \frac{t^r a^{r+1} \Gamma\left(\frac{r+1}{c}\right) \Gamma\left(k - \frac{r-1}{c}\right)}{r! c k a B \left[\frac{1}{c} + 1, k - \frac{1}{c} \right] \Gamma(k)}, r = 1, 2, \dots, \quad (4.34)$$

$\Gamma(\beta) = \int_0^{\infty} t^{\beta-1} e^{-t} dt$ is the gamma function.

Proof. Suppose Y is a random variable having the ER-BurrXII PDF in equation (4.29).

We obtain the MGF by using the following formula

$$M(t) = E[e^{tY}] = \int_0^{\infty} e^{ty} g(y) dy. \quad (4.35)$$

Applying Taylors series expansion,

$$\begin{aligned} M_Y(t) &= \int_{-\infty}^{\infty} \sum_{r=0}^{\infty} \frac{t^r y^r}{r!} g(y) dy \\ &= \sum_{r=0}^{\infty} \frac{t^r}{r!} \int_{-\infty}^{\infty} y^r g(y) dy \\ &= \sum_{r=0}^{\infty} \frac{t^r}{r!} \mu_r' \end{aligned}$$

substituting the r^{th} moment in Proposition (1.1) yields

$$M_Y(t) = \sum_{r=0}^{\infty} \frac{t^r a^{r+1} \Gamma(\frac{r+1}{c}) \Gamma(k - \frac{r-1}{c})}{r! c k a B \left[\frac{1}{c} + 1, k - \frac{1}{c} \right] \Gamma(k)}, r = 1, 2, \dots,$$

Characteristic Function

Proposition 4.14. Given a random number Y having the ER-BurrXII distribution,

Then the characteristic function for the ER-BurrXII distribution is given by

$$C_Y(t) = \sum_{r=0}^{\infty} \frac{(it)^r a^{r+1} \Gamma(\frac{r+1}{c}) \Gamma(k - \frac{r-1}{c})}{r! c k a B \left[\frac{1}{c} + 1, k - \frac{1}{c} \right] \Gamma(k)}, r = 1, 2, \dots, \quad (4.36)$$

$\Gamma(\beta) = \int_0^{\infty} t^{\beta-1} e^{-t} dt$ is the gamma function.

Proof. Suppose Y is a random variable having the ER-BurrXII. We obtain the charac-

teristics function by using the following formula

$$M(t) = E[e^{itY}] = \int_0^\infty e^{ity} g(y) dy, \quad (4.37)$$

where $E[Y]$ denotes the mathematical expectation. Applying Taylors series expansion,

$$\begin{aligned} M_Y(t) &= \int_{-\infty}^\infty \sum_{r=0}^\infty \frac{(it)^r y^r}{r!} g(y) dy \\ &= \sum_{r=0}^\infty \frac{(it)^r}{r!} \int_{-\infty}^\infty y^r g(y) dy \\ &= \sum_{r=0}^\infty \frac{(it)^r}{r!} \mu_r' \end{aligned}$$

substituting the r^{th} moment in Proposition (1.1) yields

$$M_Y(t) = \sum_{r=0}^\infty \frac{(it)^r a^{r+1} \Gamma(\frac{r+1}{c}) \Gamma(k - \frac{r-1}{c})}{r! c k a B[\frac{1}{c} + 1, k - \frac{1}{c}] \Gamma(k)}.$$

This completes the proof.

Mean Deviations

In the discipline of statistical analysis, mean deviations from the mean and median are crucial. The measure of spread in a population from the centre is measured by the use of mean deviations.

Proposition 4.15. The mean deviation of a random variable Y having the ER-BurrXII from its mean is

$$D(y) = 2\mu G(\mu) - 2ck a B^* \left(k^*; \frac{r+1}{c}, k - \frac{r-1}{c} \right) \left[B \left(\frac{1}{c} + 1, k - \frac{1}{c} \right) \right]^{-1}, \quad (4.38)$$

where $k^* = \left(1 + \left(\frac{\mu}{a}\right)^c\right)^{-1}$ and $\mu = \mu'_1$ is the mean of Y .

Proof. The mean deviation about the mean is defined as

$$D(y) = \int_0^\infty |y - \mu|g(y)dy. \quad (4.39)$$

Then the mean deviation measure of the ER-BurrXII can be calculated using the relationship,

$$\begin{aligned} D_1(y) &= \int_0^\mu (\mu - y)g(y)dy + \int_\mu^\infty (y - \mu)g(y)dy \\ &= \mu G(\mu) - \int_0^\mu yg(y)dy - \mu\{1 - G(\mu)\} + \int_\mu^\infty yg(y)dy \\ &= 2\mu G(\mu) - 2 \int_0^\mu yg(y)dy. \end{aligned}$$

Let $Y \sim \text{ER-BurrXII}(a, c, k)$. Then,

$$D(y) = 2\mu G(\mu) - 2ckaB^* \left(k^*; \frac{r+1}{c}, k - \frac{r-1}{c}\right) \left[B\left(\frac{1}{c} + 1, k - \frac{1}{c}\right)\right]^{-1},$$

where $\int_0^\mu ydy$ is derived using the first incomplete moment.

Proposition 4.16. The mean deviation of a random variable Y having the ER-BurrXII from its median is

$$D_2(y) = \mu - 2ckaB^* \left(k^{**}; \frac{r+1}{c}, k - \frac{r-1}{c}\right) \left[B\left(\frac{1}{c} + 1, k - \frac{1}{c}\right)\right]^{-1},$$

where $k^{**} = \left(1 + \left(\frac{M}{a}\right)^c\right)^{-1}$ and M is the median of Y .

Proof. The mean deviation about the mean is defined as

$$D_2(y) = \int_0^\infty |y - M|g(y)dy.$$

Then the mean deviation measure of the ER-BurrXII can be calculated using the relationship,

$$\begin{aligned} D_2(y) &= \int_0^M (M - y)g(y)dy + \int_M^\infty (y - M)g(y)dy \\ &= MG(M) - \int_0^M yg(y)dy - M\{1 - G(M)\} + \int_M^\infty yg(y)dy \\ &= \mu - 2 \int_0^M yg(y)dy. \end{aligned}$$

Let $Y \sim \text{ER-BurrXII}(a, c, k)$. Then,

$$D_2(y) = \mu - 2ckaB\left(k^{**}; \frac{r+1}{c}, k - \frac{r-1}{c}\right) \left[B\left(\frac{1}{c} + 1, k - \frac{1}{c}\right)\right]^{-1},$$

where $\int_0^M ydy$ is simplified using the first incomplete moment.

Beforroni and Lorenz curve

Proposition 4.17. The Lorenz curve for a random variable having the ER-BurrXII distribution is,

$$L_F(y) = \frac{B^*\left((1 + (\frac{y}{a})^c)^{-1}; \frac{2}{c}, k\right)}{ck\mu B\left(\frac{1}{c} + 1, k - \frac{1}{c}\right)}, \quad (4.40)$$

where $B(u, v; y)$ is the upper incomplete beta function and $B(u, v)$ is the beta function.

Proof. The Lorenz curve of the distribution of a random variable is defined as

$$L_G(y) = \frac{1}{\mu} \int_0^y xg(x)dx.$$

From the definition, the Lorenz curve is simply the product of the first incomplete moment and the reciprocal of the mean of the random variable. Hence,

$$L_G(y) = \frac{B^* \left(\left(1 + \left(\frac{y}{a}\right)^c\right)^{-1}; \frac{2}{c}, k \right)}{ck\mu B \left(\frac{1}{c} + 1, k - \frac{1}{c} \right)}.$$

The proof is therefore complete.

Proposition 4.18. The Bonferroni curve of a random variable having the ER-BurrXII distribution is

$$B_G(y) = \frac{B^* \left(\left(1 + \left(\frac{y}{a}\right)^c\right)^{-1}; \frac{2}{c}, k \right)}{ck\mu G(y) B \left(\frac{1}{c} + 1, k - \frac{1}{c} \right)},$$

where $B(u, v; y)$ is the upper incomplete beta and $F_{(1,2)}(a, b, c; z)$ is the Gauss hypergeometric function.

Proof. The Bonferroni curve by definition is given as

$$B_G(y) = \frac{L_G(y)}{G(y)}.$$

Thus, substituting the Lorenz curve into the definition of the Bonferroni curve and CDF of the ER-BurrXII distribution completes the proof.

Order Statistics

Let $Y_{(1)}, Y_{(2)}, \dots, Y_{(n)}$ denote the order statistics of a random sample $Y_1, Y_2, \dots, Y_n$ from a continuous population with CDF $G_Y(y)$ and PDF $g_Y(y)$, then the PDF of the largest

order statistics Y_p is given by,

$$g_{p:n} = U_{r:n} [G(y)]^{p-1} [1 - G(y)]^{n-p} g(y),$$

for $r = 1, 2, \dots, n$, where $U_{r:n} = \frac{n!}{(p-1)!(n-p)!} = [B(p, n-p+1)]^{-1}$ is the beta function.

The PDF of the largest order ER-BurrXII random variable $Y_{(p)}$ is given by

$$\begin{aligned} f_{p:n}(y) = & U_{r:n} \left[\frac{y F_{(1,2)}\left(\frac{1}{c}, k, 1 + \frac{1}{c}; -\left(\frac{y}{a}\right)^c\right)}{kaB\left[\frac{1}{c} + 1, k - \frac{1}{c}\right]} \right]^{p-1} \left[1 - \frac{y F_{(1,2)}\left(\frac{1}{c}, k, 1 + \frac{1}{c}; -\left(\frac{y}{a}\right)^c\right)}{kaB\left[\frac{1}{c} + 1, k - \frac{1}{c}\right]} \right]^{n-p} \\ & \times \frac{\left(1 + \left(\frac{y}{a}\right)^c\right)^{-k}}{kaB\left[\frac{1}{c} + 1, k - \frac{1}{c}\right]}. \end{aligned}$$

Therefore, the PDF of the n^{th} ER-BurrXII statistic $Y_{n:p}$ is given by,

$$f_{n:p}(y) = n \left[\frac{F_{(1,2)}\left(\frac{1}{c}, k, 1 + \frac{1}{c}; -\left(\frac{y}{a}\right)^c\right)}{kaB\left[\frac{1}{c} + 1, k - \frac{1}{c}\right]} \right]^{n-1} \frac{\left(1 + \left(\frac{y}{a}\right)^c\right)^{-k}}{kaB\left[\frac{1}{c} + 1, k - \frac{1}{c}\right]}$$

and the PDF of the first ER-BurrXII statistic $Y_{1:p}$ is given by,

$$f_{1:p}(y) = n \left[1 - \frac{F_{(1,2)}\left(\frac{1}{c}, k, 1 + \frac{1}{c}; -\left(\frac{y}{a}\right)^c\right)}{kaB\left[\frac{1}{c} + 1, k - \frac{1}{c}\right]} \right]^{n-1} \left[1 + \left(\frac{y}{a}\right)^c \right]^{-k} \left[kaB\left(\frac{1}{c} + 1, k - \frac{1}{c}\right) \right]^{-1}.$$

Entropy

Proposition 4.19. Given a random variable Y having the ER-BurrXII distribution, the

Rényi entropy of the ER-BurrXII distribution is given by,

$$I_R(\delta) = (1 - \delta)^{-1} \log \left\{ \frac{aB\left(k\delta - \frac{1}{c}, \frac{1}{c}\right)}{c \left[ckaB\left(\frac{1}{c} + 1, k - \frac{1}{c}\right) \right]^\delta} \right\}, \quad (4.41)$$

where $\delta > 0$ and $\delta \neq 0$.

Proof. By definition,

$$I_R(\delta) = (1 - \delta)^{-1} \log \left\{ \int_0^\infty g(y)^\delta dy \right\}, \quad (4.42)$$

where $\delta > 0$ and $\delta \neq 0$. Substituting the density of the MEC distribution into equation (4.42), we obtain

$$I_R(\delta) = (1 - \delta)^{-1} \log \left\{ \int_0^\infty \left[\frac{\left(1 + \left(\frac{y}{a}\right)^c\right)^{-k}}{ckaB\left(\frac{1}{c} + 1, k - \frac{1}{c}\right)} \right]^\delta dy \right\}.$$

Let

$$v(y) = \int_0^\infty \left[\frac{\left(1 + \left(\frac{y}{a}\right)^c\right)^{-k}}{ckaB\left(\frac{1}{c} + 1, k - \frac{1}{c}\right)} \right]^\delta dy.$$

We simplify $v(y)$ as follows,

$$v(y) = \frac{1}{[ckaB\left(\frac{1}{c} + 1, k - \frac{1}{c}\right)]^\delta} \int_0^\infty \left(1 + \left(\frac{y}{a}\right)^c\right)^{-k\delta} dy.$$

Let $u = \left(1 + \left(\frac{y}{a}\right)^c\right)^{-1}$. As $y \rightarrow 0, u \rightarrow 1$ and as $y \rightarrow \infty, u \rightarrow 0$. Also, $y = a \left(\frac{1-u}{u}\right)^{1/b}$ and $dy = -\frac{a^c}{cy^{c-1}u^2} du$. Thus,

$$\begin{aligned} v(y) &= \frac{1}{[ckaB\left(\frac{1}{c} + 1, k - \frac{1}{c}\right)]^\delta} \frac{a}{c} \int_0^1 (1-u)^{\frac{1}{c}-1} u^{k\delta - \frac{1}{c}-1} du \\ &= \frac{1}{[ckaB\left(\frac{1}{c} + 1, k - \frac{1}{c}\right)]^\delta} \left[\frac{a}{c} B\left(k\delta - \frac{1}{c}, \frac{1}{c}\right) \right]. \end{aligned}$$

Hence ,

$$I_R(\delta) = (1 - \delta)^{-1} \log \left\{ \frac{aB\left(k\delta - \frac{1}{c}, \frac{1}{c}\right)}{c [ckaB\left(\frac{1}{c} + 1, k - \frac{1}{c}\right)]^\delta} \right\}.$$

The proof is therefore complete.

Table 4.12 displays the Rényi entropy values for various value of a , c and k . It can be observed that for various increasing parameter values, the values of the Rényi entropy is decreases. Also, as the values of a and c increases with a constant k , the entropy values tends to negative. This indicates the flexibility of the amount of randomness of the ER-BurrXII distribution.

Table 4.12: Numerical values of the Rényi entropy of the MEC distribution at different values of a , c , and k .

Parameter			Entropy			
a	c	k	0.1	1.2	2.0	3.0
1.0	20.5	2.1	0.5325	-6.4122	-2.1530	-1.6134
1.2	20.5	2.2	0.8098	-8.1351	-2.5550	-1.8681
1.4	20.5	2.4	1.0322	-10.0095	-3.0371	-2.1759
2.0	20.5	2.5	1.5936	-12.9399	-3.6733	-2.5253
2.1	20.5	2.6	1.6626	-13.6354	-3.8607	-2.6490
2.2	10.1	3.0	2.0611	-15.2429	-4.3943	-3.0585
2.5	10.7	3.0	2.2225	-16.1633	-4.5747	-3.1452
3.0	10.8	3.0	2.5081	-17.4782	-4.8371	-3.2759
3.1	10.9	3.0	2.5541	-17.7144	-4.8837	-3.2987
3.2	20.2	3.0	2.6050	-17.9435	-4.9295	-3.3216
0.4	30.1	4.1	-1.1231	-5.6427	-2.8006	-2.4517
0.4	30.4	4.2	-1.1242	-5.8523	-2.8710	-2.5046
0.4	30.5	4.3	-1.1244	-6.0573	-2.9400	-2.5567
0.4	30.6	4.5	-1.1241	-6.4533	-3.0736	-2.6574
0.4	30.7	4.6	-1.1241	-6.6447	-3.1381	-2.7059

4.3.2 Estimation of Parameters of ER-Burr XII Distribution

We use the approach of maximum likelihood estimation to determine the estimation of the parameter of the ER-BurrXII distribution in this section.

Maximum Likelihood Estimation

Let $Y_1, Y_2, Y_3, \dots, Y_n$ be random sample of sample size n from the ER-BurrXII distribution, then the likelihood function of the ER-BurrXII distribution is defined by,

$$\begin{aligned} L(y; a, c, k) &= \prod_{i=1}^n \frac{\left(1 + \left(\frac{y}{a}\right)^c\right)^{-k}}{kaB\left[\frac{1}{c} + 1, k - \frac{1}{c}\right]} \\ &= \frac{1}{kaB\left[\frac{1}{c} + 1, k - \frac{1}{c}\right]} \prod_{i=1}^n \left(1 + \left(\frac{y_i}{a}\right)^c\right)^{-k}. \end{aligned}$$

Using the fact that,

$$B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)},$$

the likelihood function of the ER-BurrXII can be rewritten as,

$$L(y; a, c, k) = \frac{(\Gamma(1+k))^n}{(ka\Gamma\left(\frac{1}{c} + 1\right)\Gamma\left(k - \frac{1}{c}\right))^n} \prod_{i=1}^n \left(1 + \left(\frac{y_i}{a}\right)^c\right)^{-k}.$$

The log likelihood function is

$$\begin{aligned} \ln(L(x; a, c, k)) &= n \left(\ln \Gamma(1+k) - \ln \Gamma\left(\frac{1}{c} + 1\right) - \ln \Gamma\left(k - \frac{1}{c}\right) \right) - n \ln k - n \ln a \\ &\quad - k \sum_{i=0}^n \ln \left(1 + \left(\frac{y_i}{a}\right)^c\right). \end{aligned} \tag{4.43}$$

The maximum likelihood estimators of a , c and k are obtained by maximization of the log likelihood function defined by

$$\begin{aligned}
\frac{\partial \ln L(y; a, c, k)}{\partial a} &= -\frac{n}{a} - \sum_{i=0}^n \frac{kc \left(\frac{y_i}{a}\right)^c}{a \left(1 + \left(\frac{y_i}{a}\right)^c\right)} = 0 \\
\frac{\partial \ln L(y; a, c, k)}{\partial c} &= -n \left\{ \Psi \left(\frac{1}{c} + 1 \right) + \Psi \left(k - \frac{1}{c} \right) \right\} - \sum_{i=0}^n \frac{k \left(\frac{y_i}{a}\right)^c \ln \left(\frac{y_i}{a}\right)}{a \left(\frac{y_i}{a}\right)^c + 1} = 0 \quad (4.44) \\
\frac{\partial \ln L(y; a, c, k)}{\partial k} &= -\frac{n}{k} + n \left\{ \Psi(1 + k) + \Psi \left(k - \frac{1}{c} \right) \right\} - k \sum_{i=0}^n \ln \left(1 + \left(\frac{y_i}{a}\right)^c \right) = 0
\end{aligned}$$

where $\Psi(z) = \frac{d}{dz} \ln(\Gamma(z))$ is the digamma function. Equation 4.44 does not have explicit solutions, hence numerical approximation will be employed in computing the estimates of parameters a , c and k respectively.

4.4 Summary of Theoretical Results

In this Chapter we introduce the MEC and ER-BurrXII distributions and some of its important statistical properties are studied. The maximum likelihood, ordinary least squares and weighted least method for estimating the parameters for the MEC distribution are discussed. The maximum likelihood estimate approach is also presented for deriving the estimators of parameters for the ER-BurrXII.

CHAPTER 5

SIMULATIONS AND APPLICATIONS

5.1 Introduction

In this chapter, we presents results on Monte Carlo simulations and applications of the developed models to varied data sets.

5.2 Monte Carlo Simulation Study of the Modified Extended Chen Distribution

We provide the results of simulation study in this section to evaluate the performance of the various estimators presented in Chapter 4. The main goal of this section is to compare the performance of the different estimation methods for the parameters of the MEC distribution by computing their ABs and RMSEs. The experiment was performed with three different set of parameter values $(a, b, \rho) = (0.8, 0.3, 0.5), (0.4, 8.2, 6.5)$ and $(2.5, 1.8, 4.5)$. The simulation steps are:

- i. Specify the value of the parameters a, b, ρ and sample size n .
- ii. Generate random observations of size $n = 30, 60, 100, 150, 200, 300, 500$ from the MEC distribution using the quantile function.
- iii. Compute MLE, OLS and WLS of parameters a, b and ρ according to section (4.2.3).
- iv Replicate steps *ii* – *iii* for $N = 10,000$ times.
- v. To examine the performance of the estimators, we compute the AB and RMSE.

The summary of the results using the Monte-Carlo simulation study are reported in Figures (5.1-5.3).

The results in Figure 5.1 shows the performance of the MLE, OLS and WLS estimators of parameter a . It can be observed that as n increases, the resulting ABs and RMSEs trend to zero for the various estimators. In addition, it can be seen that as the sample size grows larger the MLE decays faster compared to the other estimation approaches. It is important to note that the MLE, OLS and WLS estimators for the parameter a have smaller RMSE and AB compared to that of the other parameters. In the case of parameter a , the MLE displays a better performance.

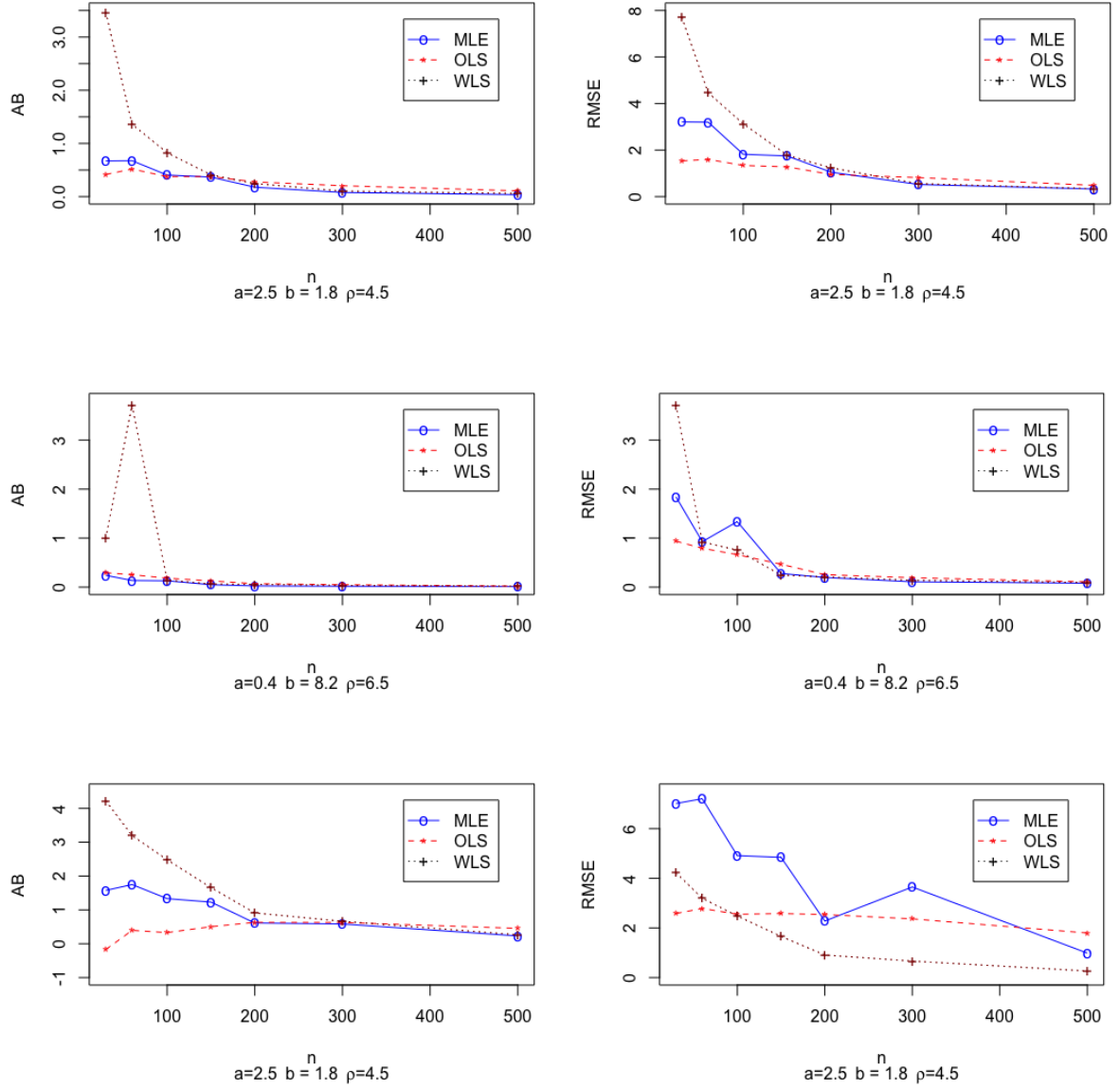


Figure 5.1: Plots of the AB (Left) and RMSE (Right) of the different estimators of a for MLE, OLS, and WLS with different sample sizes 30, 60, 100, 150, 200, 300 and 500

Figure 5.2 displays the numerical efficiency of the MLE, OLS and WLS estimators of parameter b . It can be shown that as the sample size grows, the resulting ABs and RMSEs decline to zero for each estimator. For parameter values $a = 2.5, b = 1.8, \rho = 4.5$ and $a = 0.4, b = 8.2, \rho = 6.5$ we observe that the RMSE of the WLS estimator shows exhibits and irregular behaviour. Also, for the different estimators, it can be noted that

both the MLE and WLS estimators decays towards zero as sample size increases for both AB and RMSE. However, the MLE decays faster compared to the WLS and OLS estimators.

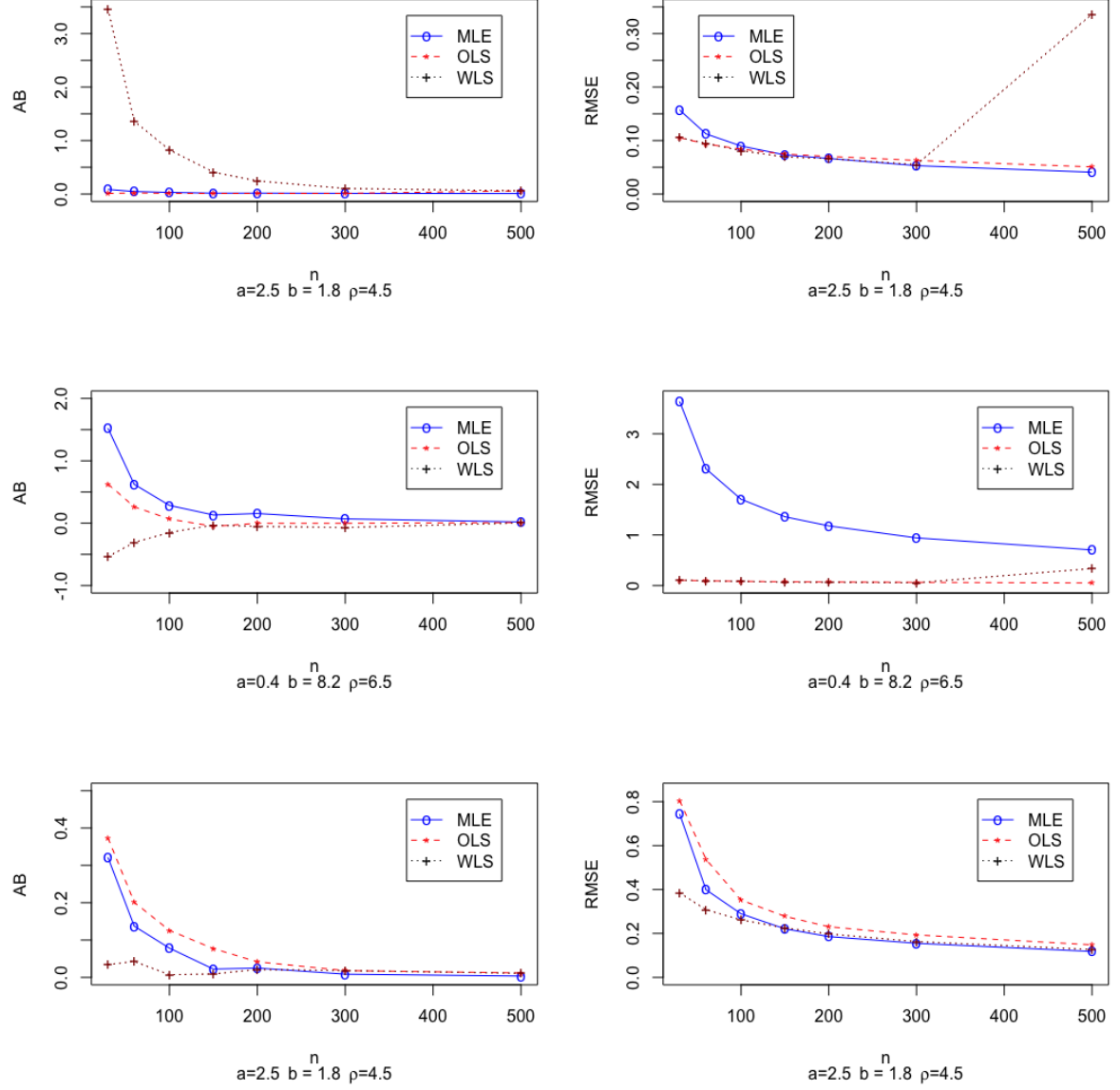


Figure 5.2: Plots of the AB(Left) and RMSE(Right) of the different estimators of b for MLE, OLS, and WLS with different sample sizes 30, 60, 100, 150, 200, 300 and 500

Figure 5.3 displays the numerical efficiency of the MLE, OLS and WLS estimators of parameter ρ . It can be shown that as the sample size grows, the resulting ABs and RMSEs decline to zero for each estimator. The MLE and the OLS estimators for parameter values

$a = 0.4, b = 8.2, \rho = 6.5$ and $a = 2.5, b = 1.8, \rho = 4.5$ have, respectively, the largest AB and RMSE among all the considered estimators for parameter ρ . However, the AB and RMSE of the MLE estimator decays faster towards zero compared the other estimators as the sample size grows. Combining the findings of the estimator simulation research, we conclude that the MLE technique is a very competitive approach for estimating the MEC distribution when compared to the OLS and WLS methods.

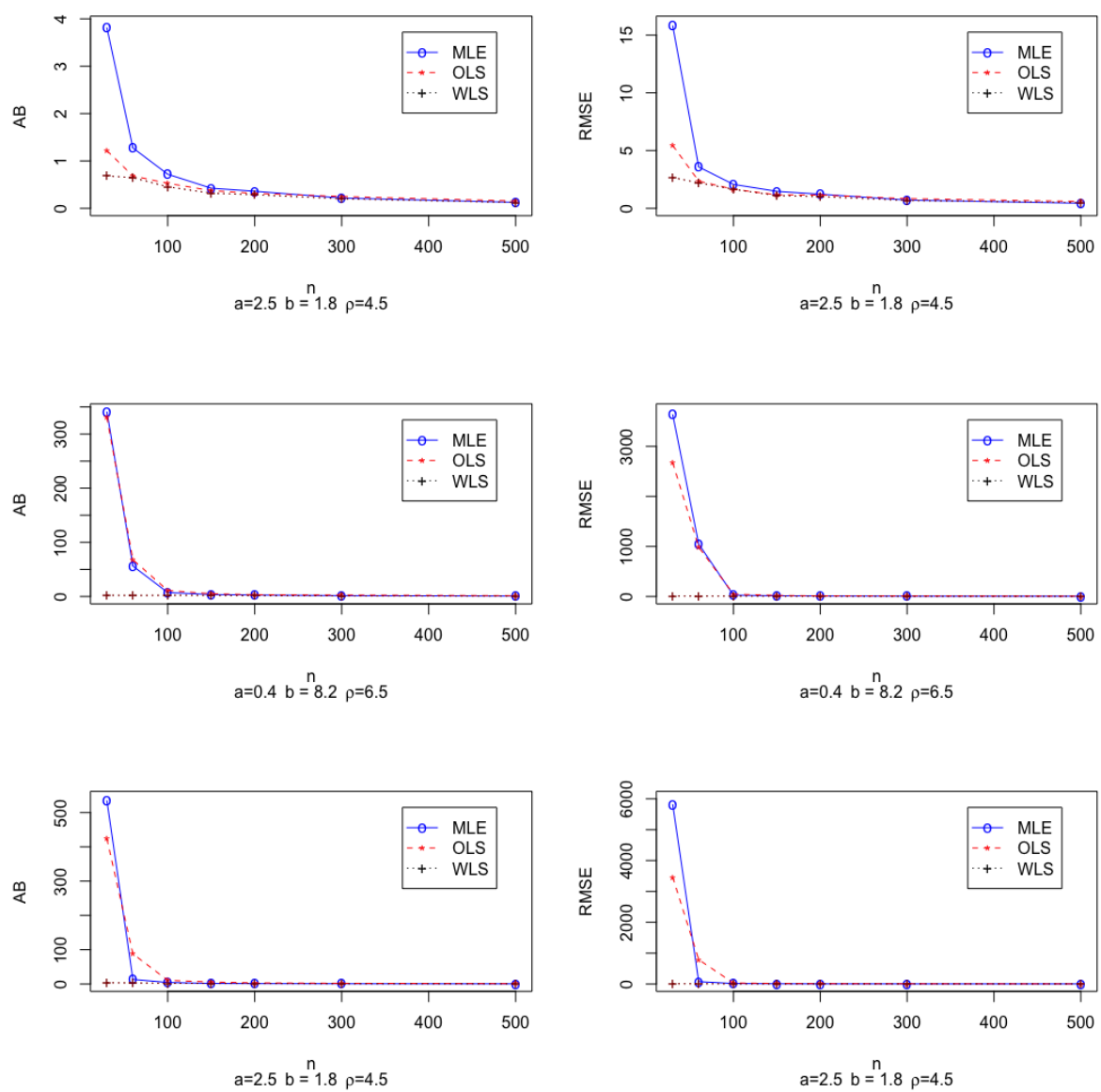


Figure 5.3: Plots of the AB(Left) and RMSE (Right)of the different estimators of ρ for MLE, OLS, and WLS with different sample sizes 30, 60, 100, 150, 200, 300 and 500

5.3 Applications of the Modified Extended Chen Distribution

To illustrate the applications of the MEC model, annual maximum rainfall data in Ghana for five different locations (Eastern, Greater-Accra, Upper East, Central and Volta Regions) were modeled by applying the MEC distribution. The performance of the MEC distribution is compared with other competing distributions such as Log-Logistic (LL) (Singh, 1998), Inverse Weibull (IW) (Akgül *et al.*, 2016), Cauchy (Carrillo *et al.*, 2010), Burr XII (3P) (Burr, 1968), Nakagami (NAK) (Nakagami, 1960), Gumbel (GUM) (Gumbel, 1935), Extended Burr XII (EB) (Shao *et al.*, 2004) and Generalised Inverse Weibull (GIW) (De Gusmao *et al.*, 2011) for maximum annual rainfall data from all five locations in Ghana.

In comparing the distributions, the most appropriate model is chosen as the one having highest Log-likelihood (L) and lowest AIC, BIC, CAIC and HQIC. The A, W and KS statistics for goodness-of-fit were also computed to compliment the selection of the best distribution for the rainfall data.

5.3.1 Maximum Annual Rainfall Data

Figure 5.4 represents maximum annual rainfall at all five locations over the period 1980-2019, showing extreme rainfall over the locations using block maxima method. The outstanding feature is that the highest and lowest magnitude of rainfall occurred in the year 1982 and 1983 respectively in Cape Coast. It can also be observed that for all locations, the scatter plots does not exhibit a linear relationship.

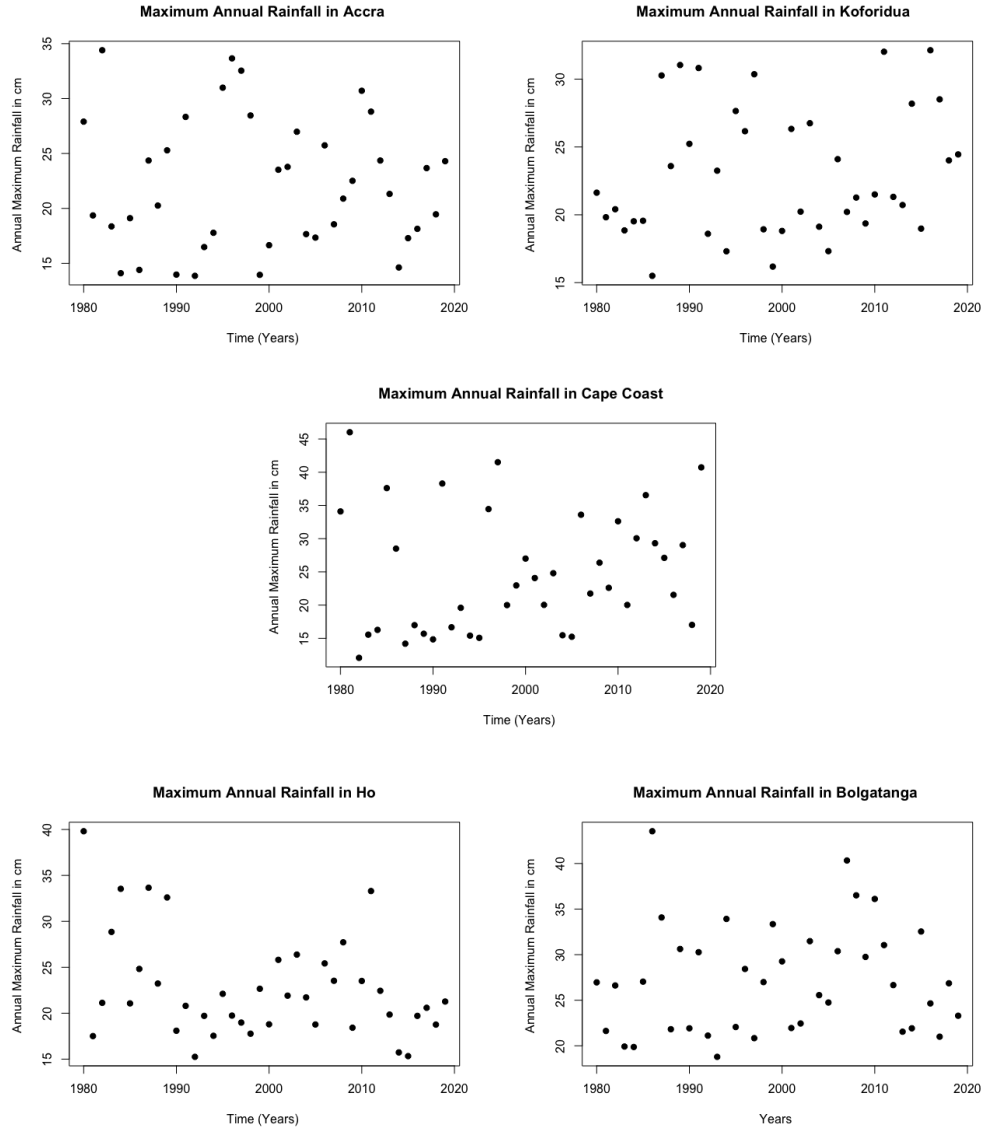


Figure 5.4: Scatter plots for annual maxima rainfall

The statistical characteristics of each location are given in Table 5.1. The minimum and maximum rainfall were 12.06 *cm* and 46.02 *cm* in locations Cape Coast and Bolgatanga respectively.

From table 5.1 it is observed that Cape Coast is characterized by a high annual maximum of average monthly precipitation compared with the other locations; after that is Bolgatanga in the second position, then Ho in the third position, followed by Accra and finally Koforidua with the lowest rainfall average.

Also, the normality assumption is rejected for all the locations since the skewness for each one is greater than zero, and the same applies to the kurtosis which is different for all the locations. However, location Ho is highly skewed (skewness = 1.198021) compared to other locations. On the basis of the Coefficient of Variation (CV) for all locations, we observe that all locations have low variation since the CV gives values between 0.2 – 0.4.

Table 5.1: Statistical properties of the annual maximum of monthly precipitation (in cm) for the five locations.

Locations	Minimum	Maximum	Median	Mean	Skewness	Kurtosis	Coefficient of Variation
Accra	13.86	34.40	21.11	22.10	0.4081619	-0.8192587	0.2683296
Koforidua	15.50	32.13	21.41	23.00	0.5183654	-0.8867075	0.2059671
Ho	15.25	39.81	21.20	22.70	0.5944017	-0.6633967	0.2472855
Cape Coast	12.06	46.02	22.78	24.77	0.5718753	2.270334	0.3639055
Bolgatanga	18.79	43.54	26.76	27.20	0.7680319	0.1516706	0.2194923

Maximum Annual Rainfall Data-Koforidua

The first data represents the maximum annual rainfall of location Koforidua in the Eastern region of Ghana.

Table 5.2 displays the maximum likelihood estimates of the parameters, standard errors and p-values for the fitted distributions. From Table 5.2, it was observed that all parameters of the MEC, CAUCHY, NAK, LL, GUM and GIW distributions were significant at 5% significance level. Moreover, parameters a and b for EB and IW is significant at 5% whiles parameter ρ is significant at 10% significance level. In addition, for BURR(3P) we observe that both parameters a and ρ were significant at 5% significance level and parameter b is statistical significant at 10% significance level.

Table 5.2: MLEs of parameters, standard errors, p-values for maximum annual rainfall for Koforidua

Distribution	Parameters	Estimates	Standard error	Z-statistic	p-value
MEC	$\hat{a}$	4.2719×10^3	4.2427×10^{-6}	1.0069×10^9	$< 2.2 \times 10^{-16*}$
	$\hat{b}$	5.7431	5.3614×10^2	1.0712×10^2	$< 2.2 \times 10^{-16*}$
	$\hat{\rho}$	7.9190×10^3	2.2882×10^6	3.4608×10^9	$< 2.2 \times 10^{-16*}$
LL	$\hat{a}$	3.3763	1.2553	2.6897	0.007152*
	$\hat{b}$	8.8843	3.0661	2.8975	0.003761*
	$\hat{\rho}$	13.0823	2.7824	4.7018	$2.579 \times 10^{-6*}$
IW	$\hat{a}$	11.4855	1.8517411	6.2025	$5.556 \times 10^{-10*}$
	$\hat{b}$	2.34849	0.6112334	3.8422	0.0001219*
	$\hat{\rho}$	0.00633	0.0115126	0.5500	0.5823502
CAUCHY	a	2.79732	0.63396	4.4125	$1.022 \times 10^{-5*}$
	$\hat{b}$	21.07031	0.76005	27.7222	$< 2.2 \times 10^{-16*}$
BURR(3P)	$\hat{a}$	15.70295	6.46572	2.4286	0.01516*
	$\hat{b}$	0.31387	0.18549	1.6921	0.09062
	$\hat{\rho}$	18.89884	1.14131	16.5588	$< 2 \times 10^{-16*}$
NAK	$\hat{a}$	525.3651	31.8421	16.499	$< 2.2 \times 10^{-16*}$
	$\hat{b}$	6.3686	1.4068	4.527	$5.983 \times 10^{-6*}$
GUM	$\hat{a}$	22.07684	0.99129	22.2708	$< 2.2 \times 10^{-16*}$
	$\hat{b}$	5.34725	1.14468	4.6714	$2.991 \times 10^{-6*}$
EB	$\hat{a}$	20.3492	1.0707	19.0062	$< 2 \times 10^{-16*}$
	$\hat{b}$	15.6814	6.4590	2.4278	0.01519*
	$\hat{\rho}$	-3.1802	1.8810	-1.6907	0.09090
GIW	$\hat{a}$	6.933899	0.869737	7.9724	$1.556 \times 10^{-15*}$
	$\hat{b}$	5.743041	0.699394	8.2114	$< 2.2 \times 10^{-16*}$
	$\hat{\rho}$	500.508617	0.002098	$2.3856 \times 10^{+5}$	$< 2.2 \times 10^{-16*}$

*: Means significant at the 5% significance level.

The log-likelihood, information criterion, and goodness-of-fit statistics for Koforidua's maximum annual rainfall are presented in Table 5.3. It was evident that the MEC had the greatest L value and the lowest AIC, CAIC, BIC, and HQIC values among the nine models stated.

The MEC distribution, on the other hand, has the smallest W, A, and KS values as well as the biggest associated p-values. In light of this information, the MEC distribution clearly outperforms the other eight alternative models in terms of fitting Koforidua's rainfall data.

Table 5.3: Comparison criterion of maximum annual rainfall for Koforidua

Distribution	L	AIC	CAIC	BIC	HQIC	W	A	KS
MEC	-116.9126	239.8252	240.0752	244.8918	13.9890	0.06529 (0.7836)	0.4658 (0.7809)	0.083376 (0.922)
LL	-117.3418	240.6836	240.9336	245.7502	15.7045	0.080829 (0.6897)	0.5226 (0.7228)	0.086735 (0.8988)
IW	-119.3193	244.6386	244.8886	249.7052	15.88815	0.074055 (0.7297)	0.67369 (0.5802)	0.12105 (0.5599)
CAUCHY	-127.2068	258.4136	258.6636	261.7913	14.15573	0.34134 (0.1035)	2.2725 (0.0657)	0.16974 (0.1776)
BURR(3P)	-117.5054	241.0109	241.2609	246.0775	13.99899	0.081822 (0.684)	0.55103 (0.6944)	0.098983 (0.7919)
NAK	-117.9936	239.9872	240.2372	243.365	18.74573	0.13265 (0.4486)	1.0388 (0.3371)	0.1109 (0.6679)
GUM	-120.1498	244.2996	253.5742	247.6773	14.04295	0.46115 (0.04954)	2.4763 (0.05128)	0.21283 (0.04547)
EB	-117.5054	241.0108	241.2608	246.0775	13.99899	0.081828 (0.684)	0.55094 (0.6945)	0.098974 (0.792)
GIW	-116.9129	239.8258	240.0758	244.8924	13.98901	0.065297 (0.7835)	0.46588 (0.7808)	0.083383 (0.9219)

Note: P-value in bracket.

The estimated PDFs and CDFs of all fitted distributions using the estimates derived in Table 5.2 are shown in Figure 5.9. The MEC distribution fits the maximum rainfall data for Koforidua better than the other eight models, as seen in Figure 5.5.

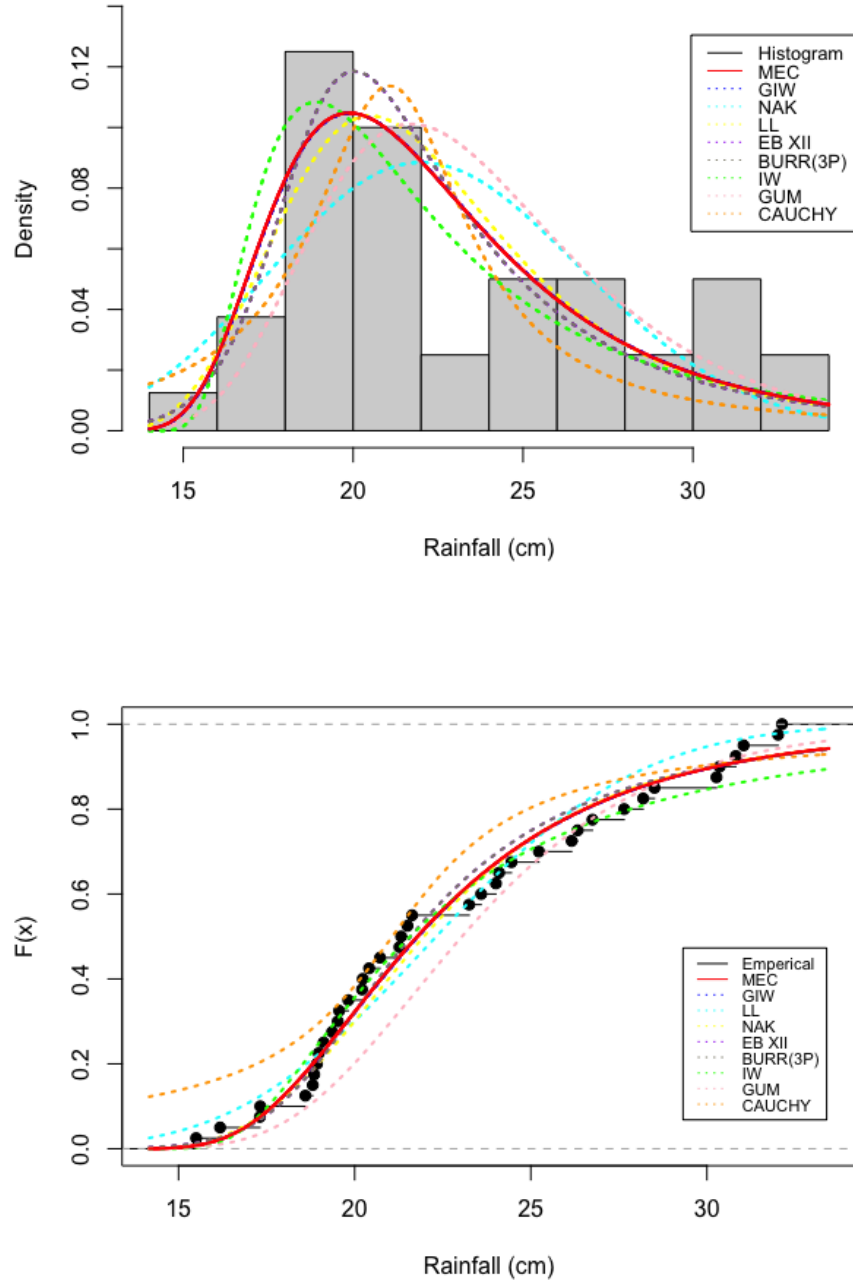


Figure 5.5: The estimated PDFs (top panel) and the estimated CDFs (bottom panel) for annual monthly rainfall maxima from Koforidua, Ghana.

Maximum Annual Rainfall Data-Ho

The second data represents the maximum annual rainfall of location Ho in the Volta region of Ghana.

Table 5.4 presents the parameter estimates and the corresponding standard errors, Z-statistic and p-values. The parameter estimates. In addition, we observe that the p-values of the MEC, LL, CAUCHY, NAK, GUM, EB, BURR(3P) and GIW are significant at 5% significance level.

However, for the IW distribution the parameters a and ρ were significant at the 5% significance level and that of the b is not significant at 5% significance level but rather significant at the 10% significance level.

Table 5.4: MLEs of parameters, standard errors, p-values for maximum annual rainfall for Ho.

Distribution	Parameter	Estimates	Standard Errors	Z-Statistic	p-value
MEC	$\hat{a}$	3.8605×10^3	4.4915×10^{-6}	9.1590×10^8	$< 2.2 \times 10^{-16*}$
	$\hat{b}$	5.4267	5.4141×10^{-2}	1.0023×10^2	$< 2.2 \times 10^{-16*}$
	$\hat{\rho}$	2.3366×10^3	6.7855×10^{-6}	4.0121×10^8	$< 2.2 \times 10^{-16*}$
LL	$\hat{a}$	3.09214	0.95964	3.2222	0.0012721*
	$\hat{b}$	8.26783	2.26688	3.6472	0.0002651*
	$\hat{\rho}$	13.06573	2.00823	6.5061	$7.713 \times 10^{-11*}$
IW	$\hat{a}$	10.8067895	2.9312220	3.6868	0.0002271*
	$\hat{b}$	2.3259777	0.9225447	2.5213	0.0116934*
	$\hat{\rho}$	0.0064566	0.0182139	0.3545	0.7229722
CAUCHY	$\hat{a}$	2.50604	0.52527	4.771	$1.833 \times 10^{-6*}$
	$\hat{b}$	20.81511	0.61554	33.816	$2.2 \times 10^{-16*}$
BURR(3P)	$\hat{a}$	13.84146	4.40400	3.1429	0.001673*
	$\hat{b}$	0.35089	0.16908	2.0753	0.037959*
	$\hat{\rho}$	18.59741	1.08282	17.1750	$< 2.2 \times 10^{-16*}$
NAK	$\hat{a}$	308.94474	40.21912	7.6815	$1.572 \times 10^{-14*}$
	$\hat{b}$	1.66544	0.57775	2.8826	0.003944*
GUM	$\hat{a}$	21.2747	1.1885	17.9011	$2.2 \times 10^{-16*}$
	$\hat{b}$	6.4848	2.2379	2.8978	0.003758*
EB	$\hat{a}$	20.05901	0.94989	21.1172	$< 2.2 \times 10^{-16*}$
	$\hat{b}$	13.84132	4.40385	3.1430	0.001672*
	$\hat{\rho}$	-2.84997	1.37328	-2.0753	0.037959*
GIW	$\hat{a}$	7.8123	8.4888×10^{-1}	9.2031	$< 2.2 \times 10^{-16*}$
	$\hat{b}$	5.4256	6.6386×10^{-1}	8.1728	$3.013 \times 10^{-16*}$
	$\hat{\rho}$	1.5988	7.6451×10^{-3}	20912.5952	$< 2.2 \times 10^{-16*}$

*: Means significant at the 5% significance level.

Table 5.5, displays the goodness-of-fit, log-likelihood and information criteria for the maximum annual rainfall for location Ho. It is obvious from Table 5.5 that MEC provides a

better fit as compared to the LL, IW, CAUCHY, BURR(3P), NAK, GUMBEL, EB and GIW since it has highest value of L and lowest value AIC, CAIC ,BIC and HQIC. Also, the MEC distribution has the highest p-values in the associated W, A, and KS tests, indicating that it gives a better fit than other competing models.

Table 5.5: Comparison criterion of maximum annual rainfall for Ho

Distribution	L	AIC	CAIC	BIC	HQIC	W	A	KS
MEC	-118.8895	243.7791	244.0291	248.8457	14.02212	0.021933 (0.9953)	0.22094 (0.9836)	0.059639 (0.9972)
LL	-118.9523	243.9046	244.1546	261.759	14.02316	0.023603 (0.9929)	0.24645 (0.9721)	0.063671 (0.9969)
IW	-120.7117	247.4233	247.6733	252.4900	14.05216	0.064223 (0.7903)	0.54868 (0.6967)	0.087788 (0.9175)
CAUCHY	-126.0075	256.0150	256.265	259.3927	14.13700	0.1689 (0.3378)	1.2341 (0.2547)	0.13468 (0.4626)
BURR(3P)	-119.0473	244.0947	244.3447	249.1613	14.02474	0.016452 (0.9994)	0.17299 (0.9961)	0.059535 (0.9989)
NAK	-123.4889	250.9778	251.2278	254.3555	14.41734	0.1954 (0.2772)	1.1943 (0.2695)	0.14819 (0.3434)
GUM	-125.4797	254.9594	255.2094	258.3371	14.12871	0.56339 (0.02707)	2.7836 (0.03556)	0.23151 (0.02284)
EB	-119.0473	244.0947	244.3447	249.1613	14.02474	0.026165 (0.988)	0.27008 (0.9586)	0.070718 (0.9882)
GIW	-118.8897	243.7794	244.0294	248.8461	14.022121	0.021947 (0.9953)	0.22099 (0.9836)	0.061725 (0.9963)

Note: P-value in bracket.

The density and empirical cumulative plots are shown in the Figure 5.6. From Figure 5.6, it is noted that the MEC distribution appears to follow the pattern of both the histogram and empirical cumulative plots better. It can be observed that the data is skewed to the right, indicating that the MEC distribution has the ability to fit right-skewed data.

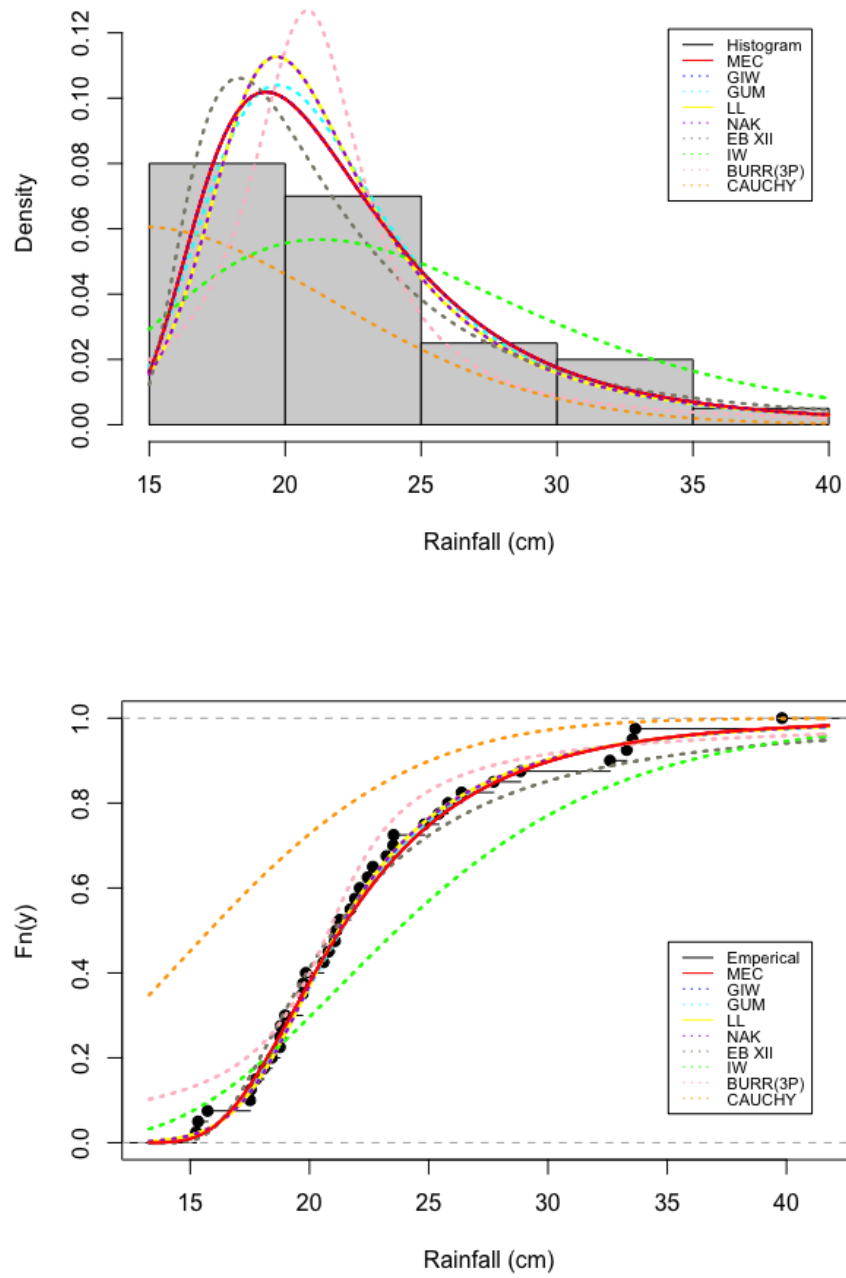


Figure 5.6: The estimated PDFs (top panel) and the estimated CDFs (bottom panel) for annual monthly rainfall maxima from Ho, Ghana.

Maximum Annual Rainfall Data-Accra

The third data represents the maximum annual rainfall of location Accra in the Greater-Accra region of Ghana.

Table 5.6 presents the parameter estimates and the corresponding standard errors, Z-statistic and p-values. The P-values of the BURR(3P), NAK, GUM, CAUCHY and GIW are statistically significant at 5% significance level.

We observe that the MEC and LL distributions are significant at 5% and 10% significance level respectively for only parameter b . In addition, for the IW distribution, the parameters a and b were significant at 5% significance level. The parameter ρ of IW distribution is not significant at 5% but at 10% significance level. The EB distribution is significant at only 5% significance level for parameter a and parameter b and ρ are both significant at 10% significance level.

Table 5.6: MLEs of parameters, standard errors, p-values for maximum annual rainfall for Accra

Distribution	Parameter	Estimates	Standard Estimates	Z-Statistic	p-value
MEC	$\hat{a}$	430.03933	300.44838	1.4313	0.1523
	$\hat{b}$	4.13516	0.49828	8.2989	$<2 \times 10^{-16*}$
	$\hat{\rho}$	422.85300	301.55429	1.4022	0.1608
LL	$\hat{a}$	3.7939	2.3594	1.6080	0.10784
	$\hat{b}$	13.0129	7.6657	1.6976	0.08959
	$\hat{\rho}$	8.0263	7.1783	1.1181	0.26351
IW	$\hat{a}$	8.3646902	0.0628004	133.1948	$<2 \times 10^{-16*}$
	$\hat{b}$	2.2028769	0.2553028	8.6285	$<2 \times 10^{-16}$
	$\hat{\rho}$	0.0064008	0.0034636	1.8480	0.0646
CAUCHY	$\hat{a}$	4.06692	0.78699	5.1677	2.37×10^{-7}
	$\hat{b}$	20.82905	1.20118	17.3405	$<2.2 \times 10^{-16*}$
BURR(3P)	$\hat{a}$	96.8083447	0.0000418	2.3160×10^6	$<2.2 \times 10^{-16*}$
	$\hat{b}$	0.0236819	0.0038028	6.2275	$4.739 \times 10^{-10*}$
	$\hat{\rho}$	13.7815275	0.1724790	7.9903×10^1	$<2.2 \times 10^{-16*}$
NAK	$\hat{a}$	358.74467	36.04506	9.9527	$<2.2 \times 10^{-16*}$
	$\hat{a}$	2.44167	0.70466	3.4650	0.0005301*
GUM	$\hat{a}$	20.5562	1.5824	12.9904	$<2 \times 10^{-16*}$
	$\hat{b}$	8.3028	3.5855	2.3156	0.02058*
EB	$\hat{a}$	14.47373	0.55482	26.0874	$<2 \times 10^{-16}$
	$\hat{b}$	71.74213	63.36079	1.1323	0.2575
	$\hat{\rho}$	-30.92685	27.72494	-1.1155	0.2646
GIW	$\hat{a}$	9.56592	0.73126	13.0814	$<2.2 \times 10^{-16*}$
	$\hat{b}$	4.13142	0.50183	8.2327	$<2.2 \times 10^{-16*}$
	$\hat{\rho}$	15.93386	0.10626	149.9487	$<2.2 \times 10^{-16*}$

*: Means significant at the 5% significance level.

Table 5.7 shows the goodness of fit, log-likelihood and information criteria values for all nine fitted models. From the findings in Table 5.7, it is evident that, the MEC distribution has the highest L value and lowest values of the AIC, BIC, CAIC, HQIC, A, KS and W and highest p-values of the KS, A and W statistic. Hence, it can be concluded that the proposed MEC distribution, could be chosen as a better model than proposed models namely LL, IW, CAUCHY, BURR(3P), NAK, GUM, EB and GIW.

Table 5.7: Comparison criterion of maximum annual rainfall for Accra

Distribution	-L	AIC	CAIC	BIC	HQIC	W	A	KS
MEC	-127.4893	260.9785	261.2285	266.0452	14.16011	0.083686 (0.6734)	0.65537 (0.5962)	0.12868 (0.5218)
LL	-127.5129	261.0258	261.2758	266.0924	14.16048	0.13938 (0.4251)	1.012 (0.3505)	0.13389 (0.4702)
IW	-129.2508	264.5015	264.7515	269.5682	14.18724	0.12399 (0.4813)	0.96402 (0.3761)	0.1258 (0.5513)
CAUCHY	-137.5277	279.0554	279.3054	282.4332	14.3100	0.16764 (0.341)	1.3042 (0.2309)	0.16815 (0.2081)
BURR(3P)	-130.4125	266.8249	267.0749	271.8916	14.20493	0.45627 (0.05101)	2.3099 (0.06275)	0.20546 (0.0683)
NAK	-136.2867	276.5733	276.8233	279.9511	14.29207	1.4978 (0.0001404)	8.8262 (5.67e-05)	0.27664 (0.004387)
GUM	-133.4749	270.9497	271.1997	274.3275	14.25083	0.45036 (0.05286)	2.6581 (0.04125)	0.18431 (0.13200)
EB	-130.7633	267.5267	267.7767	272.5933	14.21024	0.47012 (0.04695)	2.3595 (0.05907)	0.20749 (0.06385)
GIW	-127.491	260.9821	261.2321	266.0487	14.16011	0.084021 (0.6715)	0.65753 (0.5943)	0.12884 (0.5202)

Note: P-value in bracket.

Figure 5.7 presents the estimated PDFs and estimated CDFs of all fitted distributions using the estimators derived in Table 5.6 in order to select the best model. The MEC distribution fits the maximum rainfall data for Accra better than the other eight models, as seen in Figure 5.7.

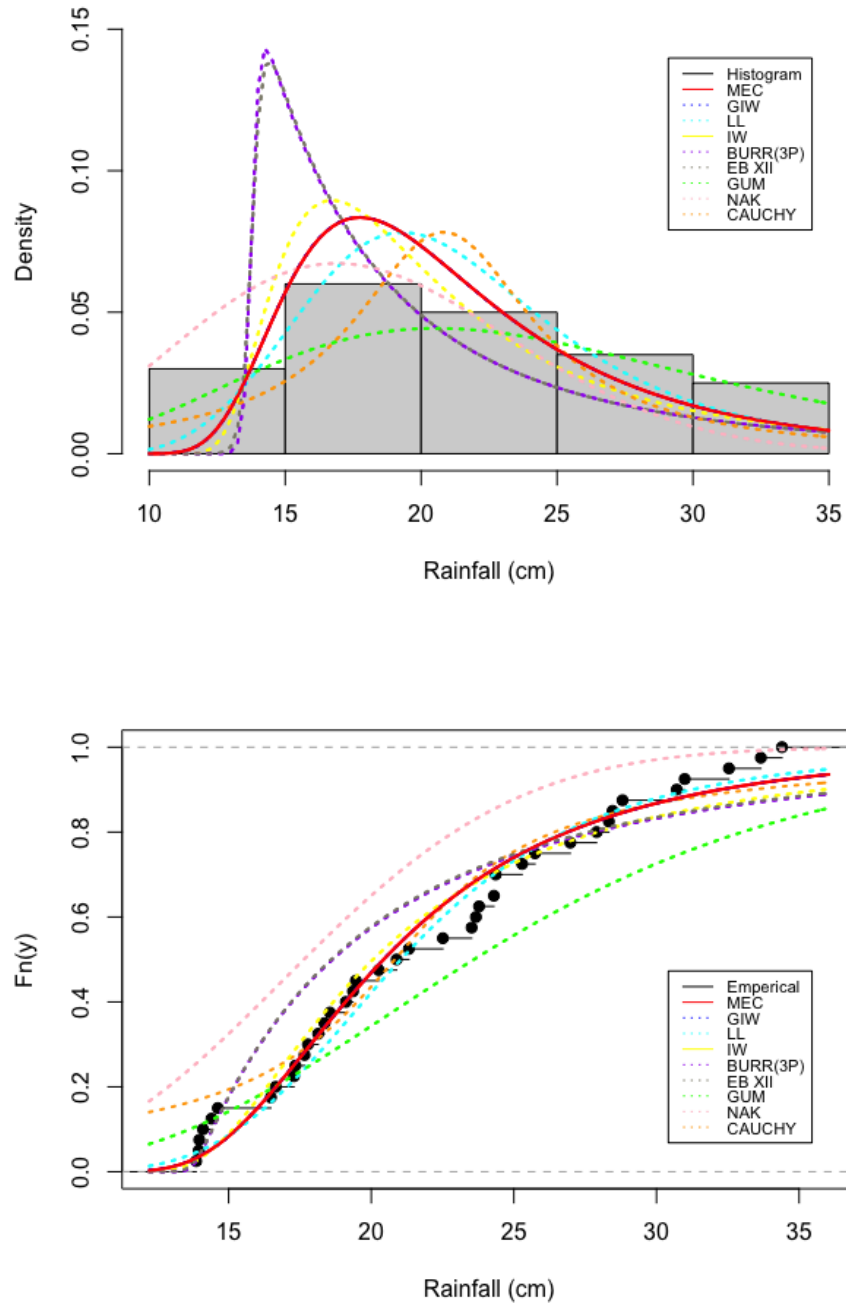


Figure 5.7: The estimated PDFs (top panel) and the estimated CDFs (bottom panel) for annual monthly rainfall maxima from Accra , Ghana.

Maximum Annual Rainfall Data-Bolgatanga

The third data represents the maximum annual rainfall of location Bolgatanga in the Upper East region of Ghana. The maximum likelihood estimators with standard errors,

z-statistics and corresponding p-values of the MEC model with other competing models are presented in Table 5.8. It can be observed that, the estimates of the MEC, LL, CAUCHY, NAK and GUM are all significant at $\alpha = 0.05$.

However, for the IW distribution the parameters a and b were significant at the 5% significance level and that of the ρ is not significant at 5% but rather significant at the 10% significance level. The parameters of EB distribution are significant at only 5% significance level for parameter a and parameter b and ρ are both significant at 10% significance level.

Table 5.8: MLEs of parameters, standard errors, p-values for maximum annual rainfall for Bolgatanga

Distribution	Parameters	Estimates	Standard Errors	Z-Statistics	p-value
MEC	$\hat{a}$	3.2486×10^4	5.0122×10^{-7}	6.4813×10^{10}	$<2.2 \times 10^{-16*}$
	$\hat{b}$	5.6626	5.0805×10^{-2}	1.1146×10^2	$<2.2 \times 10^{-16}$
	$\hat{\rho}$	2.0377×10^3	7.9904×10^{-6}	2.5502×10^8	$<2.2 \times 10^{-16*}$
LL	$\hat{a}$	2.63085	0.97936	2.6863	0.007225
	$\hat{b}$	8.65700	2.93384	2.9507	0.003170*
	$\hat{\rho}$	17.05216	2.45172	6.9552	$3.521 \times 10^{-12*}$
IW	$\hat{a}$	14.6460918	3.2126504	4.5589	$5.143 \times 10^{-6*}$
	$\hat{b}$	2.1516879	0.8680282	2.4788	0.01318*
	$\hat{\rho}$	0.0091091	0.0248331	0.3668	0.71376
CAUCHY	$\hat{a}$	25.97236	1.09890	5.2287	$1.707 \times 10^{-7*}$
	$\hat{b}$	3.89237	0.74443	23.6349	$<2.2 \times 10^{-16*}$
BURR(3P)	$\hat{a}$	27.76516	14.23223	1.9509	0.05107
	$\hat{b}$	0.13719	0.08392	1.6347	0.10211
	$\hat{\rho}$	20.60493	0.73966	27.8573	$<2 \times 10^{-16*}$
NAK	$\hat{a}$	241.2609	52.4775	14.9039	$<2.2 \times 10^{-16*}$
	$\hat{b}$	5.6714	1.2333	4.5984	$4.257 \times 10^{-6*}$
GUM	$\hat{a}$	25.51418	0.96347	26.4817	$<2.2 \times 10^{-16*}$
	$\hat{b}$	5.55434	0.89640	6.1963	$5.782 \times 10^{-10*}$
EB	$\hat{a}$	22.1287	1.0372	21.3355	$<2 \times 10^{-16*}$
	$\hat{b}$	27.8345	14.2507	1.9532	0.0508
		-7.3100	4.4650	-1.6372	0.1016
GIW	$\hat{a}$	72.6741113	0.0555038	1309.3538	$<2.2 \times 10^{-16*}$
	$\hat{b}$	5.1793923	0.9563309	5.4159	$6.098 \times 10^{-8*}$
	$\hat{\rho}$	0.0034450	0.0039713	0.8675	0.3857

*: Means significant at the 5% significance level.

The L values together with AIC, BIC, CAIC and HQIC values, and corresponding p-

values of the KS, A, W are presented in Table 5.9. By comparing the information criteria and goodness of fit statistics among the nine distributions in Table 5.9, it was observed that all distributions provide good fits. The MEC provides the best fit since the MEC has the lowest values of the AIC, BIC, CAIC, HQIC, A, W and KS. This application suggests that MEC fits right skewed data very well.

Table 5.9: Comparison criterion of maximum annual rainfall for Bolgatanga.

Distribution	L	AIC	CAIC	BIC	HQIC	W	A	KS
MEC	-124.6587	255.3173	240.0758	247.6773	14.11573	0.11621 (0.513)	0.63387 (0.6155)	0.12932 (0.5153)
LL	-125.3462	256.6923	256.9423	261.7590	14.1266	0.11429 (0.5212)	0.66985 (0.5835)	0.13346 (0.4744)
IW	-126.1552	258.3104	258.5604	263.3770	14.13932	0.12322 (0.4843)	0.74239 (0.5235)	0.14131 (0.4014)
CAUCHY	-136.0585	276.1169	276.3669	261.7913	14.28875	0.16985 (0.3353)	1.3073 (0.2299)	0.15808 (0.2703)
BURR(3P)	-125.8196	257.6392	241.2609	262.7059	14.13405	0.17943 (0.312)	0.91272 (0.4058)	0.17309 (0.1819)
NAK	-126.6781	257.3561	240.2372	260.7339	14.14749	0.11576 (0.515)	0.74836 (0.5189)	0.14674 (0.3551)
GUM	-126.0355	256.0709	256.3209	259.4487	14.13744	0.22732 (0.2205)	1.36700 (0.2116)	0.1738 (0.1783)
EB	-125.8196	257.6391	257.8891	262.7058	14.13405	0.1795 (0.3118)	0.91311 (0.4056)	0.17313 (0.1817)
GIW	-124.9136	255.8272	256.0772	260.8938	14.11977	0.087447 (0.6525)	0.5709 (0.6749)	0.13402 (0.469)

Note: P-value in bracket.

Figure 5.8 shows the theoretical and empirical PDF and CDF of the MEC distribution using the maximum annual rainfall data for Bolgatanga and it is observed that the MEC is the best-fitted line as compared to others.

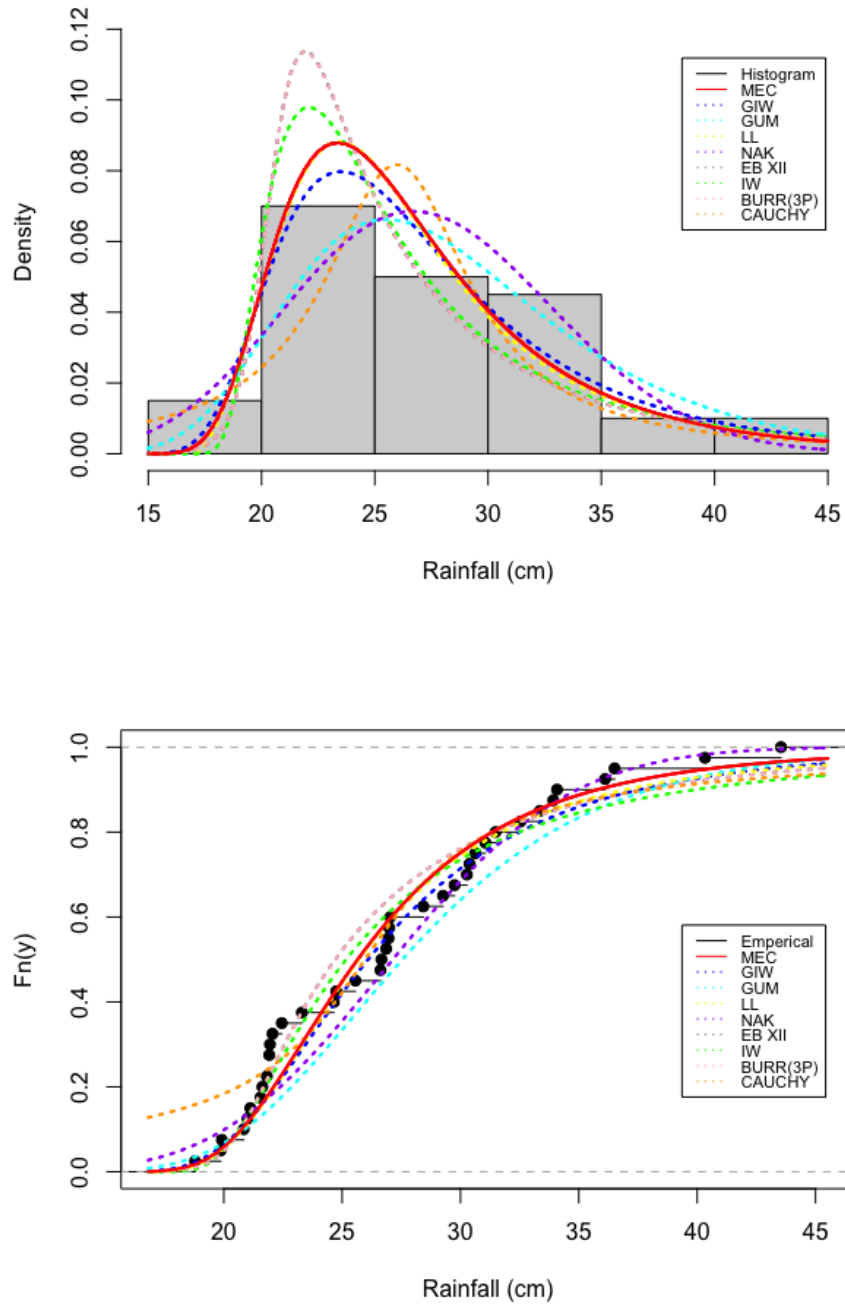


Figure 5.8: The estimated PDFs (top panel) and the estimated CDFs (bottom panel) for annual monthly rainfall maxima from Bolgatanga, Ghana.

Maximum Annual Rainfall Data-Cape Coast

The fifth data represents the maximum annual rainfall of location Cape Coast in the Central region of Ghana. Table 5.10 displays the maximum likelihood estimates of the

parameters, standard errors and p-values for the compared distributions. From Table 5.10, it also be observed that the parameter estimates of the LL, CAUCHY, NAK, GIW and GUM were significant at the 5% significance level. Parameters a and b were significant at the 5% significance level and that of the ρ is not significant at 5% but rather significant at the 10% significance level for the MEC distribution.

However, for the IW distribution, parameters a and ρ were significant at the 10% significance level and that of the b is not significant at 5%. For the BURR(3P), parameters a and b are significant at 10% significance level while parameter ρ is significant at 5% significance level. The EB distribution of b and ρ parameters are significant at 10% significance level and parameter a is significant at 5% significance level.

Table 5.10: MLEs of parameters, standard errors, p-values for maximum annual rainfall for Cape Coast

Distribution	Parameter	Estimates	Standard error	Z-Statistic	p-value
MEC	$\hat{a}$	1638.80000	0.04417	37102.4901	$<2.2 \times 10^{-16*}$
	$\hat{b}$	3.18997	0.39446	8.0870	$6.117 \times 10^{-16*}$
	$\hat{\rho}$	7.96994	8.99409	0.8861	0.3755
LL	$\hat{a}$	2.38744	0.72117	3.3105	0.0009313*
	$\hat{b}$	12.01639	3.30573	3.6350	0.0002780*
	$\hat{\rho}$	10.26471	2.55645	4.0152	$5.939 \times 10^{-5*}$
IW	$\hat{a}$	6.796415	3.477481	1.9544	0.050653
	$\hat{b}$	2.041814	0.677570	3.0134	0.002583*
	$\hat{\rho}$	0.006030	0.013904	0.4337	0.664515
CAUCHY	$\hat{a}$	5.9170	1.1778	5.0238	$5.067 \times 10^{-7*}$
	$\hat{b}$	22.0349	1.6754	13.1524	$<2.2 \times 10^{-16*}$
BURR(3P)	$\hat{a}$	15.09062	10.55420	1.4298	0.1528
	$\hat{b}$	0.14399	0.11896	1.2104	0.2261
	$\hat{\rho}$	14.88027	1.19011	12.5033	$<2 \times 10^{-16*}$
NAK	$\hat{a}$	579.3763	57.7134	10.0389	$<2.2 \times 10^{-16*}$
	$\hat{b}$	2.0012	0.4368	4.5816	$4.615 \times 10^{-6*}$
GUM	$\hat{a}$	21.8021	2.0737	10.5138	$<2 \times 10^{-16*}$
	$\hat{b}$	11.9467	5.1737	2.3091	0.02094*
EB	$\hat{a}$	16.9582	1.9213	8.8266	$<2 \times 10^{-16}$
	$\hat{b}$	14.8170	10.5008	1.4110	0.1582
	ρ	-6.8057	5.7172	-1.1904	0.2339
GIW	$\hat{a}$	16.9582	1.9213	8.8266	$<2 \times 10^{-16}$
	$\hat{b}$	14.8170	10.5008	1.4110	0.1582
	$\hat{\rho}$	-6.8057	5.7172	-1.1904	0.2339

*: Means significant at the 5% significance level.

Table 5.3 presents the log-likelihood, information criteria and goodness-of-fit statistics for the maximum annual rainfall for Cape Coast data. It was obvious that, for the nine mentioned models, the MEC has the smallest values of the L, AIC, CAIC, BIC and HQIC. On the other hand, it can be observed that the MEC distribution has the smallest W, A and KS value and the largest corresponding p-values. This confirms that the MEC distribution appears to be a very competitive model to the maximum rainfall data of Cape Coast than the other eight models.

Table 5.11: Comparison criterion of maximum annual rainfall for Cape Coast

Distribution	-L	AIC	CAIC	BIC	HQIC	W	A	KS
MEC	-141.7993	289.5986	289.8486	294.6652	14.37056	0.10861 (0.5464)	0.70327 (0.5551)	0.11181 (0.6581)
LL	-142.1333	290.2666	290.5166	295.3333	14.37518	0.11232 (0.5298)	0.84196 (0.4510)	0.14565 (0.3315)
IW	-142.8083	291.6166	291.8666	296.6832	14.38456	0.11254 (0.5288)	0.75483 (0.5139)	0.10761 (0.7031)
CAUCHY	-153.6206	311.2412	311.4912	314.619	14.52906	0.2294 (0.2173)	1.6749 (0.1399)	0.1807 (0.1294)
BURR(3P)	-143.3844	292.7689	293.0189	297.8355	14.6338	0.19757 (0.2728)	1.0894 (0.3132)	0.13682 (0.4059)
NAK	-143.8845	291.7691	292.0191	295.1468	14.39942	0.19602 (0.2759)	1.6196 (0.1505)	0.13136 (0.4564)
GUM	-148.5592	301.1184	301.3684	304.4962	14.46271	0.36194 (0.09089)	2.2971 (0.06375)	0.14979 (0.30000)
EB	-143.384	292.7680	293.018	297.8347	14.39252	0.19664 (0.2747)	1.0854 (0.3150)	0.13671 (0.4068)
GIW	-141.7994	289.5987	289.8488	294.6664	14.37082	0.10897 (0.5400)	0.7050 (0.5441)	0.11190 (0.6471)

Note: P-value in bracket.

In order to discover the best model fit, the results are graphically evaluated by plotting fitted densities with histograms and fitted CDFs with data for all nine models in Figure 5.7. The plots also demonstrates that the proposed MEC distribution is a good match for the data set under consideration.

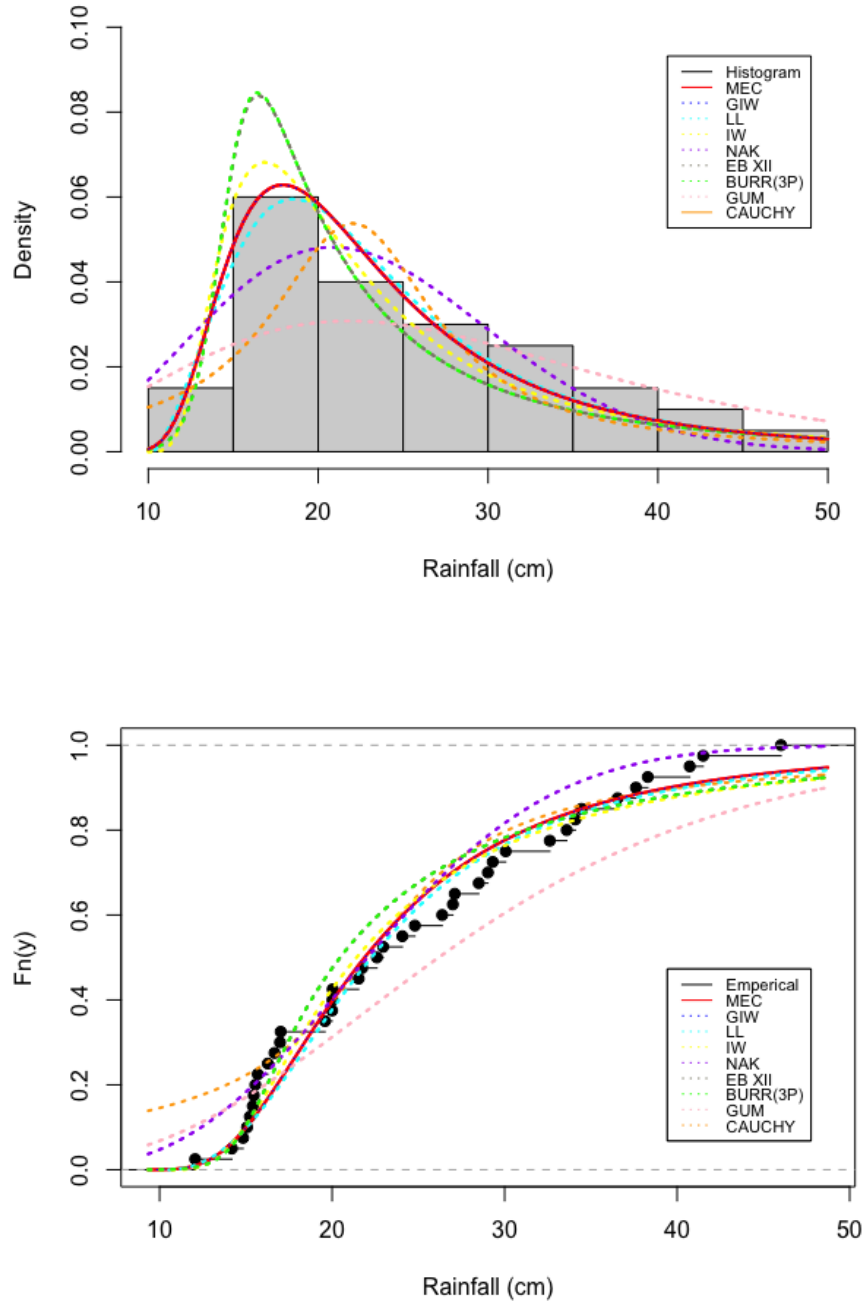


Figure 5.9: The estimated PDFs (top panel) and the estimated CDFs (bottom panel) for annual daily rainfall maxima from Cape Coast, Ghana.

5.3.2 Return level estimation for MEC model

The return level can be defined as the level which is associated to be exceeded one every $1/S(y_T)$ period, where $S(y_T)$ is the exceedance probability which is obtained from the

survival function, which is known as a return period. The return level of the MEC distribution is defined by,

$$y_T = \left[\log \left(\frac{\left(1 - \frac{1}{T}\right)^{-\frac{1}{a}} - 1}{\rho} + 1 \right) \right]^{-\frac{1}{b}},$$

where $T \leq 1$ and the return period is:

$$T = \frac{1}{1 - \left(\rho \left(e^{y_T^{-b}} - 1 \right) + 1 \right)^{-a}}.$$

A summary of the extreme values for all location using the MEC distribution are presented in Table 5.12. To know the approximate range of maximum annual rainfall for various return periods in each location, the MEC distribution was applied to analyze rainfall locations. The estimated results from the return levels can help explain the rarity of the extreme rainfall from a location. Generally, we observe that as the periods increase in years, the return levels of the maximum annual rainfall also increases. Considering a return period of 250 years, the minimum value is reached in Koforidua with an estimated value of 53.50542 *cm*; and the maximum value is obtained in Cape Coast reaching an extreme rainfall value of 110.28806 *cm*.

Table 5.12: Return level results of all locations

Location	Return Period (Years)								
	2	5	10	20	50	100	150	200	250
Koforidua	21.81348	26.57290	30.28221	34.32587	40.37202	45.59103	48.94042	51.46188	53.50542
Accra	20.43402	26.88123	32.23124	38.36042	48.05553	56.89499	62.78192	67.31883	71.05996
Ho	21.29653	26.24343	30.13546	34.41007	40.85569	46.46543	50.08565	52.82060	55.04292
Bolgatanga	25.65947	31.34561	35.78748	40.63858	47.90699	54.19365	58.23369	61.27769	63.74629
Cape Coast	21.88983	31.24114	39.53684	49.55694	66.38758	82.64979	93.91367	102.81330	110.28806

A summary of the extreme periods are presented in Table 5.13. The idea of return

periods can aid guide planners and policymakers. They can more easily make decisions for the length of time using the expected maximum annual rainfall determined by the return period. The estimated results indicate an increasing trend in the return periods as the return levels increases. Considering a minimum accumulated rainfall of 29 *cm* annually, the return period for Koforidua, Accra, Ho, Bolgatanga, and Cape Coast is approximately 8, 7, 8, 3 and 4 years, respectively. Also, with a maximum accumulated rainfall of 52 *cm* an annually, the return period for Koforidua, Accra, Ho, Bolgatanga, and Cape Coast is approximately 212, 69, 184, 79 and 23 annually, respectively. If this type of rainfall accumulation hits Cape Coast in the future, with an accumulated rainfall of up to 52 *cm* a month, Cape Coast will experience extreme rainfall with a return period of approximately 23 years.

Table 5.13: Return period results of all locations

Location	Return Levels (cm)							
	29	32.5	34.37	39.32	43.54	46.02	49	52
Koforidua	7.914349	14.749113	20.144348	41.085880	76.886298	105.497982	151.0482	212.2815
Accra	6.646036	10.331237	12.885893	21.379317	33.416752	41.889104	54.14825	69.09079
Ho	8.21642	14.80616	19.87713	38.96458	70.41476	94.93259	133.2377	183.7490
Bolgatanga	3.413794	6.015052	8.060820	15.948207	29.312236	39.926524	56.74442	79.24263
Cape Coast	4.058631	5.598979	6.588153	9.595786	13.412663	15.903379	19.31050	23.23039

5.4 Monte Carlo Simulation Study of the Equilibrium Renewal Burr XII Distribution

The major goal of this section is to investigate the ABs and RMSEs of the maximum likelihood estimators for the ER-BurrXII distribution's parameters. The experiment was performed with four different set of parameter values $(a, b, k) = (20.9, 1.5, 500)$ and $(40, 10, 100)$. The simulation steps are;

- i. Specify the value of the parameters a, c, k and sample size n .
- ii. Generate random observations of size $n = 10, 30, 60, 100, 150, 200, 300, 500$ from the ER-BurrXII distribution using the quantile function.
- iii. Compute MLE of parameters a, b and k according to section (4.3.2).
- iv Replicate steps *ii – iii* for $N = 10,000$ times.
- v. To examine the performance of the estimates, we calculated the AB and RMSE using the formulas,

$$AB = \frac{1}{N} \sum_{i=0}^N (\hat{\theta}_i - \theta) \quad \quad RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{\theta}_i - \theta)^2},$$

for $\theta = a, c, k$.

The ABs of the maximum likelihood estimators of parameter values $(a, c, k) = (20.9, 1.5, 500)$ are displayed in Figure 5.10. The ABs for parameter a and c are generally positive, except for parameter k which exhibits an initial AB values of negative. Although parameter c and k appear intermittent in nature, the AB for each parameter reduces to zero as n increases. The ABs appear largest and smallest for the parameter a and k respectively.

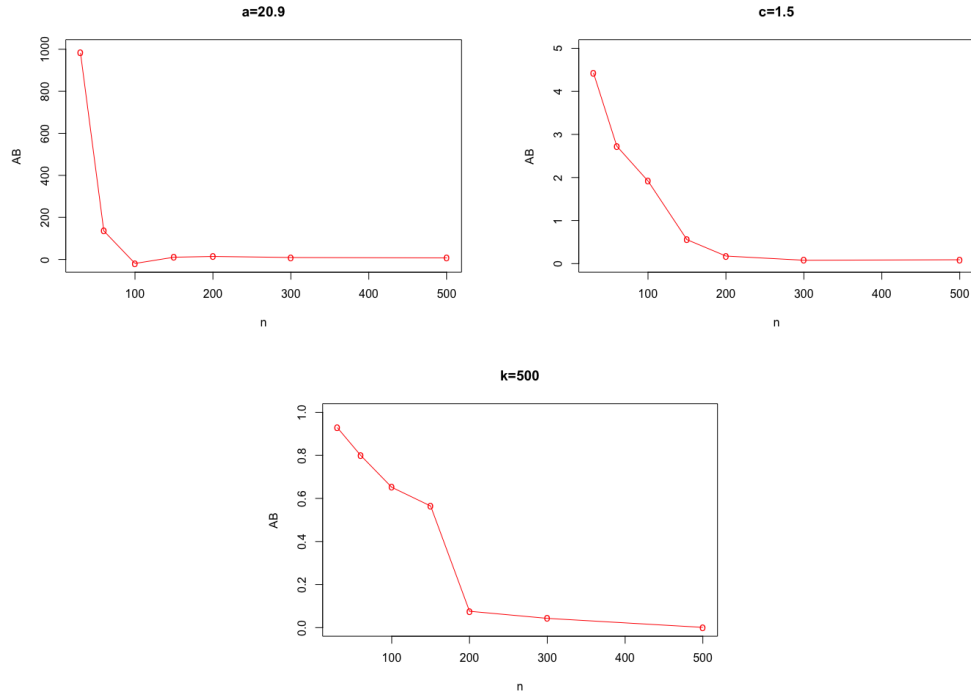


Figure 5.10: AB for Estimators of the ER-BurrXII distribution

The RMSEs of the maximum likelihood estimators of parameter values $(a, c, k) = (20.9, 1.5, 500)$ are displayed in Figure 5.11. Although the RMSEs appear volatile, the RMSE for each parameter decrease to zero as n becomes larger. The RMSEs is largest for parameters a and k except for parameter c which displays lesser RMSEs.

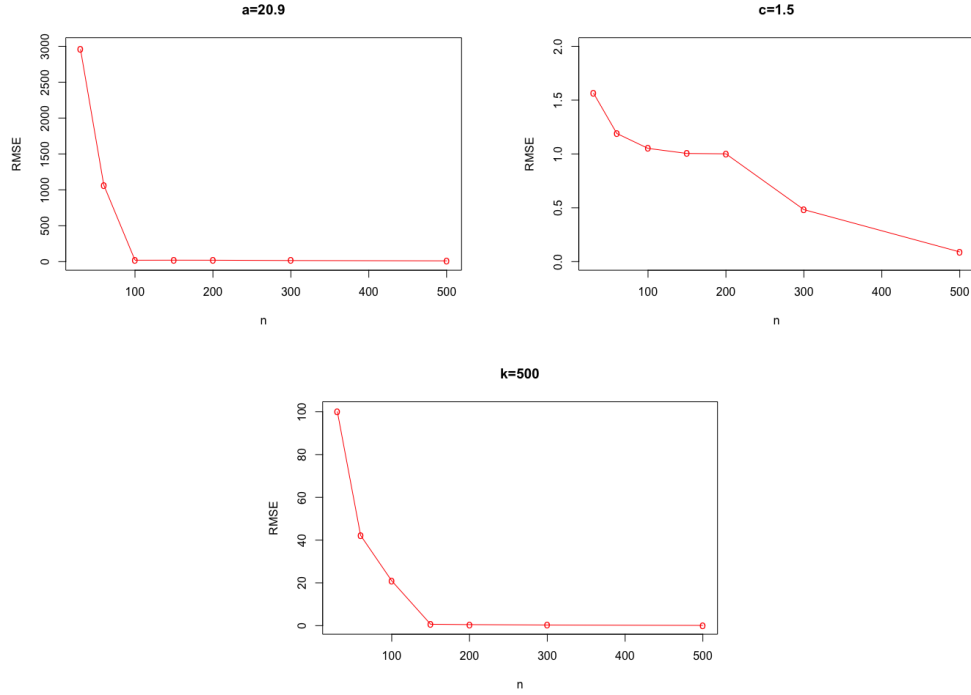


Figure 5.11: RMSE for Estimators of the ER-BurrXII distribution

Figure 5.12 displays the ABs of the maximum likelihood estimators of parameter values $(a, c, k) = (40, 10, 100)$. The ABs for parameter a and k are generally positive, except for parameter c which exhibits initial ABs values of negative. We observe that, the ABs of parameter a decreases faster compared to parameter c and k . Although parameter c appears intermittent in nature, the AB decrease to zero as n increases. The ABs appear largest for the parameter a .

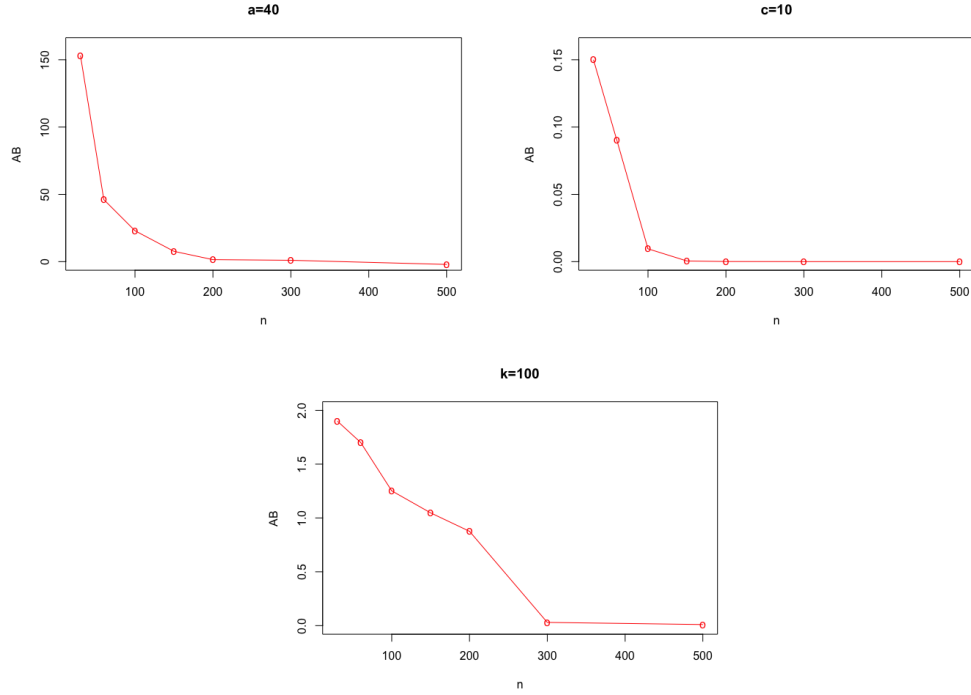


Figure 5.12: AB for Estimators of the ER-BurrXII distribution

The RMSEs of the maximum likelihood estimators of parameter values $(a, c, k) = (40, 10, 100)$ are displayed in Figure 5.13. Although, the RMSE for parameters c and k appear sporadic in nature, we observe that the RMSE for all parameters decreases as n becomes larger. We also observe that the RMSE for parameter a decreases.

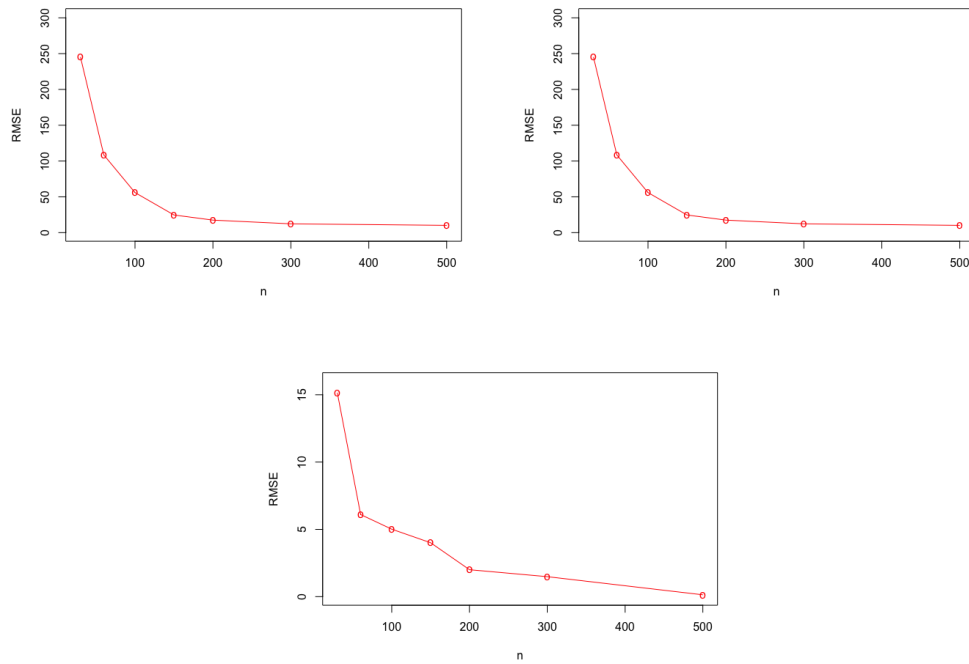


Figure 5.13: RMSE for Estimators of the ER-BurrXII distribution

5.5 Applications of the Equilibrium Renewal Burr XII Distribution

The illustration of the applicability of the ER-BurrXII distribution in three areas in the health sector namely remission and survival times of patients data and the length of stay of patients in the maternity wards from the Northern region of Ghana are discussed.

The parameters are estimated using maximum likelihood method of estimation by employing numerical optimization routine and their standard errors are computed. The known model selection criteria is the L and AIC, BIC, CAIC and HQIC are used for selection of the best model. For a visual comparison, we fit the distributions in the form of histogram with the fitted densities and fitted CDFs.

5.5.1 Remission and survival times data

To illustrate the importance and potentiality of the proposed ER-BurrXII distribution, the remission and survival times of patients data is applied and compared with some other existing distributions namely, the Generalised Pareto (Van Montfort and Witter, 1986), Cauchy (Carrillo *et al.*, 2008), Exponential Pareto (Al-Kadim and Boshi, 2013), Weibull Fr échet (Afify *et al.*, 2016), Exponentiated Inverse Rayleigh (Rao and Mbwambo, 2019), Weibull (2P) (Hallinan Jr, 1993), Generalized Rayleigh (Dey *et al.*, 2014), Burr XII (2P) (Zimmer *et al.*, 1998) and Topp-Leon Burr XII (Reyad and Othman, 2017).

Remission times of bladder cancer data

Table 5.14 displays some descriptive statistics of the bladder cancer patients data. The average survival time is approximately 10 months, the minimum and maximum survival times were approximately 1 month and 39 months respectively. We observe that the survival times are right skewed with a skewness measure of 1.4728. The distribution of the dataset has coefficient of variation of 0.7327 which indicates that the distribution has a high level of dispersion around the mean.

Table 5.14: Descriptive statistics

Minimum	Maximum	Median	Mean	Skewness	Kurtosis	Coefficient of variation
0.8000	38.500	8.100	9.877	1.4728	5.5403	0.7327

Table 5.15 lists the MLEs, standard error, Z-Statistics and p-value of the parameters. It is observed that all parameter values are statistically significant at 5% significance level, except for parameter c and k of the ER-BurrXII distribution which is statistically significant at 10% significant level.

Table 5.15: MLEs of parameters, standard errors, p-values for remission times of the gall bladder cancer patients data.

Distribution	Parameters	Estimates	Standard Error	Z-Statistics	P-Value
ER-BurrXII	$\hat{a}$	10.3475	3.0825	3.3569	0.0008*
	$\hat{c}$	2.9035	1.4876	1.9517	0.0510
	$\hat{k}$	1.0779	0.7575	1.4230	0.1547
Generalized Pareto	$\hat{c}$	8.6840	1.0291	8.4386	2×10^{-16} *
	$\hat{k}$	0.0716	0.0792	0.9048	0.0046*
Cauchy	$\hat{c}$	3.1964	0.3305	8.4005	$< 2.2 \times 10^{-16}$ *
	$\hat{k}$	5.4423	0.4518	12.0459	$< 2.2 \times 10^{-16}$ *
Exponential Pareto	$\hat{c}$	4.6574	0.6855	6.7947	1.088×10^{-11} *
	$\hat{k}$	1.0877	0.0858	12.6760	$< 2.2 \times 10^{-16}$ *
Weibull Fréchet	$\hat{a}$	11.3549	2.5524	4.4487	8.638×10^{-6} *
	$\hat{c}$	0.6521	0.1149	5.6748	1.388×10^{-8} *
	$\hat{k}$	0.3928	0.0277	14.1635	$< 2.2 \times 10^{-16}$ *
	$\hat{d}$	231.1295	0.0677	344.7469	$< 2.2 \times 10^{-16}$ *
Exponential Inverse Rayleigh	$\hat{c}$	1.6071	0.1713	9.3819	$< 2.2 \times 10^{-16}$ *
	$\hat{k}$	2.8988	0.1957	14.8124	$< 2.2 \times 10^{-16}$ *
Weibull (2-P)	$\hat{c}$	0.6740	0.0596	11.3137	$< 2.2 \times 10^{-16}$ *
	$\hat{k}$	9.3656	1.0083	9.2885	$< 2.2 \times 10^{-16}$ *
General Rayleigh	$\hat{c}$	2.4761	0.2649	9.3471	$< 2.2 \times 10^{-16}$ *
	$\hat{k}$	0.1378	0.0085	16.1548	$< 2.2 \times 10^{-16}$ *
Burr XII (2-P)	$\hat{c}$	2.3348	0.3541	6.5943	4.772×10^{-11} *
	$\hat{k}$	0.2337	0.3910	5.8488	4.952×10^{-16} *
TP Leon Burr XII	$\hat{a}$	9.2514	4.7342	1.9542	0.0007*
	$\hat{c}$	0.9507	0.3148	3.0205	0.0025*
	$\hat{k}$	0.6537	0.1662	3.9307	8.39×10^{-5} *

*: Means significant at the 5% significance level.

Table 5.16 presents the log-likelihood and information criteria for the bladder cancer patients data. We observe that out of the nine models, the ER-BurrXII has the highest value of the L and lowest value of the AIC, CAIC, BIC and HQIC. As a result, we conclude that the ER-BurrXII distribution is extremely adaptable in terms of modeling the data.

Table 5.16: Comparison criterion of remission times of the gall bladder cancer patients data.

Distribution	L	AIC	CAIC	BIC	HQIC
ER-BurrXII	-410.7559	827.5118	827.7618	836.0679	16.4839
Generalized Pareto	-413.8329	831.6658	831.9158	837.3698	16.4988
Cauchy	-442.3938	888.7876	889.0376	894.4917	16.6318
Exponential Pareto	-426.3051	856.6102	889.0376	894.4917	16.6318
Weibull Fréchet	-422.3092	852.6185	852.8685	864.0266	16.5392
Exponential Inverse Rayleigh	-598.4301	1200.8600	1201.1100	1206.5640	17.2343
Weibull (2-P)	-402.9332	809.8673	810.1173	813.5713	16.4544
General Rayleigh	-484.8450	973.6901	973.9401	979.3941	16.8145
Burr XII (2-P)	-453.5166	911.0332	911.2832	916.7372	16.6813
TP Leon Burr XII	-424.7617	855.5293	855.7793	864.0845	16.5508

In Figure 5.14, we show the estimated PDFs and estimated CDFs of all fitted distributions using the estimators obtained in Table 5.15. From Figure 5.14, it is noticed that the ER-BurrXII distribution fits the remission times of the gall bladder cancer patients data better than the other nine models.

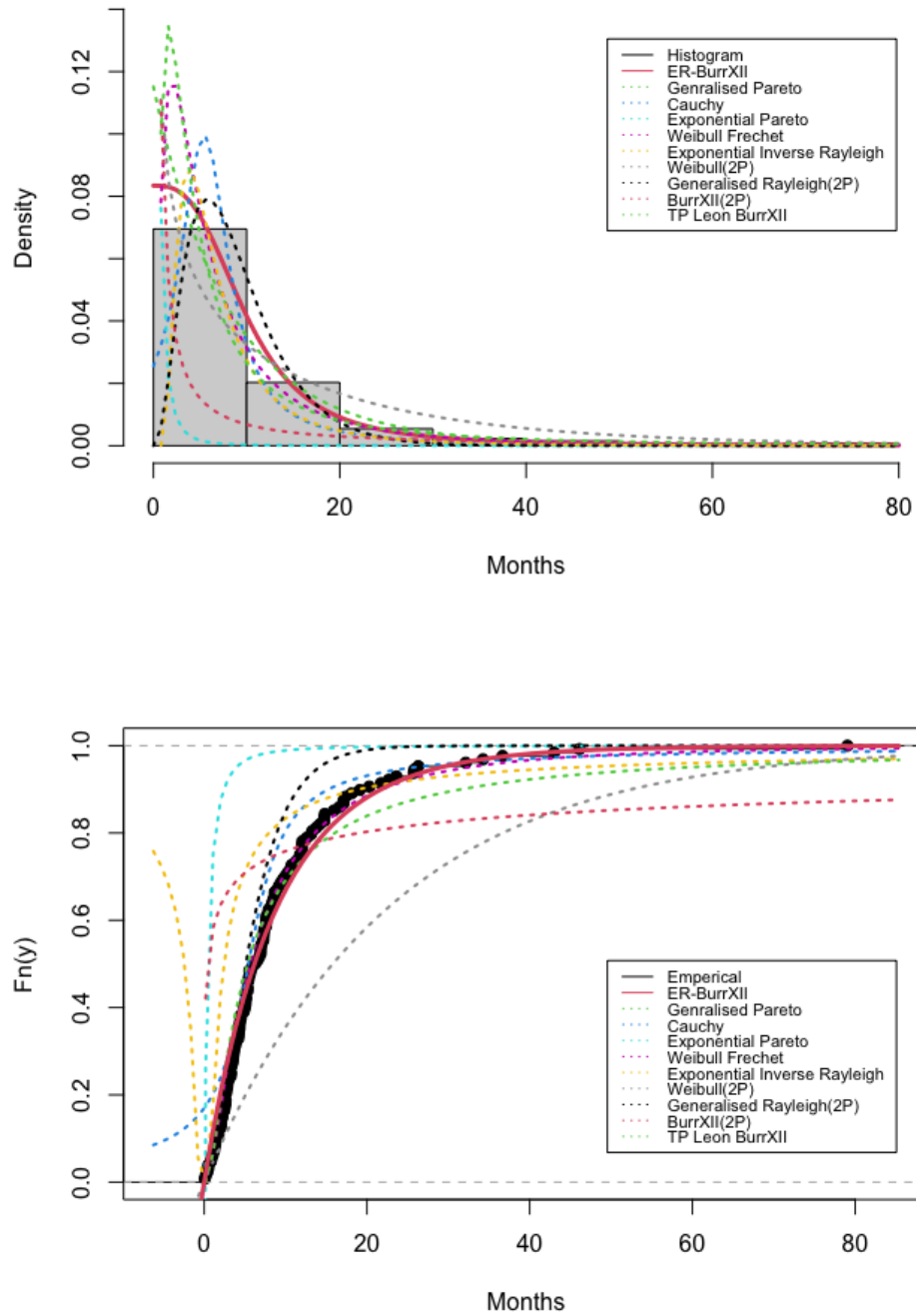


Figure 5.14: The estimated PDFs (top panel) and the estimated CDFs (bottom panel) remission times of the gall bladder cancer patients data.

Acute myelogenous leukaemia data

Table 5.17 displays some descriptive statistics of survival times of patients suffering from acute myelogenous leukaemia. The mean survival time is 40.8788 weeks, the minimum and maximum survival times were 1.0000 and 156.0000 weeks respectively. We observe that the survival times are right skewed with coefficient of skewness of 1.1646. The distribution of the dataset has coefficient of variation of 1.1425 which indicates that the distribution has a high level of dispersion around the mean.

Table 5.17: Descriptive statistics of patients suffering from acute myelogenous leukaemia.

Minimum	Maximum	Median	Mean	Skewness	Kurtosis	Coefficient of variation
1.0000	156.0000	22.0000	40.8788	1.1646	3.1221	1.1425

Table 5.18 presents the parameter estimates and the respective standard errors, Z-statistic and p-values. The parameter estimates for the Cauchy, Exponentiated Pareto, Weibull (2P) and General Rayleigh models are all statistical significant at 10% significance level. In addition, the Burr XII (2P), TP Leon Burr XII and General Pareto are all statistical significance at 5% significance level. However, parameters c and k of the ER-BurrXII are statistically significant at 10% significance level and a is statistically significant at 5% significance level. In addition, parameter c , c and k are statistically significant at 5% except for parameter d which is statistically significant at 5% significance level for the Weibul Frechét distribution.

Table 5.18: MLEs of parameters, standard errors, p-values for acute myelogenous leukaemia data.

Distribution	Parameter	Estimates	Standard Errors	Z-Statistics	P-Value
ER-BurrXII	$\hat{a}$	77.8058	84.2080	0.9240	0.3555
	$\hat{c}$	0.7393	0.2374	3.1136	0.0019*
	$\hat{k}$	4.6952	2.7809	1.6884	0.0913
Generalized Pareto	$\hat{c}$	26.1958	10.6080	2.4694	0.0135
	$\hat{k}$	0.4195	0.3733	1.1237	0.2612
Cauchy	$\hat{c}$	16.5966	4.5489	3.6485	0.0003*
	$\hat{k}$	13.6167	4.7800	2.8487	0.0044*
Exponentiated Pareto	$\hat{c}$	4.3564	1.3003	3.3502	0.0008*
		0.7199	0.1185	6.0765	1.228×10^{-9} *
Weibull Fréchet	$\hat{a}$	2.8591	0.9067	3.1533	0.0016*
	$\hat{c}$	0.4244	0.1451	2.9247	0.0035*
	$\hat{k}$	0.5065	0.0976	5.1910	2.091×10^{-7} *
	$\hat{d}$	148.9776	76.0732	1.9583	0.0502
Exponential Inverse Rayleigh	$\hat{c}$	1.0251	0.1961	5.2278	1.715×10^{07} *
	$\hat{k}$	6.0503	0.8145	7.4282	1.101×10^{-13} *
Weibull (2P)	$\hat{c}$	0.5317	0.0926	5.7446	9.216×10^{-9} *
	$\hat{k}$	40.8011	9.7224	4.1966	2.709×10^{-5} *
General Rayleigh	$\hat{c}$	1.1669	0.2233	5.2257	1.7350×10^{-7} *
	$\hat{k}$	0.0255	0.0034	7.6065	2.816×10^{-14} *
Burr XII (2P)	$\hat{c}$	12.9973	32.1562	0.4042	0.6861
	$\hat{k}$	0.0271	0.0671	0.4041	0.6861
TP Leon Burr XII	$\hat{a}$	37.8724	99.4409	0.3809	0.7033
	$\hat{c}$	1.6909	1.7733	0.9535	0.3403
	$\hat{k}$	0.3066	0.2552	1.2014	0.2296

*: Means significant at the 5% significance level.

Table 5.22 brings the proposed model as well as the following statistics: AIC, BIC, CAIC, HQIC and log-likelihood to check how the distribution fits the data. The smaller the values of the AIC, BIC, CAIC and HQIC together with the highest value of log likelihood the better the model fit. We can observe, in Table 5.22, that the results involving the Exponentiated Pareto, Cauchy, Generalised Pareto, Weibull Fréchet, Exponentiated Inverse Rayleigh, Weibull (2P), General Rayleigh, Burr XII (2P) and TP Leon Burr XII produces high values for the AIC, CAIC, BIC and HQIC for the ER-BurrXII model. Also, the ER-BurrXII model has the highest log likelihood value compared to the other models. Thus, we can say that the ER-BurrXII distribution can be applied to explain

the survival times of patients.

Table 5.19: Comparison criterion for acute myelogenous leukaemia data.

Distribution	L	AIC	CAIC	BIC	HQIC
ER-BurrXII	-154.2055	314.4109	314.6609	318.9004	14.5366
Generalized Pareto	-226.3644	456.7288	456.9788	459.7218	15.2982
Cauchy	-172.8878	349.7756	368.1964	352.7686	14.76322
Exponentiated Pareto	-156.0465	316.0929	368.1964	352.7686	14.7632
Weibull Fréchet	-159.5132	327.0263	323.2911	333.0123	14.6036
Exponential Inverse Rayleigh	-204.927	413.8541	414.1041	416.8471	15.1006
Weibull (2P)	-156.2085	315.5101	315.8011	318.9921	14.6549
General Rayleigh	-194.3337	392.6674	392.9174	395.6604	14.9952
Burr XII (2P)	-162.1310	328.2621	328.5121	331.2551	14.6359
TP Leon Burr XII	-153.4793	316.2584	328.5121	320.7479	14.5272

We can observe from Figure 5.15 the ER-BurrXII is an excellent fit to the data distribution in relation to the adequacy of the data.

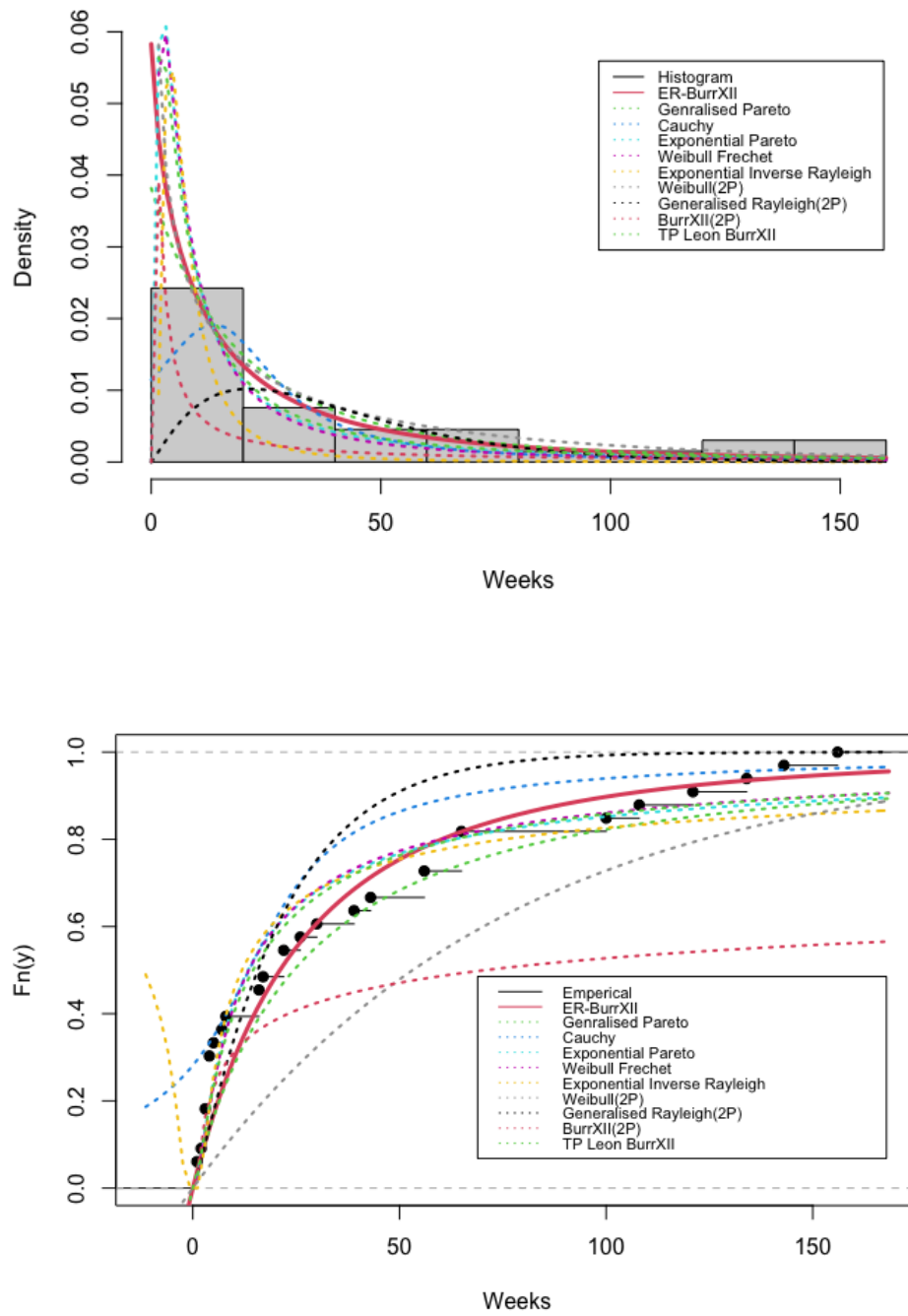


Figure 5.15: The estimated PDFs (top panel) and the estimated CDFs (bottom panel) for for acute myelogenous leukaemia data.

5.5.2 Length of stay data

In this section, we compare eight distributions namely the Generalised Pareto (Van Montfort and Witter, 1986), Exponential (Nadarajah and Kotz, 2006a), Exponentiated Pareto (Al-Kadim and Boshi, 2013), Weibull (2P) (Hallinan Jr, 1993), Nakagami (Shittu and Adepoju, 2013), (Zimmer *et al.*, 1998) and Topp-Leon Burr XII (Reyad and Othman, 2017). In addition we compare the newly proposed ER-BurrXII to two distributions proposed by Voda (2008) through the concept of equilibrium renewal process. We may call these distributions defined by (Voda, 2008) as the Equilibrium renewal Weibull (ER-Weibull) and Equilibrium renewal Generalised Exponential (ER-Generalised Exponential) distributions to the length of stay data.

The statistical characteristics of each district are given in Table 5.20. The minimum average stay of patients in the maternity ward is 2.4 hours in districts Saboba and Gushiegu. The maximum length of stay is 259.2000 hours in district Nanumba.

From table 5.20 it is observed that Saboba district is characterized by a high length of stay with a mean of 61.9100 hours compared with the other locations. It can also be observed that Gusheigu district has the lowest length of stay with a mean estimate of 39.0738 hours. From Table 5.20, it is noticed that all districts are highly skewed to the right since all skewness estimates are greater than 1 except for Gusheigu district which exhibits a moderately skewed data with an estimate of 0.6520. In relation to the kurtosis of the data, it can be observed that from Table 5.20 all districts exhibits a leptokurtic feature since all kurtosis estimates are greater than 0. On the basis of the coefficient of variation for all districts, it is observe that Saboba and Nanumba have a relatively high variation. Tamale, Kpandai and Gusheigu have moderately low variation.

Table 5.20: Descriptive statistics

Districts	Minimum	Maximum	Median	Mean	Skewness	Kurtosis	Coefficient of variation
Saboba	2.4000	235.2000	68.4000	61.9100	1.1937	3.5557	0.6271
Tamale	16.5600	136.8000	50.4000	53.6100	1.2318	2.7732	0.4350
Kpandai	28.8000	184.8000	52.8000	54.5538	4.9257	33.1594	0.3377
Gusheigu	2.4000	96.0000	36.0000	39.0738	0.6520	0.4136	0.4710
Nanumba	23.7600	259.2000	38.4000	50.7662	3.8948	16.7745	0.7873

Saboba district

The first data represents the length of stay of patients in the maternity ward of various public hospitals in the Saboba district the Northern region of Ghana.

Table 5.21 displays the MLEs of the parameters, standard errors and p-values for the fitted distributions. From Table 5.21, it is observed that all parameters of the ER-BurrXII, TP Leon Burr XII, Nakagami, Weibull, Exponentiated Pareto, Generalised Pareto and Exponential distributions were significant at 5% significance level. Moreover, parameter a for TP Leon BurrXII is significant at 10% whiles parameters c and k is significant at 5% significance level. In addition, for ER-Generalised Exponential we observe that both parameters a , and k are significant at 10% significance level and parameter k is statistical significant at 10% significance level.

Table 5.21: MLEs of parameters, standard errors, p-values for the length of stay of patients in Saboba

Distribution	Parameter	Estimates	Standard Errors	Z-Statistics	P-Value
ER-BurrXII	$\hat{a}$	95.9209	3.7900	25.3089	$< 2.2 \times 10^{-16}^*$
	$\hat{c}$	87.4523	0.0064	13656.3937	$< 2.2 \times 10^{-16}^*$
	$\hat{k}$	0.0729	0.0148	4.9427	$7.706 \times 10^{-07}^*$
TP Leon BurrXII	$\hat{a}$	146.4465	94.1524	1.5554	0.1198
	$\hat{c}$	1.3558	0.4031	3.3633	0.0001*
	$\hat{k}$	0.4730	0.1125	4.2030	$2.634 \times 10^{-5}^*$
ER-Weibull	$\hat{c}$	0.0088	0.0009	9.9875	$< 2.2 \times 10^{-16}$
	$\hat{k}$	2.5442	0.4929	5.1611	$2.455 \times 10^{-07}^*$
Nakagami	$\hat{c}$	976.9090	108.1827	9.0302	$< 2.2 \times 10^{-16}^*$
	$\hat{k}$	0.2109	0.0367	5.7461	9.131×10^{-9}
ER-Generalised Exponential	$\hat{a}$	11.430	96.708	0.1182	0.9059
	$\hat{c}$	218.3500	56.7889	2.8999	0.0051
	$\hat{k}$	21.6640	74.7270	0.2899	0.7719
Weibull(2P)	$\hat{c}$	0.0162	0.0020	7.9685	$1.606 \times 10^{-15}^*$
	$\hat{k}$	0.8141	0.0922	8.8318	$< 2.2 \times 10^{-16}^*$
Exponentiated Pareto	$\hat{c}$	47.1662	13.2234	3.5668	0.0004*
	$\hat{k}$	1.0970	0.0833	13.1736	$< 2.2 \times 10^{-16}^*$
Generalised Pareto	$\hat{c}$	76.9058	0.0012	1.0031	$1.2654 \times 10^{-9}^*$
	$\hat{k}$	0.2835	1.0043	2.1034	0.0003*
Exponential	$\hat{c}$	0.0162	0.0018	8.8318	$< 2.2 \times 10^{-16}^*$

*: Means significant at the 5% significance level.

Table 5.16 presents the log-likelihood and information criteria for the length of stay data. It can be observed that out of the eight models, the ER-BurrXII has the highest value of the L and lowest value of the AIC, CAIC, BIC and HQIC. Thus, we conclude that the ER-BurrXII distribution is quite flexible in the modeling of the proposed data.

Table 5.22: Comparison criterion for length of stay for Saboba district.

Distribution	L	AIC	CAIC	BIC	HQIC
ER-BurrXII	-383.7616	773.5231	773.7731	780.5932	16.2376
TP Leon BurrXII	-413.8329	820.4074	820.6574	827.4775	16.3557
ER-Weibull	-390.4550	784.9100	785.1600	789.6234	16.2720
Nakagami	-464.6398	933.2796	933.5296	937.9930	16.6187
ER-Generalised Exponential	-420.1653	846.3305	846.5805	853.4007	16.4321
Weibull (2P)	-598.4301	800.0610	800.3110	804.7744	16.3078
Exponentiated Pareto	-410.6535	825.3070	825.557	830.0205	16.3725
Generalised Pareto	-394.6901	793.3803	793.6303	798.0937	16.80411
Exponential	-399.8003	801.6006	801.8506	830.0205	16.4301

In Figure 5.16, we show the estimated PDFs and estimated CDFs of all fitted distributions using the estimators obtained in Table 5.21. The ER-BurrXII distribution matches the pattern of the length of stay data better than the other eight models, as shown in Figure 5.16, which supports the results in Tables 5.22.

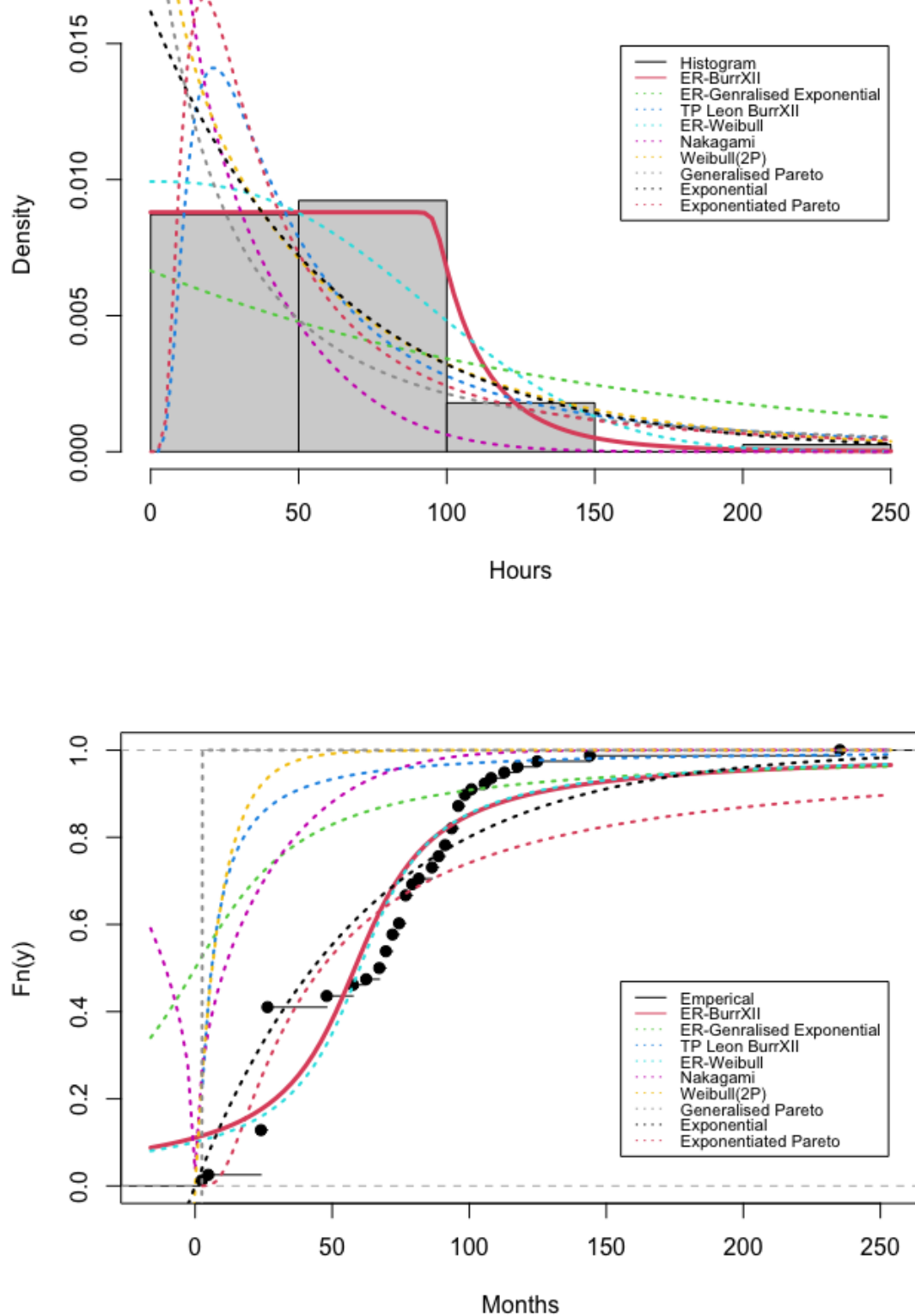


Figure 5.16: The estimated PDFs (top panel) and the estimated CDFs (bottom panel) length of stay of patients data in Saboba district.

Kpandai district

Table 5.25 presents the parameter estimates and the respective standard errors, Z-statistic and p-values. The parameter for the ER-BurrXII, TP Leon BurrXII, ER-Weibull, Nakagami, Weibull, Exponentiated Pareto, General Pareto and Exponential models are all statistical significant at 5% significance level. In addition, the parameters of the ER-Generalised Exponential are all statistical significance at 10% significance level.

Table 5.23: MLEs of parameters, standard errors, p-values for length of stay of patients in Kpandai

Distribution	Parameter	Estimates	Standard Errors	Z-Statistics	P-Value
ER-BurrXII	$\hat{a}$	61.2516	2.2990	26.6422	$<2.2 \times 10^{-16}^*$
	$\hat{c}$	94.3030	0.0136	6949.7955	$<2.2 \times 10^{-16}^*$
	$\hat{k}$	0.0659	0.0129	5.1032	$3.339 \times 10^{-07}^*$
TP Leon BurrXII	$\hat{a}$	2.3649×10^2	3.7497×10^{-04}	$6.3068 \times 10^{+5}$	$<2.2 \times 10^{-16}^*$
	$\hat{c}$	2.4447×10^{-1}	2.9696×10^{-2}	8.2325	$<2.2 \times 10^{-16}^*$
	$\hat{k}$	2.8917	3.5125×10^{-1}	8.2326	$<2.2 \times 10^{-16}^*$
ER-Weibull	$\hat{c}$	0.0110	0.0008	13.3567	$<2.2 \times 10^{-16}^*$
	$\hat{k}$	3.0207	0.3889	7.7664	$8.077 \times 10^{-15}^*$
Nakagami	$\hat{c}$	1448.3552	148.9491	9.7238	$<2.2 \times 10^{-16}^*$
	$\hat{k}$	0.8721	0.1947	4.4792	$7.492 \times 10^{-6}^*$
ER-Generalised Exponential	$\hat{a}$	11.4540	88.7960	0.1290	0.8974
	$\hat{c}$	180.0340	150.0142	3.1023	0.6121
	$\hat{k}$	-21.860	71.273	-0.3067	0.7591
Weibull(2P)	$\hat{c}$	0.0183	0.0021	8.6852	$<2.2 \times 10^{-16}^*$
	$\hat{k}$	0.9671	0.1095	8.8318	$<2.2 \times 10^{-16}^*$
Exponentiated Pareto	$\hat{c}$	177.9804	47.6794	3.7329	0.0002*
	d	1.3520	0.0723	18.7107	$<2.2 \times 10^{-16}^*$
Generalised Pareto	$\hat{c}$	67.5624	0.9010	14.5523	$2.3 \times 10^{-12}^*$
	$\hat{k}$	-0.3361	0.0012	12.0045	$1.2 \times 10^{-9}^*$
Exponential	$\hat{c}$	0.0184	0.0021	8.8318	$<2.2 \times 10^{-16}^*$

*: Means significant at the 5% significance level.

The L values and values of AIC, BIC, CAIC, HQIC are presented in Table 5.24. By comparing the information criteria and L values among the eight distributions in Table 5.24, it is observed that all distributions provides level fit to the given data. However, the ER-BurrXII provides an adequate fit since the ER-BurrXII has the lowest values of the AIC, BIC, CAIC, HQIC and highest value for the L. This application suggests that

ER-BurrXII fits the right skewed data well.

Table 5.24: Comparison criterion for length of stay for Kpandai district.

Distribution	L	AIC	CAIC	BIC	HQIC
ER-BurrXII	-349.6264	705.2529	705.5029	712.323	16.05202
TP Leon BurrXII	-364.384	734.7681	735.0181	741.8382	16.13436
ER-Weibull	-368.7787	741.1213	741.3713	745.8347	16.15706
Nakagami	-382.5919	769.1839	769.4339	773.8973	16.23149
ER-Generalised Exponential	-405.1211	816.2422	816.4922	823.3123	16.3455
Weibull (2P)	-389.8911	783.7821	784.0321	788.4956	16.26665
Exponentiated Pareto	-389.8911	783.7821	779.3986	747.7249	17.99057
Generalised Pareto	-380.4477	764.8954	784.0321	769.6088	16.21768
Exponential	-389.9367	781.8734	782.1234	747.7249	16.3803

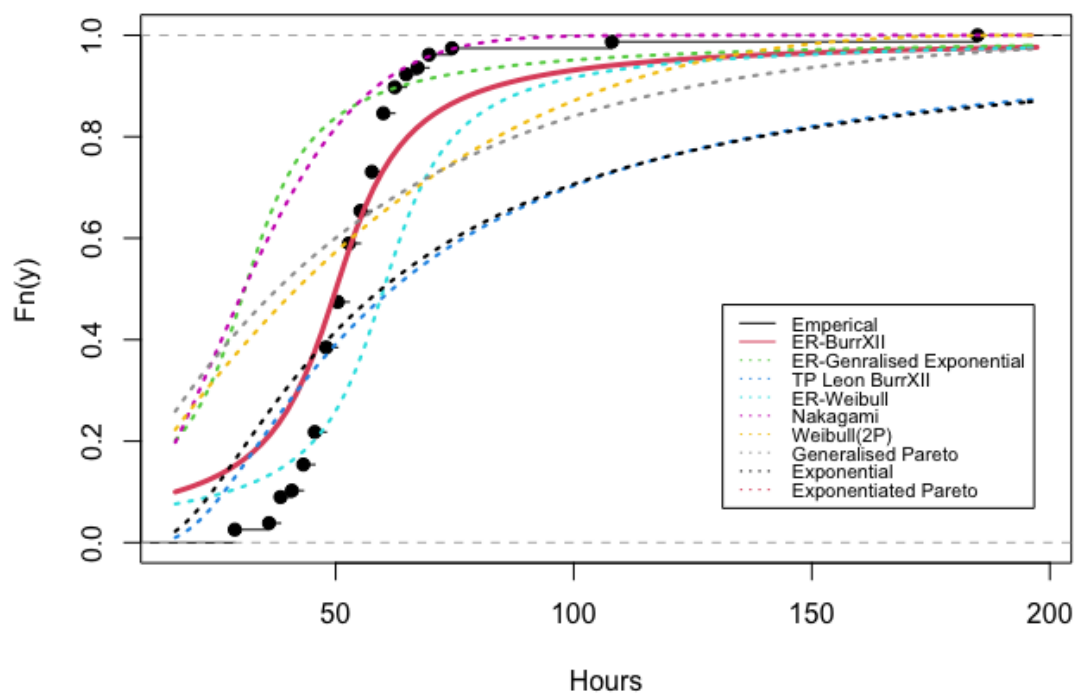
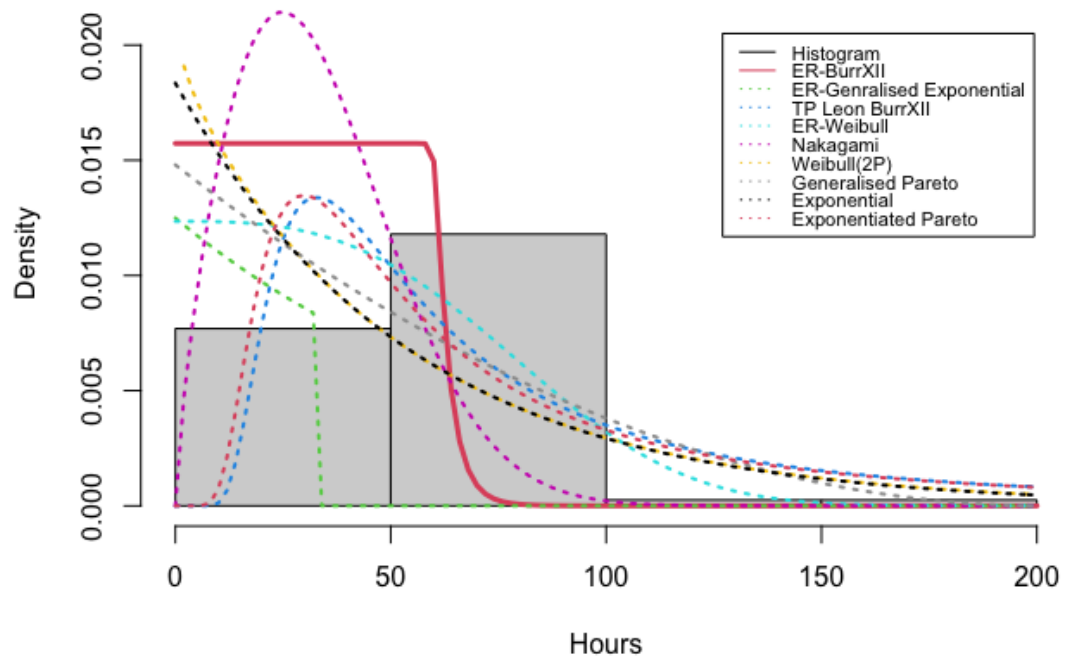


Figure 5.17: The estimated PDFs (top panel) and the estimated CDFs (bottom panel) length of stay of patients data in Kpandai district

Gusheigu district

The estimates of parameters for the ER-BurrXII distribution together with competing distributions are displayed in Table 5.25. We observe that the estimates for the ER-BurrXII, TP-Leon BurrXII, ER-Weibull, Nakagami, Weibull, Exponentiated Pareto, General Pareto and Exponential are statistically significant at 5% significance level. However, the estimates of the ER-Generalised Exponential are statistically significant at 10% significance level.

Table 5.25: MLEs of parameters, standard errors, p-values for length of stay of patients in Gushiegu

Distribution	Parameter	Estimates	Standard Errors	Z-Statistics	P-Value
ER-BurrXII		6.0389×10^1	4.1111	14.6894	$<2 \times 10^{-16*}$
	$\hat{c}$	1.3414×10^2	8.5052×10^{-3}	15771.8063	$<2 \times 10^{-16*}$
	$\hat{k}$	5.5090×10^{-2}	1.5472×10^{-2}	3.5606	0.0004*
TP Leon BurrXII	$\hat{a}$	122.2566	53.0226	2.3057	0.0001*
	$\hat{c}$	1.0501	0.2767	3.7947	0.0002*
	$\hat{k}$	0.67436	0.1508	4.4720	$7.7510 \times 10^{-6*}$
ER-Weibull	$\hat{c}$	0.0138	0.0009	15.2053	$<2.2 \times 10^{-16}$
	$\hat{k}$	4.5676	1.1050	4.1335	$3.573 \times 10^{-5*}$
Nakagami	$\hat{c}$	672.0246	74.7134	8.9947	$<2.2 \times 10^{-16*}$
	$\hat{k}$	0.2447	0.0448	5.4675	$4.564 \times 10^{-8*}$
ER-Generalised Exponential	$\hat{a}$	11.5420	62.8160	0.1837	0.8542
	$\hat{c}$	85.1060	110.8190	8.1432	0.1098
	$\hat{k}$	-21.809	57.165	-0.3815	0.7028
Weibull(2P)	$\hat{c}$	0.0256	0.0031	8.3106	$<2.2 \times 10^{-16*}$
	$\hat{k}$	0.8855	0.1003	8.8318	$<2.2 \times 10^{-16*}$
Exponentiated Pareto	$\hat{c}$	79.8477	23.0368	3.4661	0.0049*
	$\hat{k}$	1.3427	0.0931	14.4209	$<2.2 \times 10^{-16*}$
Generalised Pareto	$\hat{c}$	58.1187	5.2137	4.3217	$< 2.1 \times 10^{-17*}$
	$\hat{k}$	-0.59208	47.0421	1.1121	0.0002*
Exponential	$\hat{c}$	0.0256	0.0029	8.8318	$<2.2 \times 10^{-16*}$

*: Means significant at the 5% significance level.

From Table 5.26, the distribution that corresponds to the highest log-likelihood and lowest AIC, CAIC, BIC, and HQIC is judged to be the best. With this regards we conclude that the ER-BurrXII fits better to the given compared to the other competing distributions.

Table 5.26: Comparison criterion for length of stay for Gushiegu district.

Distribution	L	AIC	CAIC	BIC	HQIC
ER-BurrXII	-343.4126	692.8252	693.0752	699.8953	16.01631
TP Leon BurrXII	-361.7973	729.5946	729.8446	736.6647	16.12017
ER-Weibull	-368.7787	741.1213	741.3713	745.8347	16.15706
Nakagami	-382.5919	769.1839	769.4339	773.8973	16.23149
ER-Generalised Exponential	-346.7135	699.4269	699.6769	706.4971	16.03536
Weibull (2P)	-363.3051	730.6102	730.8602	735.3236	16.12571
Exponentiated Pareto	-363.7493	731.4986	731.7486	736.2120	16.13089
Generalised Pareto	-348.6781	701.3561	701.6061	706.0696	16.04375
Exponential	-363.9054	729.8107	730.0607	736.2120	16.2427

Figure 5.18 shows a plot of all competing distributions versus the histogram and ECDF of the observed data. Figure 5.18 shows the results of the analysis, which suggest that the ER-BurrXII distribution is better suited to the data than the other competing distributions.

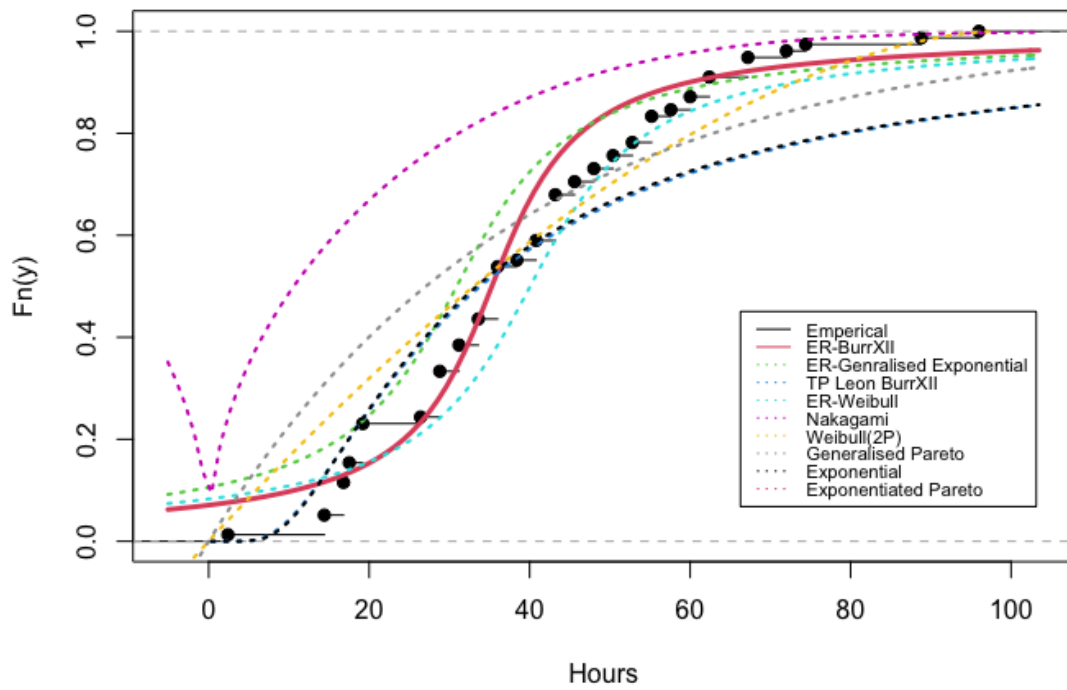
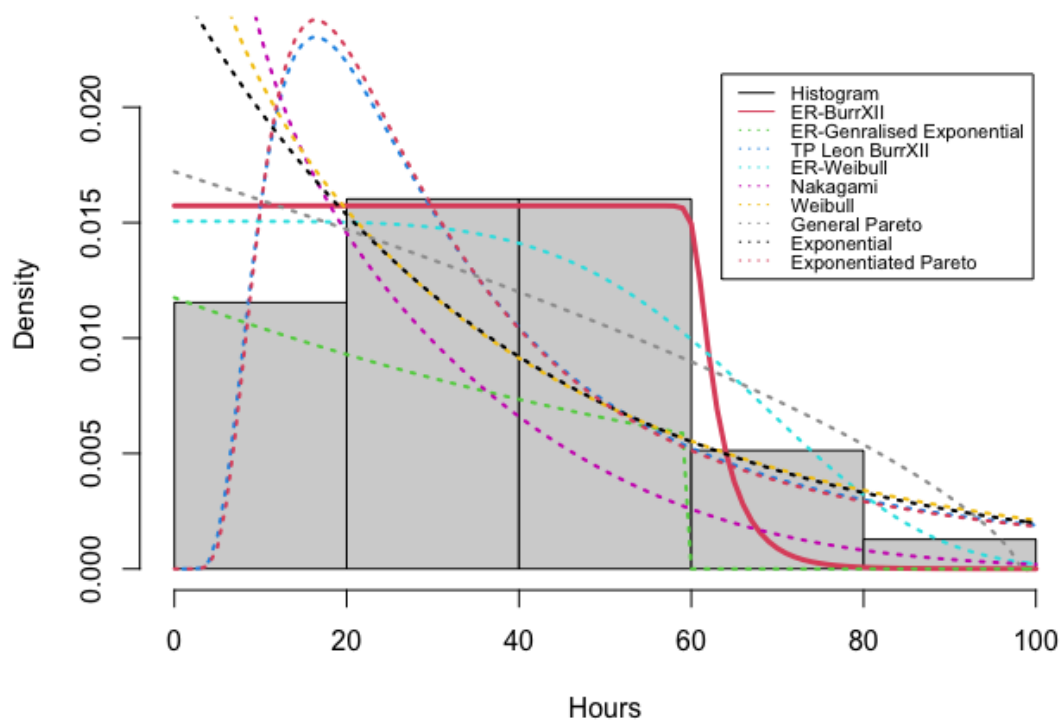


Figure 5.18: The estimated PDFs (top panel) and the estimated CDFs (bottom panel) length of stay of patients data in Gushiegu district

Nanumba district

Table 5.27 displays the MLEs of the parameters, standard errors and p-values for the fitted distributions. From Table 5.27, it is observed that all distribution parameters significant at 5% significance level except for the ER-Generalised Exponential which is significant at 10% significance level.

Table 5.27: MLEs of parameters, standard errors, p-values for length of stay of patients in Nanumba

Distribution	Parameter	Estimates	Standard Errors	Z-Statistics	P-Value
ER-BurrXII	$\hat{a}$	$4.2924 \times 10^{+1}$	4.6335	9.2639	$<2.2 \times 10^{-16}^*$
	$\hat{c}$	1.0100×10^2	2.5134×10^{-3}	40186.7606	$<2.2 \times 10^{-16}^*$
	$\hat{k}$	2.9298×10^{-2}	3.8314×10^{-3}	7.6467	$2.062 \times 10^{-14}^*$
TP Leon BurrXII	$\hat{a}$	357.2051	1.5744×10^{-3}	$1.5962 \times 10^{+5}$	$<2 \times 10^{-16}^*$
	$\hat{c}$	0.2836	4.3666×10^{-01}	7.1320×10^{-01}	0.4757
	$\hat{k}$	2.8866	3.4729	7.1320×10^{-1}	0.4757
ER-Weibull	$\hat{c}$	0.0131	0.0018	7.3692	$1.7170 \times 10^{-13}^*$
	$\hat{k}$	1.5645	0.2136	7.3251	$2.3880 \times 10^{-13}^*$
Nakagami	$\hat{c}$	$1.0160 \times 10^{+3}$	1.1033×10^2	9.2090	$<2.2 \times 10^{-16}^*$
	$\hat{k}$	2.8492×10^{-01}	4.9203×10^{-2}	5.7907	$7.0070 \times 10^{-9}^*$
ER-Generalised Exponential	$\hat{a}$	11.4680	83.9190	0.1367	0.8913
	$\hat{c}$	159.318	221.0123	0.1781	0.2109
	$\hat{k}$	-21.8750	68.7200	-0.3183	0.7502
Weibull(2P)	$\hat{c}$	0.0197	0.0024	8.2560	$<2.2 \times 10^{-16}^*$
	$\hat{k}$	0.8739	0.0989	8.8318	$<2.2 \times 10^{-16}^*$
Exponentiated Pareto	$\hat{c}$	237.1512	69.4716	3.4136	0.0001*
	$\hat{k}$	1.5115	0.0840	17.9898	$<2.2000 \times 10^{-16}^*$
Generalised Pareto	$\hat{c}$	55.5742	7.2760	7.6380	$2.2060 \times 10^{-14}^*$
	$\hat{k}$	-0.1034	0.0671	-1.5402	0.1235
Exponential	$\hat{c}$	0.0197	0.0022	8.8318	$<2.2 \times 10^{-16}^*$

*: Means significant at the 5% significance level.

From Table 5.28, it has been observed that the ER-BurrXII distribution has lesser AIC, CAIC, BIC and HQIC values in comparison to the other distributions. Hence, it can conclude that the ER-BurrXII distribution leads to a better fit than the other competing distributions.

Table 5.28: Comparison criterion for length of stay for Nanumba district.

Distribution	L	AIC	CAIC	BIC	HQIC
ER-BurrXII	-365.2754	736.5508	736.8008	743.6209	16.13923
ER-Weibull	-379.8289	763.6577	763.9077	768.3712	16.21705
Nakagami	-435.2073	874.4145	874.6645	879.1279	16.48825
ER-Generalised Exponential	-395.591	797.1819	797.4319	804.2520	16.29806
Weibull (2P)	-383.5823	771.1645	771.4145	775.8779	16.23404
Exponentiated Pareto	-355.8278	715.6556	715.9056	720.369	16.08704
General Pareto	-383.4164	770.8328	771.0828	775.5462	16.31249
TP Leon BurrXII	-370.9870	746.1464	746.2541	756.9870	16.3425
Exponential	-384.3239	770.6479	770.8979	720.369	16.35142

In Figure 5.19, we display the estimated PDFs and estimated CDFs of all fitted distributions using the estimates obtained in Table 5.27. From Figure 5.19, it is noticed that the ER-BurrXII distribution fits the length of stay data for Nanumba better than the other eight models.

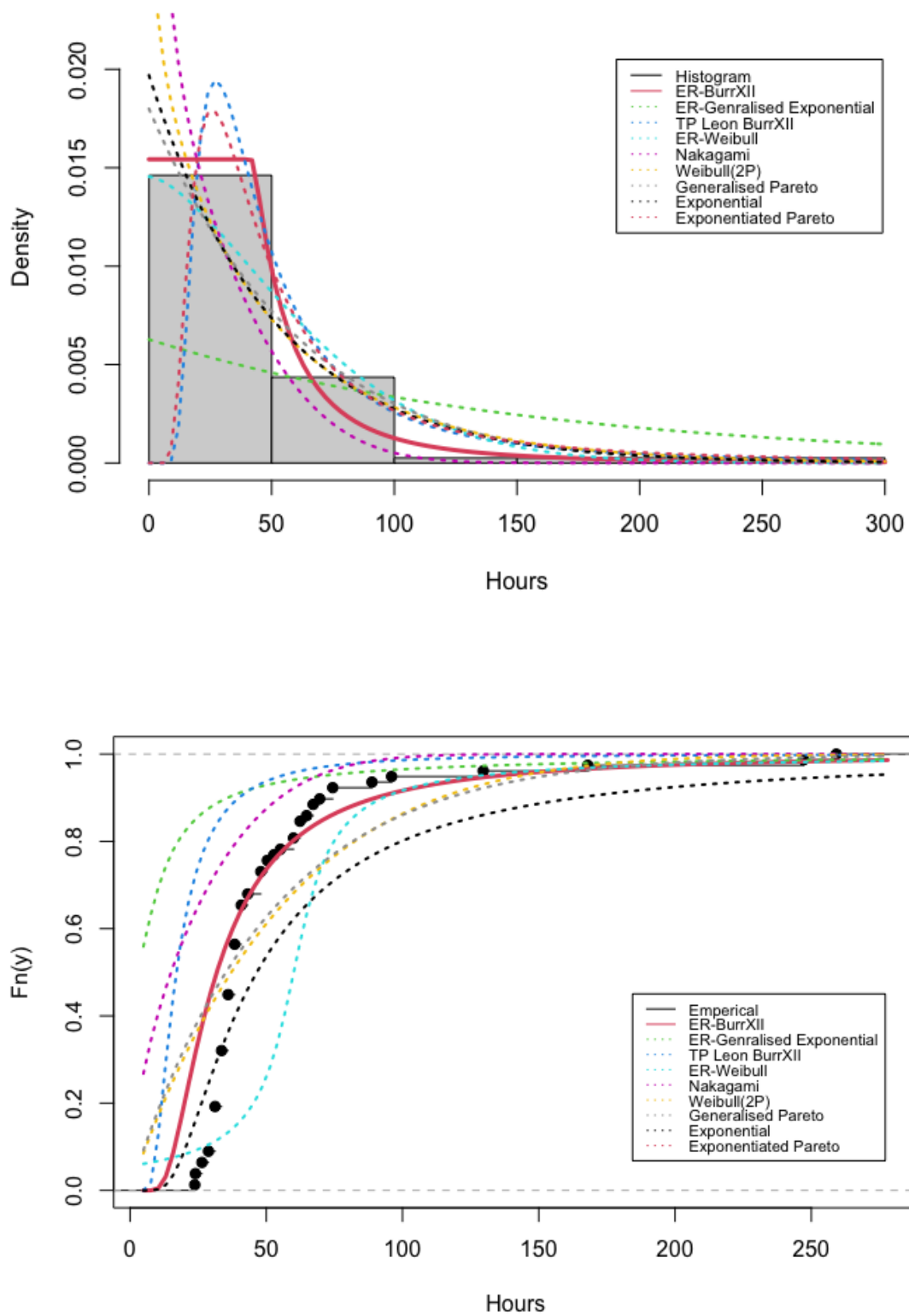


Figure 5.19: The estimated PDFs (top panel) and the estimated CDFs (bottom panel) length of stay of patients data in Nanumba district

5.6 Summary of Simulations and Applications

In this chapter, the simulation studies and applications of both the MEC distribution and ER-BurrXII distribution were presented. In the simulation study, the maximum likelihood estimation approach was considered the best technique for estimating the parameters of the MEC and ER-BurrXII distribution. Comparative data modeling application of the proposed models with some competing distributions was carried out considering eleven different data sets. The results showed that the MEC and ER-BurrXII distributions both performed well for modeling different data sets.

CHAPTER 6

FINDINGS, CONCLUSIONS AND RECOMMENDATIONS

6.1 Findings

Probability distributions are fundamental in data modeling in a variety of industries, including insurance, economics, biology, and social sciences. Because of the dynamic nature of data, new statistical distributions with better capacity, flexibility, and adaptability are frequently necessary to properly characterize data.

This study proposed the modified extended Chen (MEC) through the use of the Burr-Hatke differential equation. The hazard function of the MEC distribution exhibits versatile behaviours such as increasing, decreasing, J-shaped, reversed-J shaped, unimodal and upside-down bathtub. These features are very attractive features that render the MEC distribution suitable for modelling monotonic and non-monotonic data sets. The PDF of the MEC distribution has varied shapes suitable for modelling right-skewed, and approximately symmetric survival data and also survival data with highly varied kurtosis. The statistical properties of the MEC distributions, such as the quantile function, moments, moment generating function and Rényi entropy were derived.

The Equilibrium renewal Burr XII (ER-BurrXII) distribution is developed through the use of the equilibrium renewal process. The PDF of the ER-BurrXII distribution can be right skewed or constant. It also exhibits some interesting shapes such as reverse S and J shapes. The hazard function of the ER-BurrXII distribution exhibits a decreasing, increasing and upside down bathtub shapes. The statistical properties of the ER-BurrXII

distributions, such as the quantile function, moments, moment generating function, mean deviations, order statistics, Rényi entropy were derived.

To construct estimators for the parameters of the MEC distribution, the maximum likelihood estimation, ordinary and weighted least squares techniques were used, and simulation experiments were conducted to test the estimators' qualities. The simulation results revealed that the maximum likelihood estimation method performed best in estimating the parameters of the MEC distribution. To create estimators for the parameters of the ER-BurrXII distribution, the maximum likelihood estimation approach was used, and simulation tests were conducted to assess the estimators' characteristics. The importance of the MEC distribution is demonstrated using annual maximum rainfall data in Ghana.

The ER-BurrXII distribution was applied to the remission times of patients suffering from bladder cancer, survival times of patients suffering from myelogenous leukaemia and length of stay of patients in the maternity wards in the northern region. The best model was however selected based on smaller AIC, BIC, CAIC and HQIC values together with higher values of W, A and K-S values. From the results, both the MEC and ER-BurrXII distributions outperformed the rest of the models as it had the highest log-likelihood values and the lowest values for all the measures of the information criteria considered for all the datasets. This was further visually supported by the plots of the histograms and empirical CDF of the fitted distributions.

6.2 Conclusions

The study developed two new statistical distributions, MEC and ER-Burr XII distributions, through the use of the Burr-Hatke differential equation and equilibrium renewal

process to improve upon their flexibility in modeling univariate datasets. The hazard and density functions of both distributions exhibit various shapes of failure rates, both monotonic and non-monotonic failure rates.

Some Statistical properties such as the quantile function, moments, moment generating function, mean deviations, order statistics, Rényi entropy were derived for the developed distributions. The maximum likelihood estimation, weighted and ordinary least squares methods were used in developing the estimators of the MEC distribution. However, for the ER-Burr XII distribution, only the maximum likelihood estimation approach was used.

The Monte Carlo simulation was performed to examine the performance of the estimators for the developed distributions. The MEC distribution was applied to five (5) maximum rainfall data sets in across five locations in Ghana. In all these locations, it can be concluded that, the MEC distribution performed better in all data sets than some competing distributions since it recorded lowest AIC, BIC, CAIC and HQIC values and high KS, A and W values. Finally, the ER-Burr XII distribution was applied to six (6) data sets. The distributions of these data sets exhibited varied shapes of skewness and kurtosis. The ER-Burr XII distribution performed better compared to other competing distributions. It can be concluded that, both the MEC and ER-Burr XII distributions can be applied to data sets that are both right and left skewed, heavily tailed and approximately symmetric data sets that have upside down bathtub failure rate.

6.3 Contributions to Knowledge

Generally, the study contributes to knowledge with the establishment of two new statistical distributions. Specific contributions to knowledge are summarised as follows;

- i. The scope of developing a modification of the extended Chen distribution is to propose a new probability distribution which can be used to describe the behaviour of extremes. The challenge of using the extended Chen model for extreme values is that even though it covers the large data sets well, it fails to handle the behaviour extreme values due to the nature of the tail. In this study, the developed MEC distribution exhibits varied shapes such as right-skewed, approximately symmetric, with large upper tails for univariate extreme values.
- ii. The study also derived some of the properties of the developed distributions which are useful for undertaking inferential statistics as well as the reliability functions which are useful in the field of engineering.
- iii. Although the Burr XII distribution has several advantages, one of its most significant disadvantages is that it lacks hazard function flexibility. Based on the limitation of the Burr XII distribution, the study developed a three-parameter modification of the Burr XII distribution using the equilibrium renewal process and call it the ER-Burr XII distribution. The novel distribution, as demonstrated in this study provides different types of shapes of the PDF and hazard function. Particularly increasing, decreasing, upside down bathtubs, unimodal and constant hazard function shapes, which are not always reached for by the Burr XII distribution.

6.4 Recommendations

- i. In all the distributions discussed, the thesis used only MLE for estimating the parameters of the ER-BurrXII distribution and discussed the performance of the estimator of OLS, MLE and WLS method for the MEC distribution. However,

there are different methods available in literature including the Maximum Product Spacing, Percentile and Cramer–von Mises Method. Therefore, further research should consider using different methods estimation procedures and compare the results obtained.

- ii. The research work recommends exploring the statistical properties of the ER-Burr XII at an in-depth level and examining its efficiency with respect to the existing distributions in the field of reliability, especially in the context of the equilibrium renewal densities.
- iii. The study recommends that, policy makers should provide more drainage systems and sea defense walls to mitigate the estimated high rainfall levels in the future especially in location Cape Coast.
- iv. The MEC distribution exhibits varied shapes including a highly positive skewed shape. The research limited the applicability of the MEC distribution to model extreme rainfall data, this is due to the shape of the MEC distribution having a long and heavy upper tail. Hence, the research work recommends exploring the application of the MEC distribution to the insurance industry to model payment data due to the fact that majority of payment data sets are highly positive skewed in nature.
- v. Many situations in the health and meteorology sector require consideration of various factors of more than one variable. In this study, it is not adequate to study a univariate analysis of extreme annual rainfall, remission, survival and length of stay of patients. The MEC, and ER-Burr XII distributions can be extended to the bivariate and multivariate cases.

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APPENDICES

Appendix A Secondary Data

Table A1: Remission times of bladder cancer patients.

0.080	0.200	0.400	0.500	0.510	0.810	0.900	1.050	1.190	1.260	1.350	1.400	1.460
1.760	2.020	2.020	2.870	3.020	4.230	4.260	5.340	5.410	7.260	7.280	8.660	9.020
12.03	12.07	17.36	18.10	46.12	79.05	2.070	2.090	2.230	2.260	2.460	2.540	2.620
2.640	2.690	2.690	2.750	2.830	3.250	3.310	3.360	3.360	3.480	3.520	3.570	3.640
3.700	3.820	3.880	4.180	4.330	4.340	4.400	4.500	4.510	4.870	4.980	5.060	5.090
5.170	5.320	5.320	5.41	5.490	5.620	5.710	5.850	6.250	6.540	6.760	6.930	6.940
6.970	7.090	7.320	7.390	7.590	7.620	7.630	7.660	7.870	7.930	8.260	8.370	8.530
8.650	9.220	9.470	9.740	10.06	10.34	10.66	10.75	11.25	11.64	11.79	11.98	12.02
12.63	13.11	13.29	13.80	14.24	14.76	14.77	14.83	15.96	16.62	17.12	17.14	19.13
20.28	21.73	22.69	23.63	25.74	25.82	26.31	32.15	34.26	36.66	43.01		

Table A2: Survival times of patients suffering from acute myelogenous leukaemia.

65	156	100	134	16	108	121	4	39	143
56	26	22	1	1	5	65	56	65	17
7	16	22	3	4	2	3	8	4	3
30	4	43							

Appendix B R Code

```
## ## ## MEC distribution CDF ## ##
MEC_CDF<-function(x,a,b,rho){
  B<-(rho*(exp(x-b)-1)+1)-a
  CDF<-B
  return(CDF)
}
```

```
## ## MEC distribution PDF ## ##
MEC_PDF<-function(y,a,b,rho){
  A<-a*b*rho*y(-b-1)*exp(y-b)
  B<-(rho*(exp(y-b)-1)+1)-a-1
  PDF<-A*B
  return(PDF)
}
```

```
## ## MEC distribution Hazard ## ##
MEC_hazard<-function(y,a,b,rho){
  A<-a*b*rho*y(-b-1)*exp(x-b)
```

```

B<-1-(rho*(exp(x^(-b))-1)+1)^(-a-1)

hazard<-A/B

return(hazard)

}

```

MEC distribution Quantile

```

quantile<-function(a,b,rho,u){

  A<-1+(((u^(-1/a))-1)/rho)

  quant<-(log(A))^(-1/b)

  return(quant)

}

```

MEC Distribution Negative Loglikelihood for Optimization###

```

MEC_LL<-function(a,b,rho){

  A<-a*b*rho*x^(-b-1)*exp(x^(-b))

  B<-(rho*(exp(x^(-b))-1)+1)^(-a-1)

  PDF<-A*B

  LL<-sum(log(PDF))

  return(LL)

}

```

#####

MEC Distribution Optimization

#####

```

library (bbmle)

fit<-mle2(MEC_LL,start=list(a=a, b=b, rho=rho),data=list(x),
method="BFGS")

profile(fit)

summary(fit)


# # # # # Simulation of MEC distribution # # # # #


# # Quantile Function # #

quantile<-function(a,b,rho,u){

  A<-1+(((u^(-1/a))-1)/rho)

  quant<-(log(A))^(-1/b)

  return(quant)

}

##### Loglikelihood_MLE #####

MECH_LL<-function(par){

  a=par[1]

  b=par[2]

  rho=par[3]

  B<-a*b*rho*(x^(-b-1))*(exp(x^(-b)))

  C<-(1+rho*(exp(x^(-b))-1))^(-a-1)

  PDF<-B*C

  LL<-sum(log(PDF))

  return(LL)

```

```
}
```

```
###Algorithm for Monte Carlo Simulation Study###
```

```
library(numDeriv)
```

```
library(Matrix)
```

```
a=#
```

```
b=#
```

```
rho=#
```

```
n1=c(##,##,##,##)
```

```
for(j in 1:length(n1)){
```

```
  n=n1[j]
```

```
  N=1000
```

```
  mle_a<-c(rep(0,N))
```

```
  mle_b<-c(rep(0,N))
```

```
  mle_rho<-c(rep(0,N))
```

```
  LC_a<-c(rep(0,N))
```

```
  UC_a<-c(rep(0,N))
```

```
  LC_b<-c(rep(0,N))
```

```
  UC_b<-c(rep(0,N))
```

```
  LC_rho<-c(rep(0,N))
```

```
  UC_rho<-c(rep(0,N))
```

```
  count_a=0
```

```
  count_b=0
```

```
  count_rho=0
```

```

temp=1

HH1<-matrix(c(rep(2,9)),nrow=3,ncol=3)

HH2<-matrix(c(rep(2,9)),nrow=3,ncol=3)

for(i in 1:N)
{
  print(i)

  flush.console()

  repeat{
    x<-c(rep(0,n))

    # Generate a random variable from uniform distribution

    u<-0

    u<-runif(n,min=0,max=1)

    for(k in 1:n){
      x[k]<-quantile(a,b,rho,u[k])
    }

    mle.result<-nlminb(c(a,b,rho),MECH_LL,

    lower=c(0,0,0))

    temp=mle.result$convergence

    if(temp==0){
      temp_a<-mle.result$par[1]

      temp_b<-mle.result$par[2]

      temp_rho<-mle.result$par[3]

      HH1<-hessian(MECH_LL,

      c(temp_a,temp_b,temp_rho))
    }
  }
}

```

```

    if (sum(is.nan(HH1))==0 & (diag(HH1)[1] > 0)
        & (diag(HH1)[2] > 0) & (diag(HH1)[3] > 0)) {

        HH2<-solve(HH1)

        #print(det(HH1))

    }

    else {

        temp=1}

}

if ((temp==0) & (diag(HH2)[1] > 0) & (
diag(HH2)[2] > 0) & (diag(HH2)[3] > 0) & (sum(is.nan(HH2))==0)) {

    break

}

else {

    temp=1}

}

temp=1

mle_a[i]<-mle.result$par[1]

mle_b[i]<-mle.result$par[2]

mle_rho[i]<-mle.result$par[3]

HH<-hessian(MECH_LL,

c(mle_a[i],mle_b[i],mle_rho[i]))

H<-solve(HH)

LC_a[i]<-mle_a[i]-qnorm(0.975)*sqrt(diag(H)[1])

UC_a[i]<-mle_a[i]+qnorm(0.975)*sqrt(diag(H)[1])

```

```

if ((LC_a[i]<=a)&(a<=UC_a[i])) {

  count_a=count_a+1

}

LC_b[i]<-mle_b[i]-qnorm(0.975)*sqrt(diag(H)[2])
UC_b[i]<-mle_b[i]+qnorm(0.975)*sqrt(diag(H)[2])

if ((LC_b[i]<=b)&(b<=UC_b[i])) {

  count_b=count_b+1

}

LC_rho[i]<-mle_rho[i]-qnorm(0.975)*sqrt(diag(H)[3])
UC_rho[i]<-mle_rho[i]+qnorm(0.975)*sqrt(diag(H)[3])

if ((LC_rho[i]<=rho)&(rho<=UC_rho[i])) {

  count_rho=count_rho+1

}

}

#Calculate Average Bias
ABias_a<-sum(mle_a-a)/N
ABias_b<-sum(mle_b-b)/N
ABias_rho<-sum(mle_rho-rho)/N

print(cbind(ABias_a,ABias_b,ABias_rho))

#Calculate RMSE
RMSE_a<-sqrt(sum((a-mle_a)^2)/N)
RMSE_b<-sqrt(sum((b-mle_b)^2)/N)
RMSE_rho<-sqrt(sum((rho-mle_rho)^2)/N)

```

```

print(cbind(RMSE_a, RMSE_b, RMSE_rho))

}

##### ER-BurrXII Distribution PDF #####

ER.PDF<-function(x, a, c, k){

  A<-((x/a)^(c+1))^(-k)

  B<-((a*k)*beta(1/c+1, k-1/c))

  PDF<-A/B

  return(PDF)

}

##### importing library #####

library(hypergeo)

##### ER-BurrXII Distribution CDF #####

ER.CDF<-function(x, a, c, k){

  A<- (x*hypergeo_contfrac(1/c, k, 1+1/c, -(a/x)^(c),
    tol = 0, maxiter=200))

  B<- ((a*k)*beta(1/c+1, k-1/c))

  CDF<-A/B

  return(CDF)

```

```
}
```

```
##### ER-BurrXII Distribution Log likelihood function #####
```

```
ER_LL2<-function(a,c,k){  
  A<-((x/a)^(c+1))^(-k)  
  B<-((a*k)*beta(1/c+1,k-1/c))  
  PDF<-A/B  
  LL<-sum(log(PDF))  
  return(LL)  
}
```

```
##### ER-BurrXII Distribution Quantile function #####
```

```
quantile<-function(a,c,alpha,u){  
  f<-function(x){(x*hypergeo_shanks(1/c,alpha,1+(1/c),  
    (a/x)^(-c)))/((a*alpha)*beta(1/c+1,alpha-1/c))-u}  
  roots<-uniroot(f,lower=0,upper=1000,tol=1e-15)$root  
  return(roots)  
}
```

```
##### ER-BurrXII distribution Optimization #####
```

```
fit<-mle2(ER_LL2, start=list(a=a,c=c,k=k), data=list(x),method="BFGS")  
summary(fit)
```

```
#####
```

Simulation of ER-Burr XII distribution

#####

#####Algorithm for Monte Carlo Simulation Study #####

library(numDeriv)

library(Matrix)

a=a

c=c

k=k

n1=**c**(#, #, #, #, #)

for(j in 1:length(n1)){

 n=n1[j]

 N=100

 mle_a<-**c**(**rep**(0,N))

 mle_c<-**c**(**rep**(0,N))

 mle_alpha<-**c**(**rep**(0,N))

 LC_a<-**c**(**rep**(0,N))

 UC_a<-**c**(**rep**(0,N))

 LC_c<-**c**(**rep**(0,N))

 UC_c<-**c**(**rep**(0,N))

 LC_k<-**c**(**rep**(0,N))

 UC_k<-**c**(**rep**(0,N))

count_a=0

count_c=0

```

count_k=0

temp=1

HH1<-matrix(c(rep(2,9)),nrow=3,ncol=3)
HH2<-matrix(c(rep(2,9)),nrow=3,ncol=3)

for(i in 1:N)
{
  print(i)

  flush.console()

  repeat{
    x<-c(rep(0,n))

    # Generate a random variable from uniform distribution

    u<-0

    u<-runif(n,min=0,max=1)

    for(k in 1:n){
      x[k]<-quantile(a,c,k,u[k])
    }

    #Maximum likelihood estimation

    mle.result<-nlminb(c(a,c,k),ERBURRXII_LL,lower=c(0,0,0))

    temp=mle.result$convergence

    if(temp==0){
      temp_a<-mle.result$par[1]

      temp_c<-mle.result$par[2]

      temp_alpha<-mle.result$par[3]

      HH1<-hessian(ERBURRXII_LL, c(temp_a,temp_c,temp_alpha))
    }
  }
}

```

```

    if (sum(is.nan(HH1))==0 & (diag(HH1)[1] > 0) &
        (diag(HH1)[2] > 0) & (diag(HH1)[3] > 0)) {
        HH2<-solve(HH1)
        #print(det(HH1))
    }
    else {
        temp=1}
}

if ((temp==0) & (diag(HH2)[1] > 0) &
    (diag(HH2)[2] > 0) &
    (diag(HH2)[3] > 0) & (sum(is.nan(HH2))==0)) {
    break
}

else {
    temp=1}
}

temp=1

mle_a[i]<-mle.result$par[1]
mle_c[i]<-mle.result$par[2]
mle_k[i]<-mle.result$par[3]

HH<-hessian(ERBURRXII_LL,c(mle_a[i],mle_c[i],mle_alpha[i]))
H<-solve(HH)

LC_a[i]<-mle_a[i]-qnorm(0.975)*sqrt(diag(H)[1])
UC_a[i]<-mle_a[i]+qnorm(0.975)*sqrt(diag(H)[1])

```

```

if ((LC_a[i] <= a) & (a <= UC_a[i])) {

  count_a = count_a + 1

}

LC_c[i] <- mle_c[i] - qnorm(0.975) * sqrt(diag(H)[2])
UC_c[i] <- mle_c[i] + qnorm(0.975) * sqrt(diag(H)[2])

if ((LC_c[i] <= c) & (c <= UC_c[i])) {

  count_c = count_c + 1

}

LC_k[i] <- mle_k[i] - qnorm(0.975) * sqrt(diag(H)[3])
UC_k[i] <- mle_k[i] + qnorm(0.975) * sqrt(diag(H)[3])

if ((LC_k[i] <= k) & (k <= UC_k[i])) {

  count_k = count_k + 1

}

}

#Calculate Average Bias
ABias_a <- sum(mle_a - a) / N
ABias_c <- sum(mle_c - c) / N
ABias_k <- sum(mle_k - alpha) / N

print(cbind(ABias_a, ABias_c, ABias_k))

#Calculate RMSE
RMSE_a <- sqrt(sum((a - mle_a)^2) / N)
RMSE_c <- sqrt(sum((c - mle_c)^2) / N)
RMSE_k <- sqrt(sum((k - mle_k)^2) / N)

```

```
print(cbind(RMSE_a, RMSE_c, RMSE_k))  
}
```

Appendix C Ghana Health Service Ethical Clearance

GHANA HEALTH SERVICE ETHICS REVIEW COMMITTEE

*In case of reply the
number and date of this
Letter should be quoted.*



My Ref: GHS/RDD/ERC/Admin/App 121/528
Your Ref. No.

Research & Development Division
Ghana Health Service
P. O. Box MB 190
Accra
Digital Address: GA-050-3303
Mob: +233-50-3539896
Tel: +233-302-681109
Fax + 233-302-685424
Email: ethics.research@ghsml.org
25th November 2021

Abdulzeid Yen Anafo
University of Mines and Technology
Faculty of Engineering Sciences, Department of Mathematical Sciences
Tarkwa, Ghana

The Ghana Health Service Ethics Review Committee has reviewed and given approval for the implementation of your Study Protocol.

GHS-ERC Number	GHS-ERC: 003/10/21
Study Title	Development of Statistical Distributions from the Burr-Hatke Differential Equation and Equilibrium Renewal Process
Approval Date	25 th November, 2021
Expiry Date	24 th November, 2022
GHS-ERC Decision	Approved

This approval requires the following from the Principal Investigator

- Submission of a yearly progress report of the study to the Ethics Review Committee (ERC)
- Renewal of ethical approval if the study lasts for more than 12 months,
- Reporting of all serious adverse events related to this study to the ERC within three days verbally and seven days in writing.
- Submission of a final report after completion of the study
- Informing ERC if study cannot be implemented or is discontinued and reasons why
- Informing the ERC and your sponsor (where applicable) before any publication of the research findings.

You are kindly advised to adhere to the national guidelines or protocols on the prevention of COVID -19

Please note that any modification of the study without ERC approval of the amendment is invalid.

The ERC may observe or cause to be observed procedures and records of the study during and after implementation.

Kindly quote the protocol identification number in all future correspondence in relation to this approved protocol

SIGNED.....
Dr. Cynthia Bannerman
(GHS-ERC Chairperson)

Cc: The Director, Research & Development Division, Ghana Health Service, Accra

Appendix D List of Publications

- i. Anafo, A. Y., Brew, L., Nasiru, S. (2022). "The Modified Extended Chen Distribution: Properties and Application to Rainfall Data", *Applied Mathematics Information Sciences*, Vol. 16, No. 1, pp. 121-139.
- ii. Anafo, A. Y., Brew, L., Nasiru, S. (2021). "The Equilibrium Renewal Burr XII Distribution: Properties and Applications", *Asian Journal of Probability and Statistics*, Vol. 15, No. 2, pp. 18-40. <https://doi.org/10.9734/ajpas/2021/v15i230349>